

AI Agent for Coding Claude Code



**[https://github.com/up1/
workshop-claude-code-cowork](https://github.com/up1/workshop-claude-code-cowork)**



AI Agent for coding with Claude Code



Let's Start with AI



How to use AI ?

AI as
a **tool**

AI as
an **assistant**

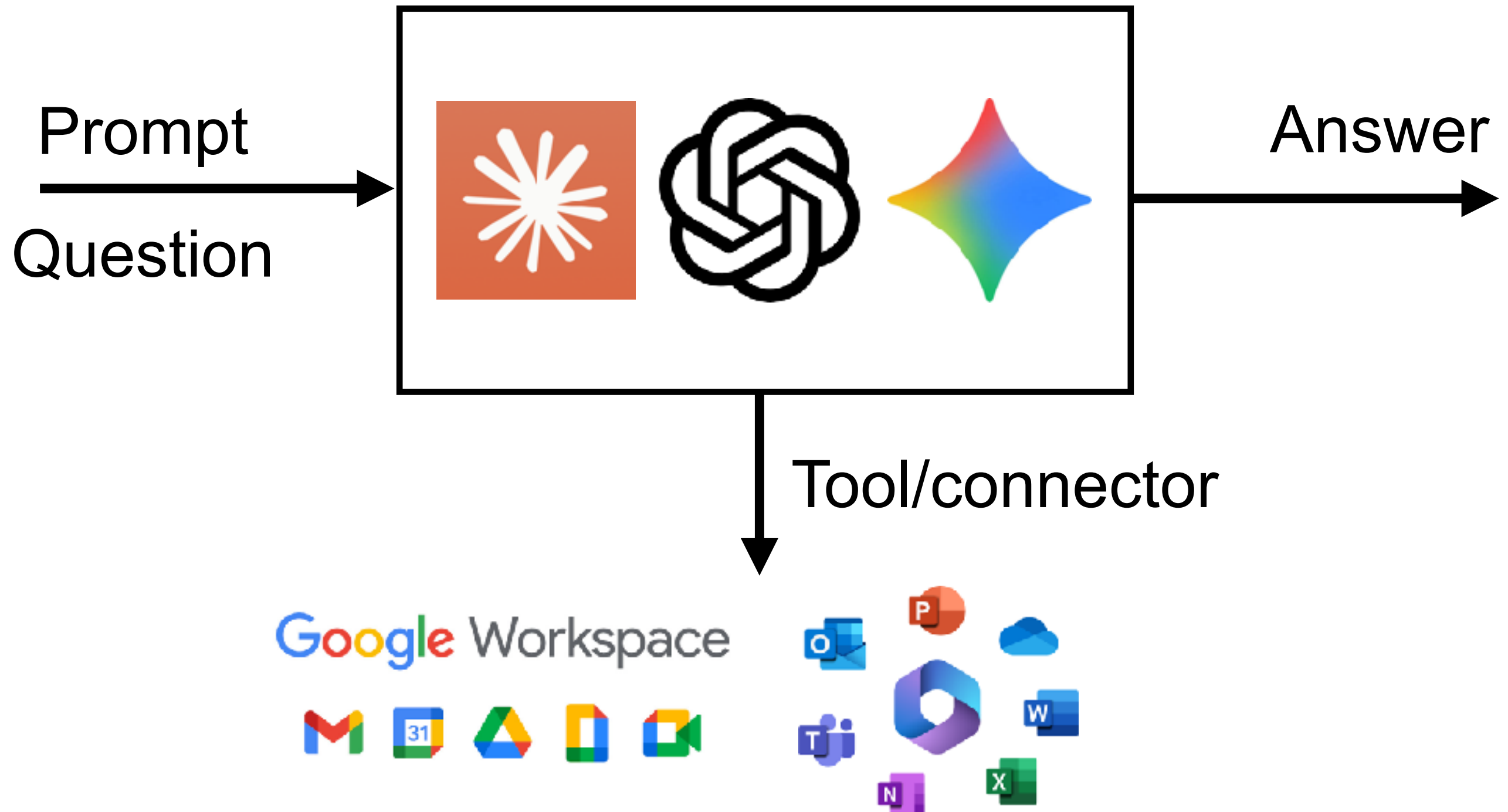
AI as
an **agent**



Chat-based

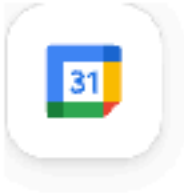


Use tool to connect other app !!



Connect to Google Calendar

[< Back](#)



Google Calendar

Manage your schedule and coordinate meetings effortlessly

[Connect](#)

Connect Google Calendar to Claude to view your schedule, manage events, and coordinate meetings. Claude can search your calendar for events, check your availability, find free time slots, create and update events, respond to invitations, and help you prepare for meetings. Useful for understanding what's coming up, scheduling new meetings by finding mutual availability, managing your calendar by creating or updating events, coordinating schedules across multiple people, or preparing for meetings by reviewing attendee lists and details.

Developed by [Google](#) 

Only use connectors from developers you trust. Anthropic does not control which tools developers make available and cannot verify that they will work as intended or that they won't change.

<https://claude.ai/directory/connectors/google-calendar-calendarmcp>



Example

1. Prompt

2. Finding tools

3. Answer

ในสัปดาห์หน้ามีงานอะไร

Pinpointed next week's date range for scheduling inquiry ~

🕒 The user is asking in Thai about upcoming events, so I need to check their Google Calendar for next week's schedule. I'll need to use the tool search first before I can access the calendar data.

🔍 Finding tools

Result

🕒 I'm noting that next week runs from Monday, June 15 through Sunday, June 21, 2026.

📅 Lists calendar events in a given calendar.

Result

🕒 Done

สัปดาห์หน้า (15-21 มิ.ย.) มี 3 รายการครับ:

อังคาร 16 มิ.ย. • 18:30-20:30 น.
[WLB26 - Livestream] สาธิตประสบการณ์การใช้งาน Agentic AI AWS KIRO ใน Software Development — มี Google Meet (จัด โดยคุณ Prathan ตอบรับแล้ว)

พุธ 17 มิ.ย. 18:00 น. - พฤหัสบดี 18 มิ.ย. 19:00 น.
Somkiat: Claude Code and SDD to build product — มีรายละเอียดเรื่องใบเสนอราคาและ tech stack (Golang, Vue.js, MongoDB ฯลฯ) อยู่ในคำอธิบายงาน

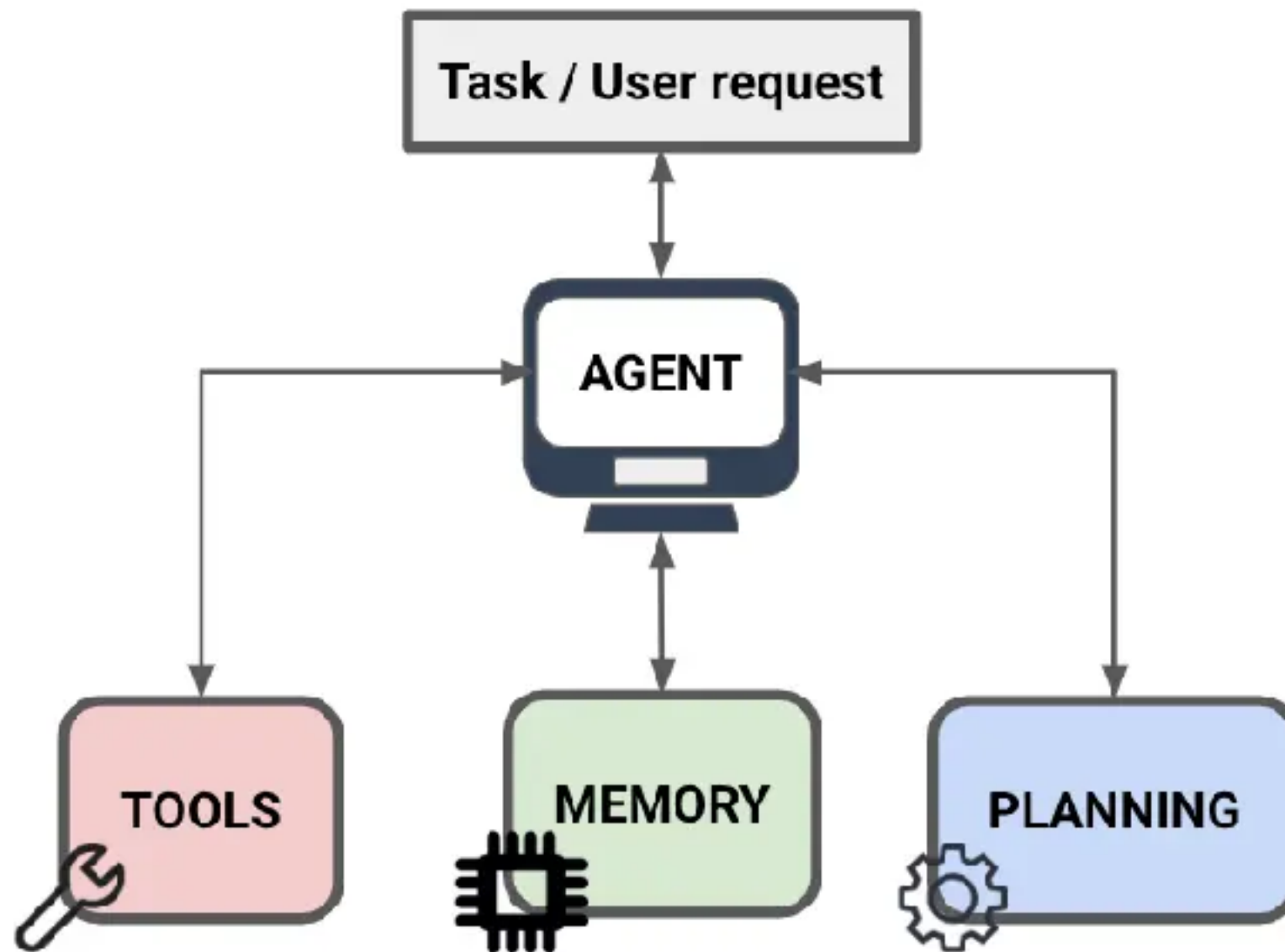
ศุกร์ 19 - จันทร์ 22 มิ.ย. (ทั้งวัน)
[somkiat] กลับบ้าน

วันจันทร์ 15 มิ.ย. ยังว่างอยู่ครับ



AI Agent

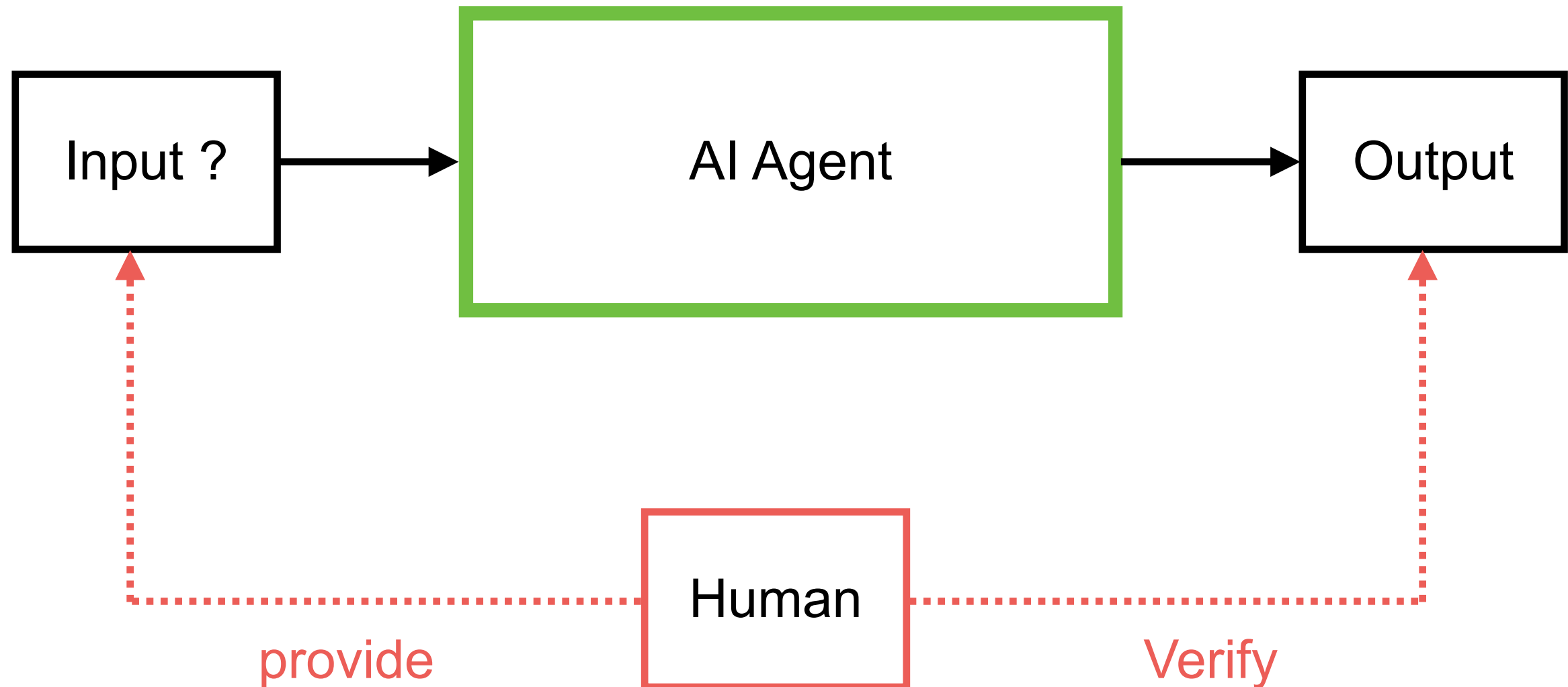
Define your goal



<https://www.promptingguide.ai/agents/introduction>



Coding with AI Agent



AI Agent for Coding



Codex



Antigravity



GitHub Copilot



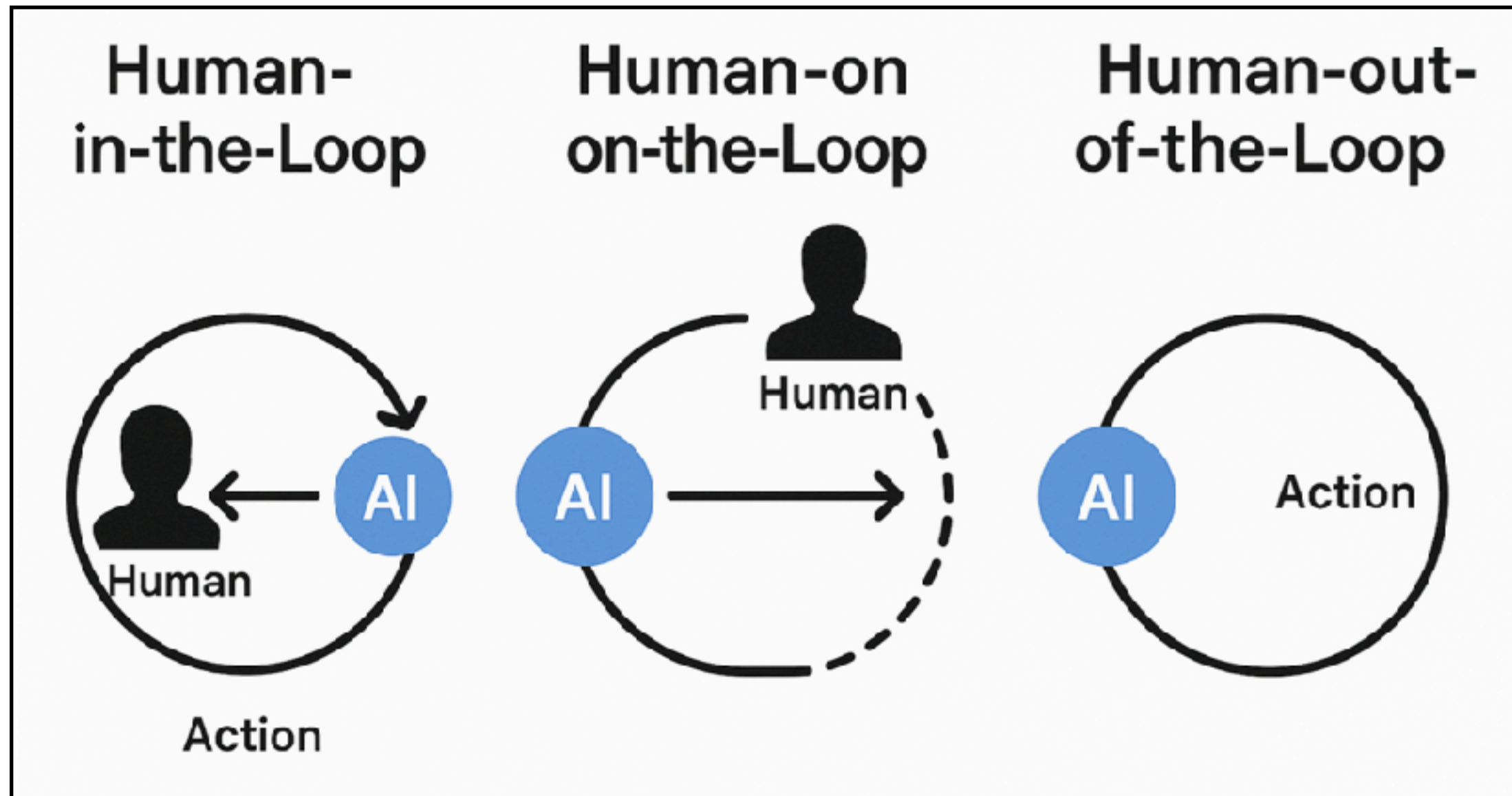
CURSOR



Cline

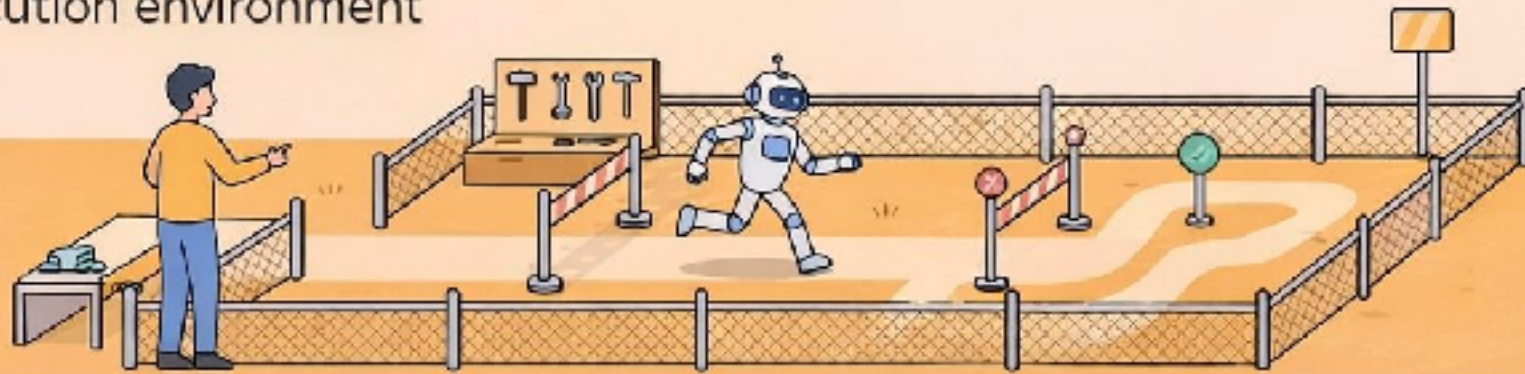


AI Agent with/without Human



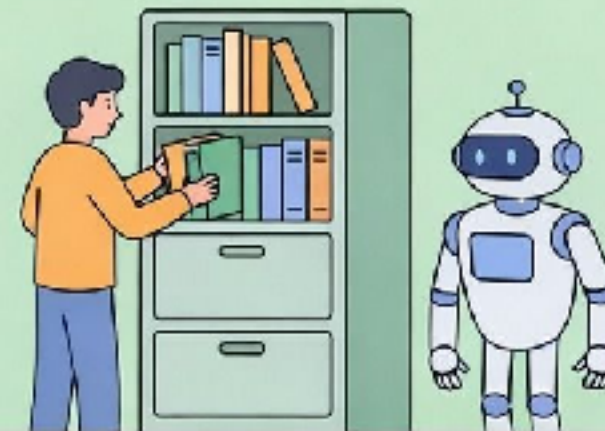
Harness Engineering — Build the World

Design the execution environment



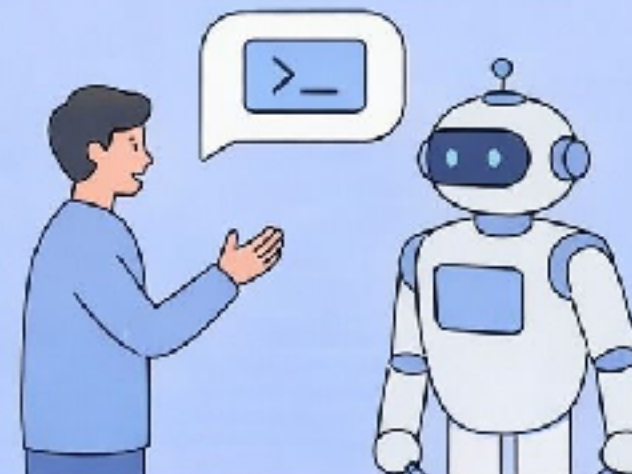
Context Engineering — What to See

Manage visible information



Prompt Engineering — What to Say

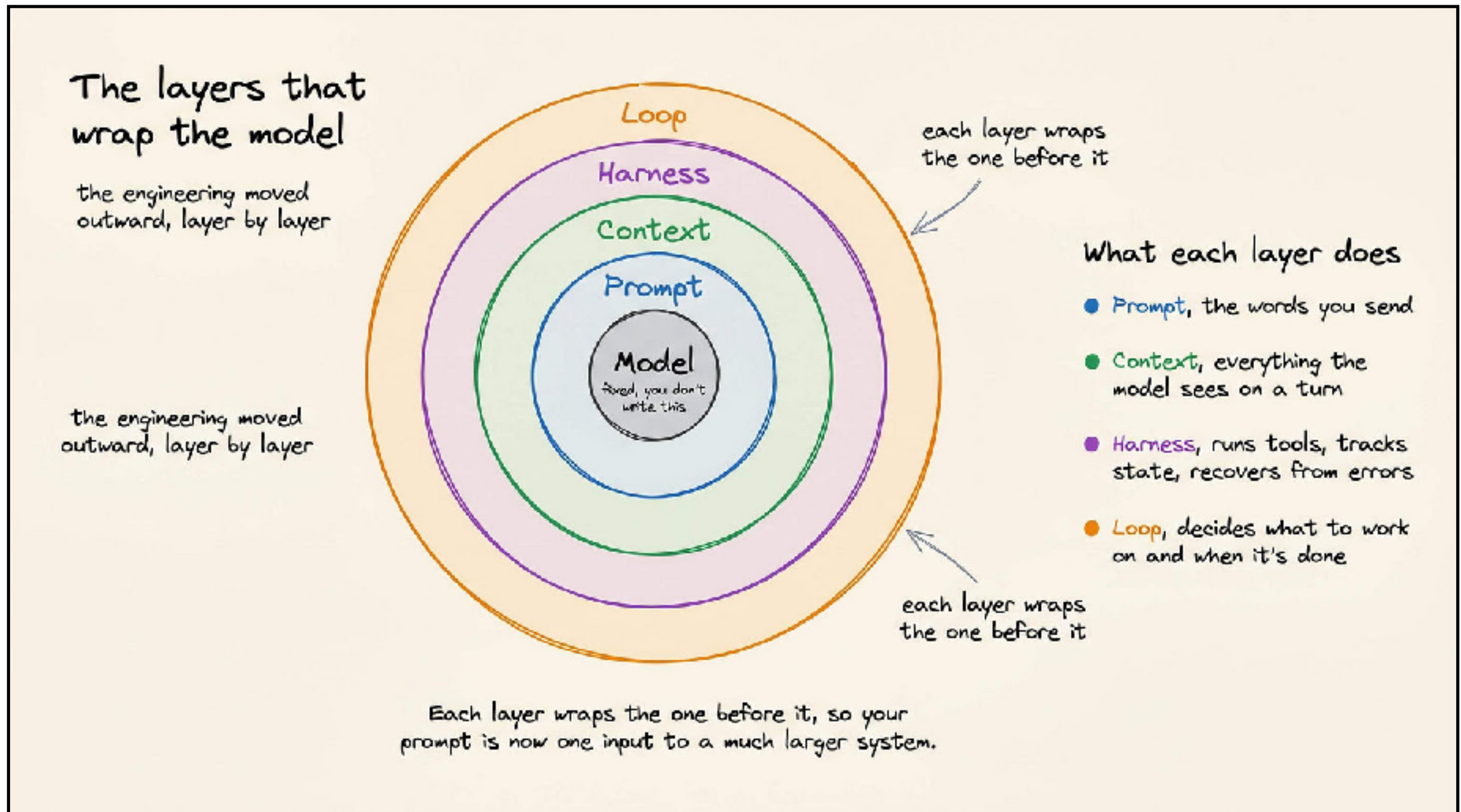
Optimize single-turn interactions



<https://milvus.io/blog/harness-engineering-ai-agents.md>



Loop Engineering



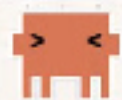
<https://www.dailydoseofds.com/p/the-anatomy-of-an-agent-harness/>



Graph Engineering

Graph *Engineering*

Orchestrate **1000** agents from a single prompt



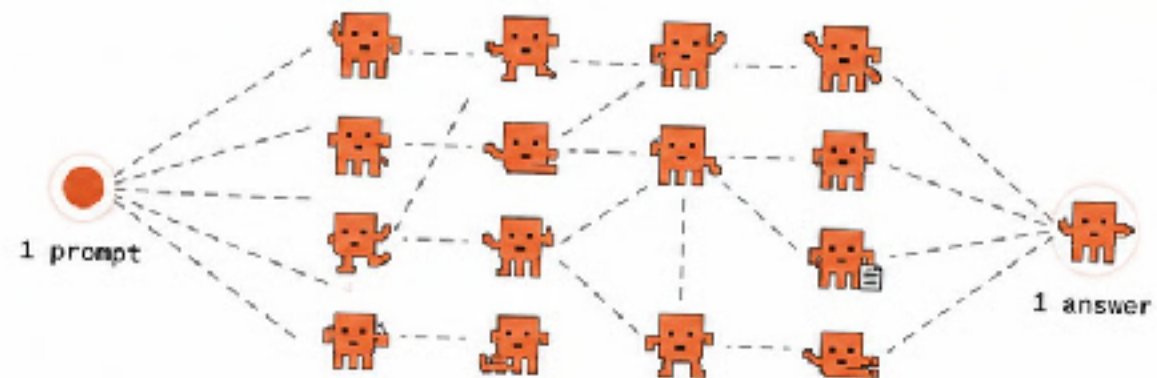
5-STEP FULL COURSE

LOOP



one at a time • slow

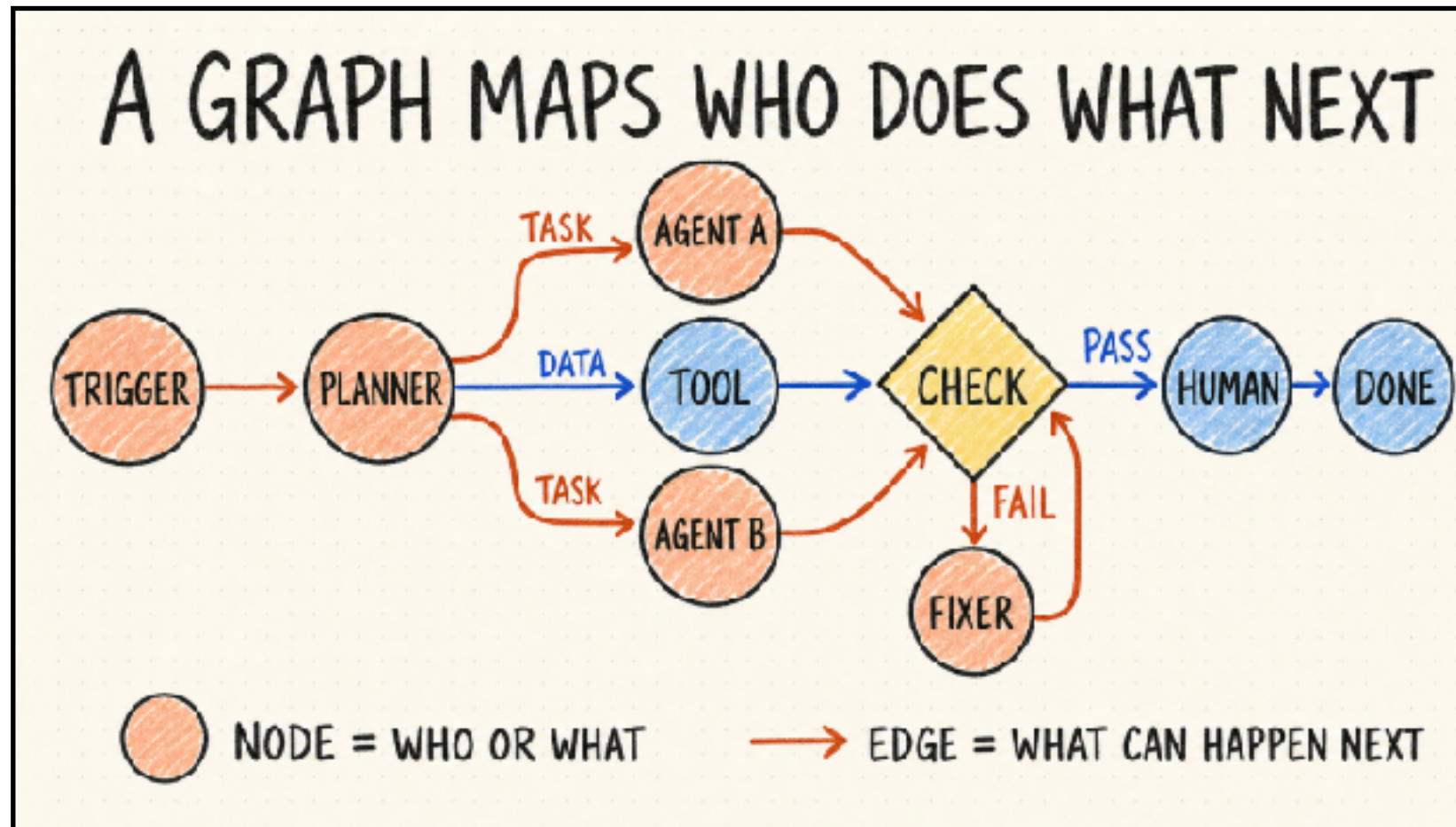
PARALLEL GRAPH



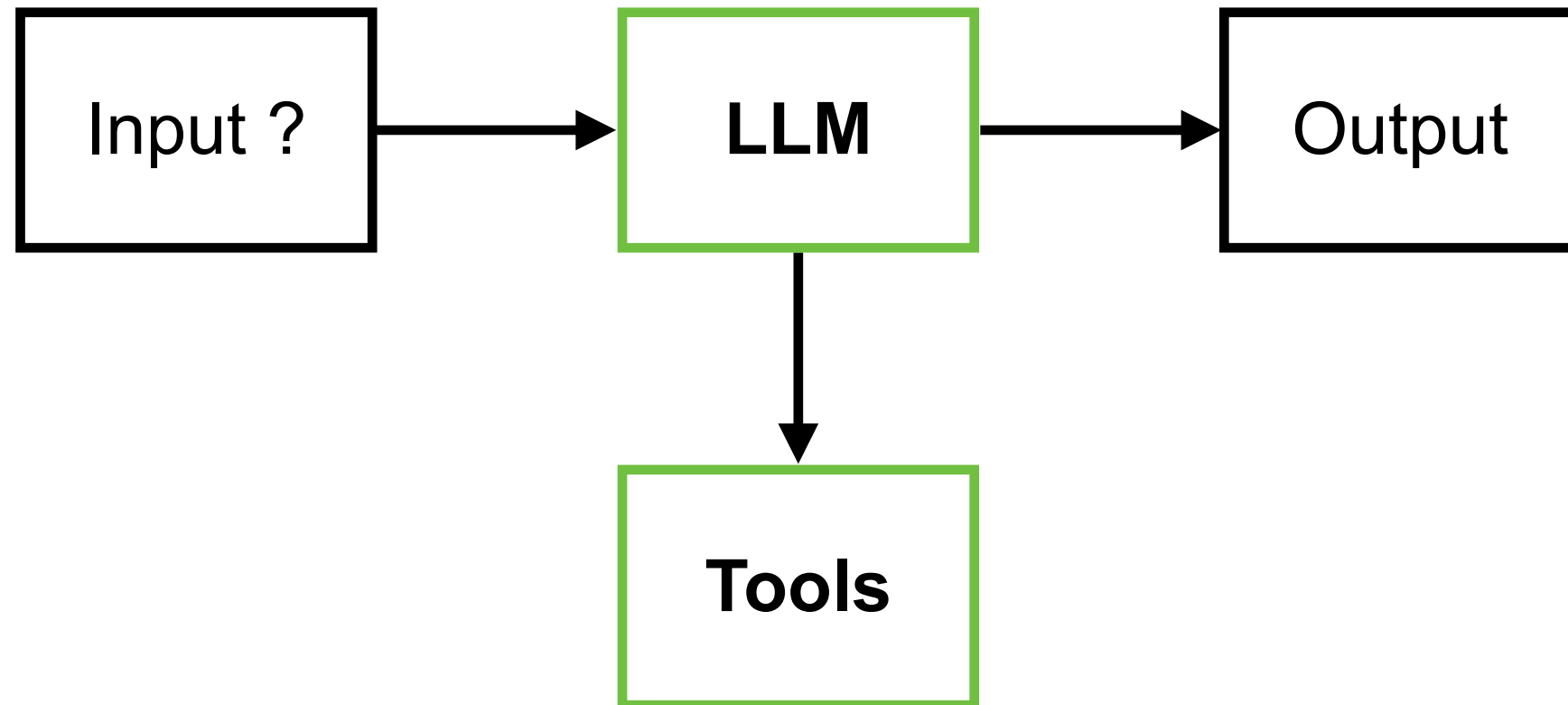
up to 1000



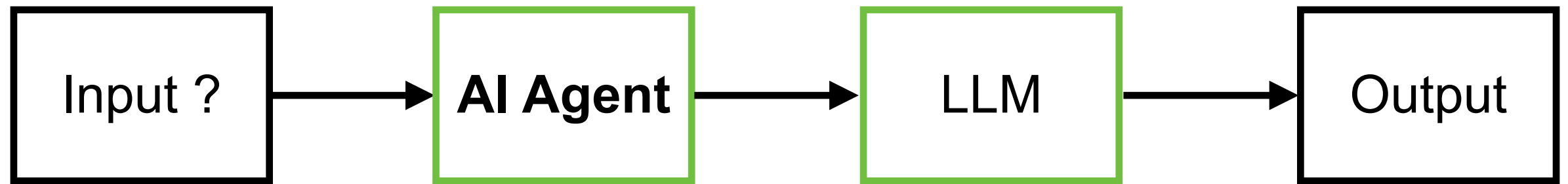
Graph Engineering



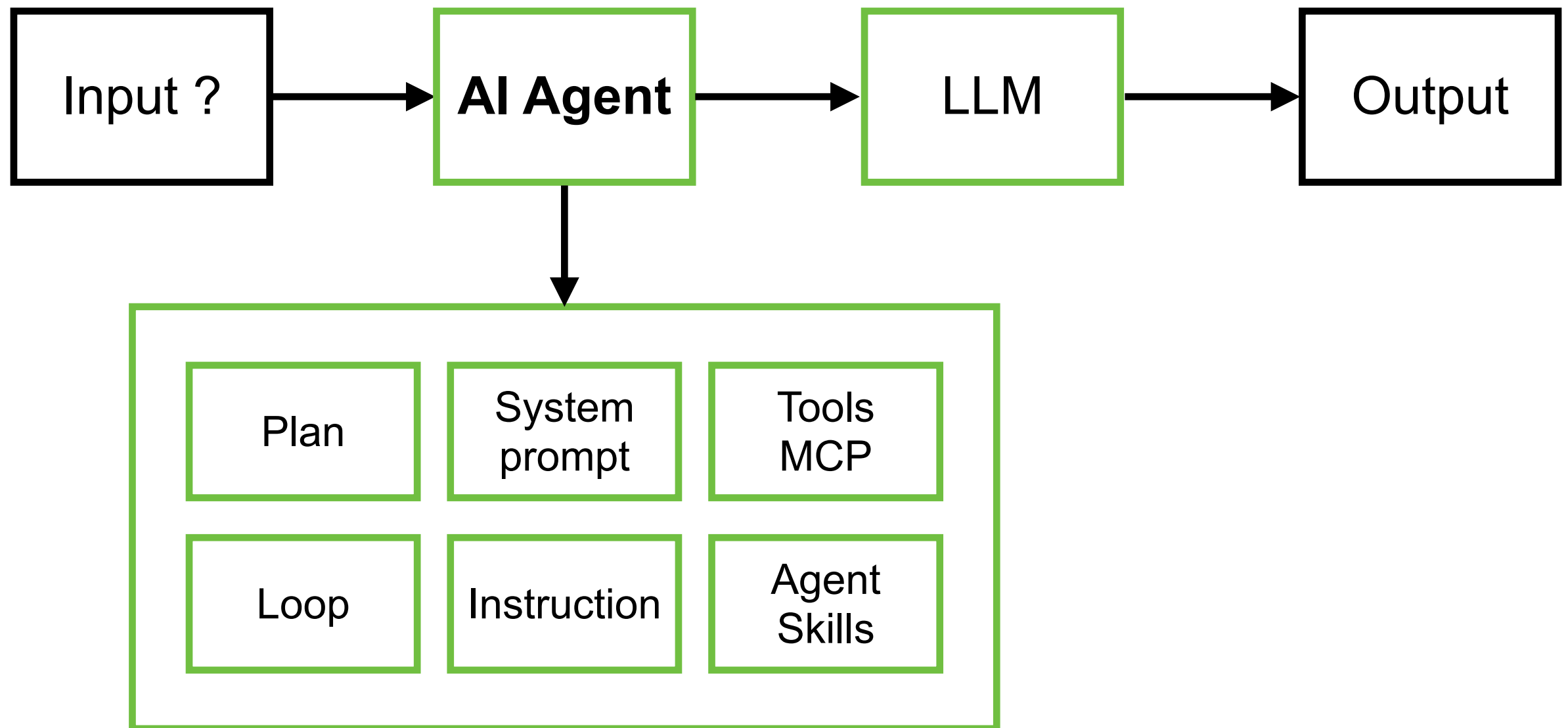
Start with Chat



Working with AI Agent



Working with AI Agent



AI Agent for Coding



Using Claude Code

CLI

Claude Desktop app

Extension in VS Code/Cursor

<https://claude.com/product/claude-code>



Claude Code in VS Code



Claude Code for VS Code

Anthropic  anthropic.com |  18,206,281 installs |  (713)

Claude Code for VS Code: Harness the power of Claude Code without leaving your IDE

[Install](#) [Trouble Installing?](#)

[Overview](#) [Version History](#) [Q & A](#) [Rating & Review](#)

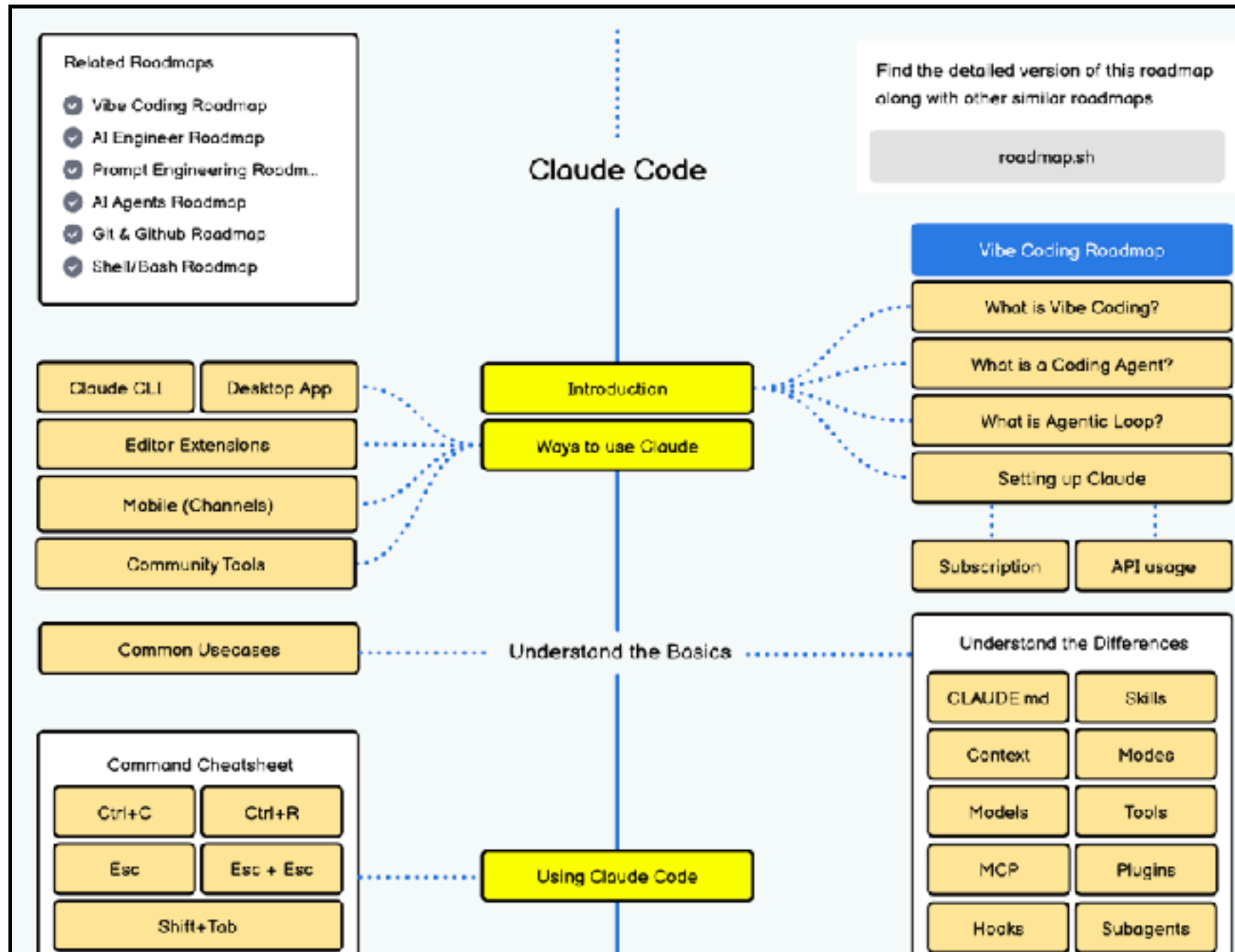
Claude Code for VS Code

Unleash Claude's raw power directly in your terminal. Search million-line codebases instantly. Turn hours-long workflows into a single command. Your tools. Your workflow. Your codebase, evolving at thought speed.

<https://marketplace.visualstudio.com/items?itemName=anthropic.claude-code>



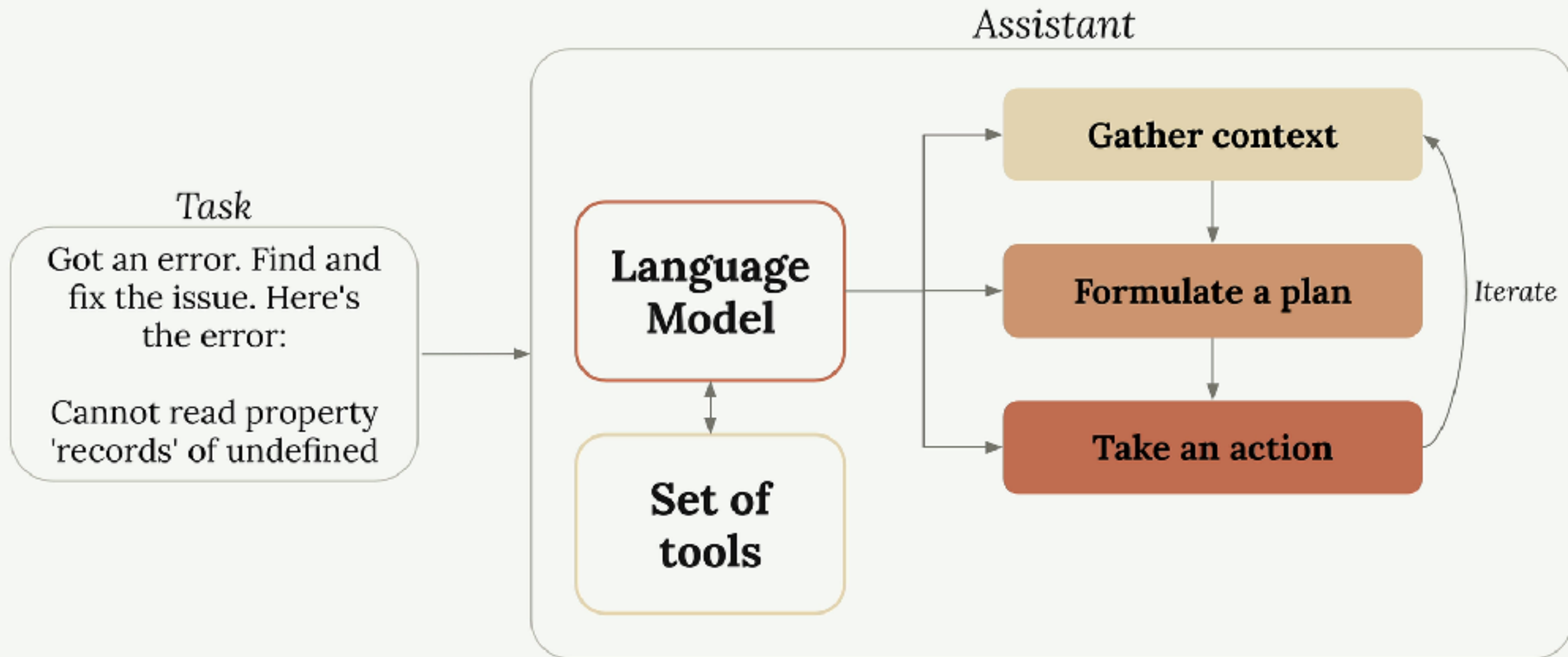
Roadmap of Claude Code



<https://roadmap.sh/claude-code>



How Coding Assistant Work ?

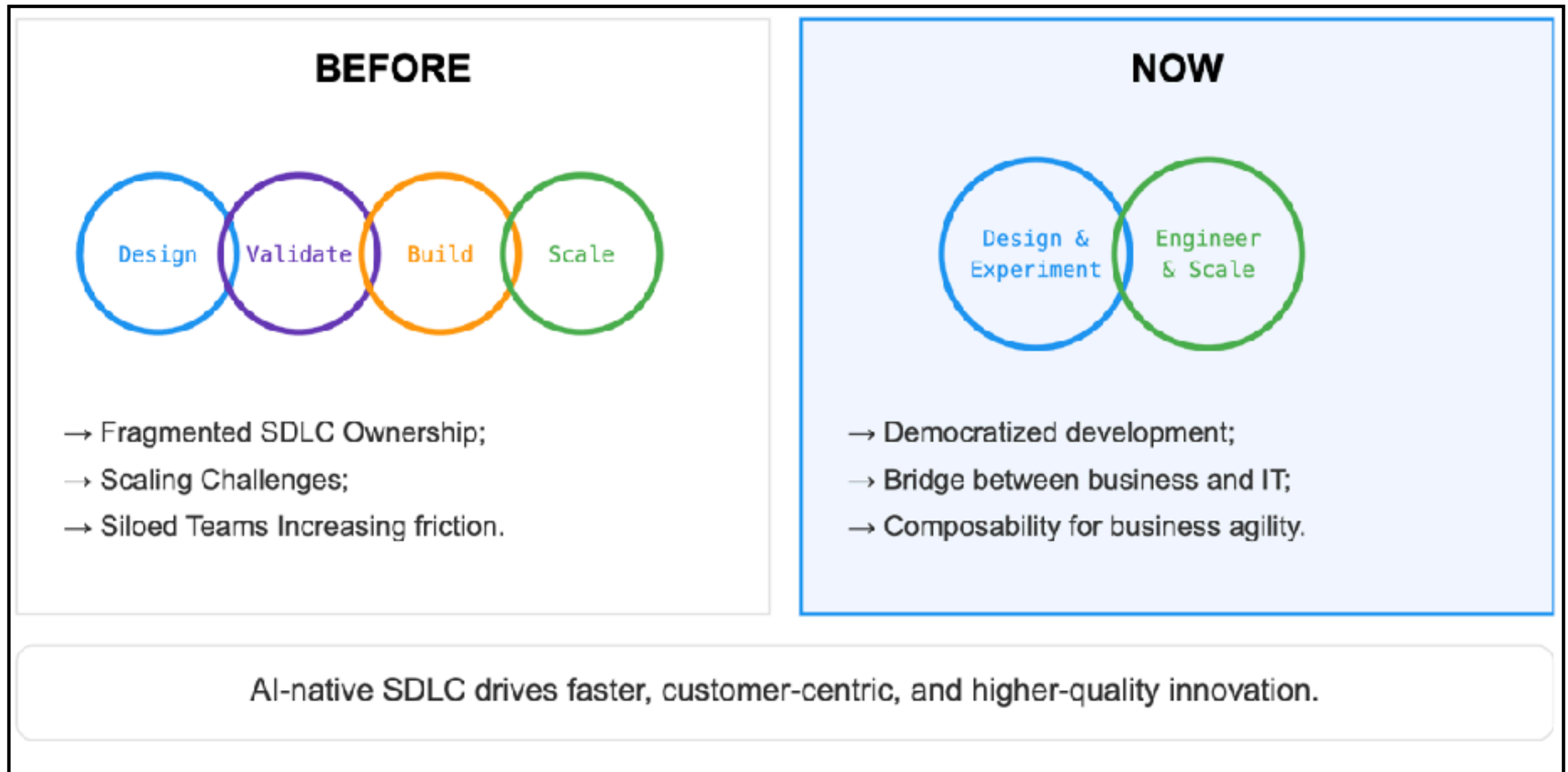


ANTHROPIC

- Most coding assistants use remotely hosted language models
- Claude Code uses the Claude series of models hosted at Anthropic, AWS, or Google Cloud (configurable)



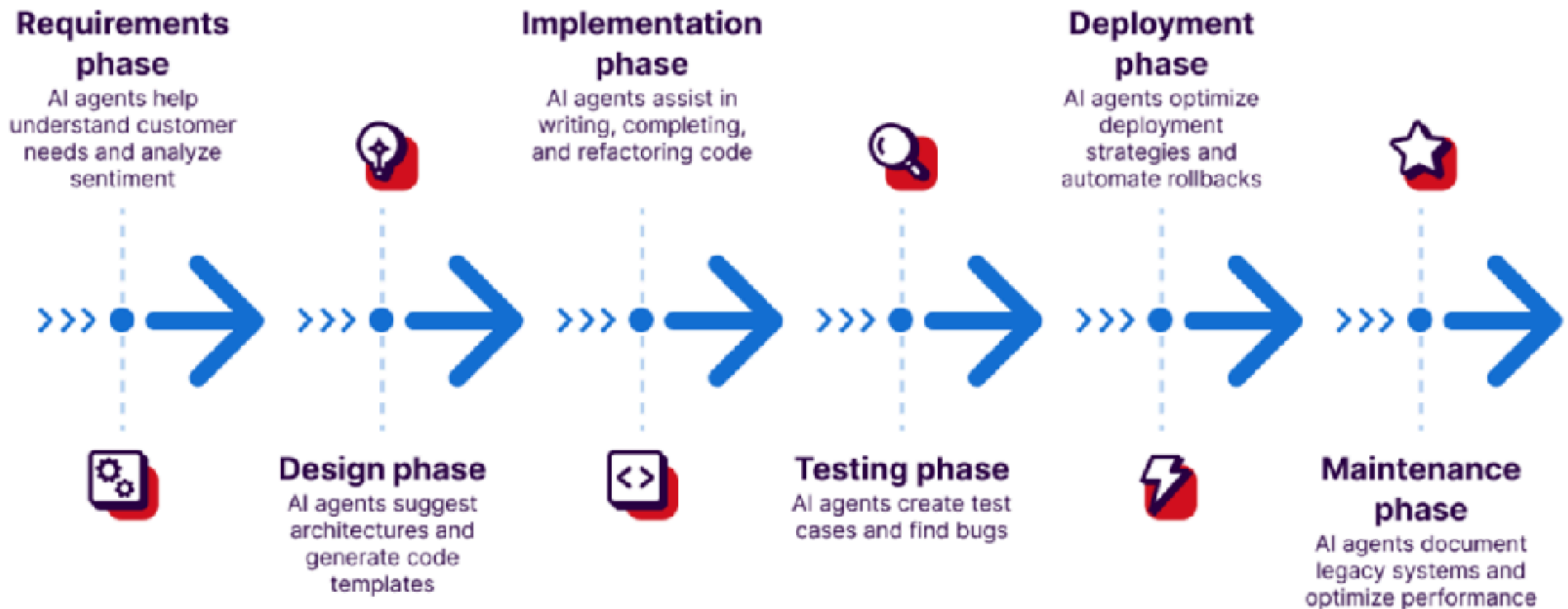
AI in Software Development



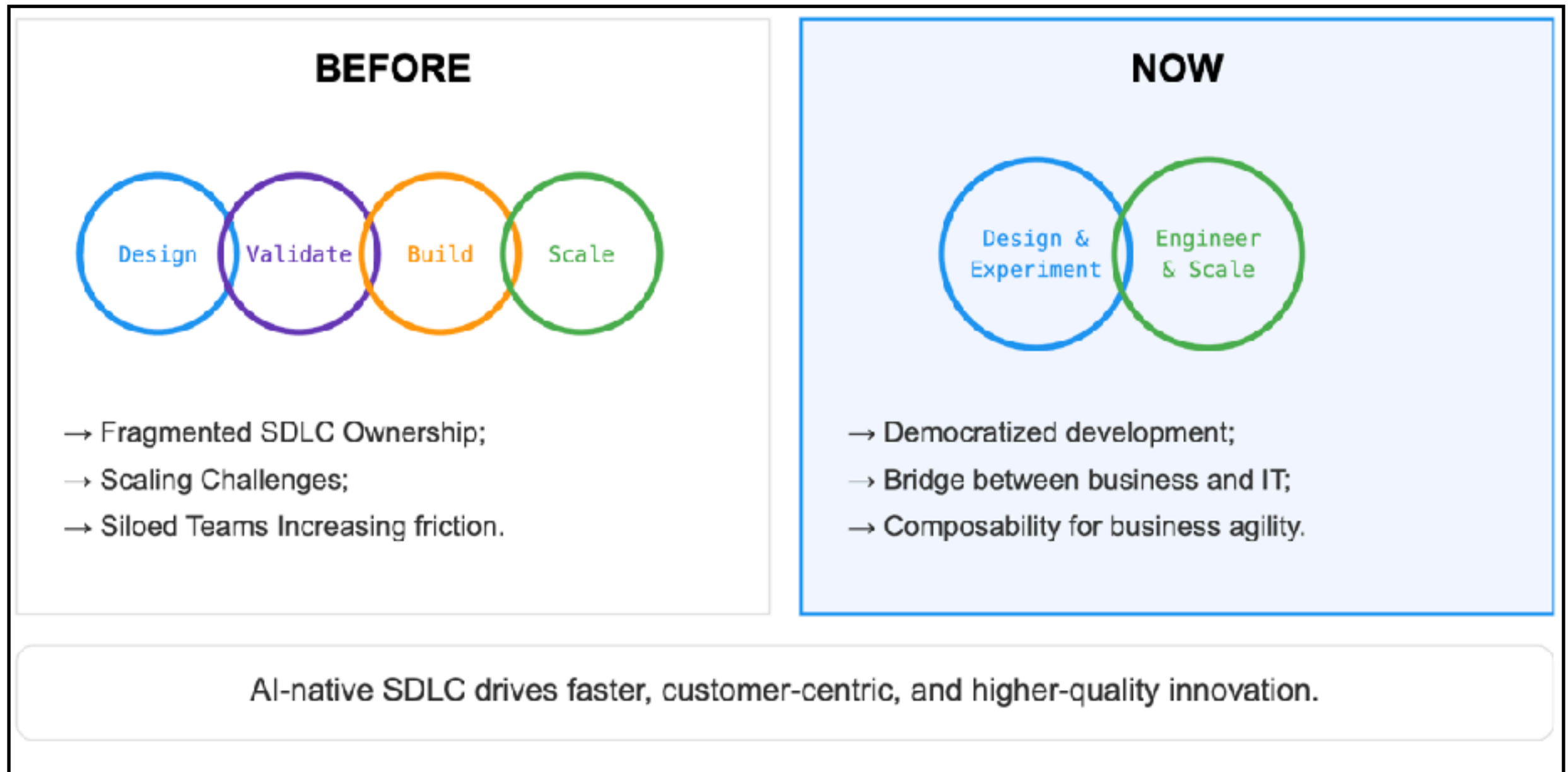
AI is the new normal in software development



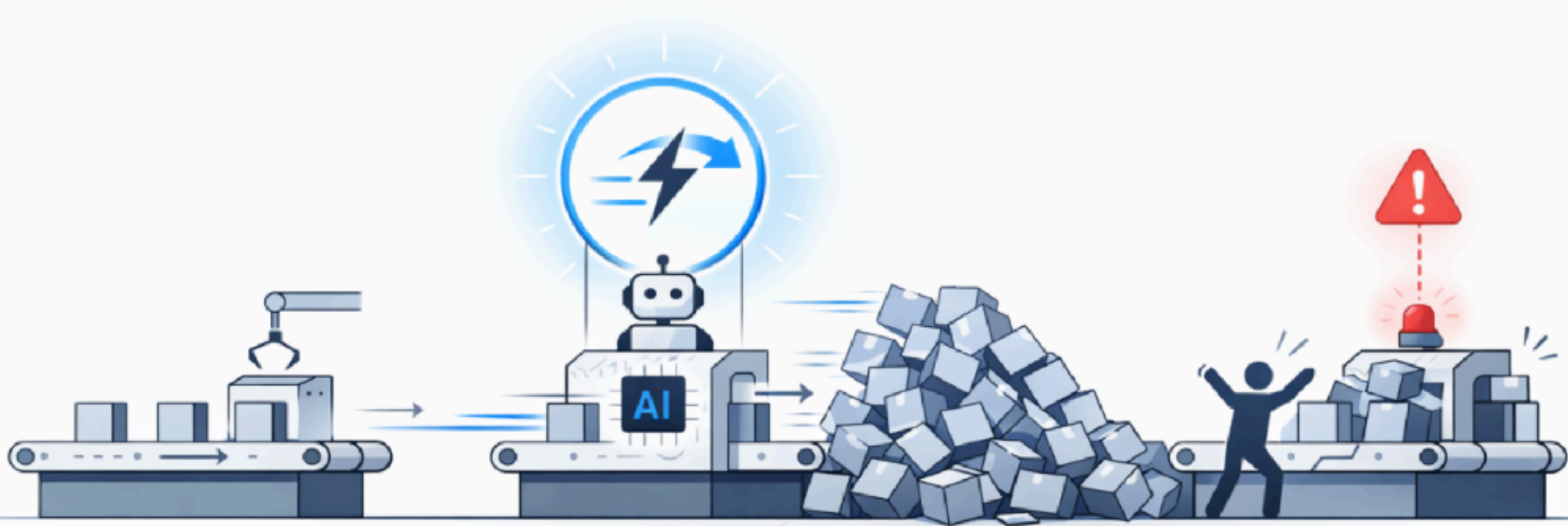
AI in Software Development



AI in Software Development

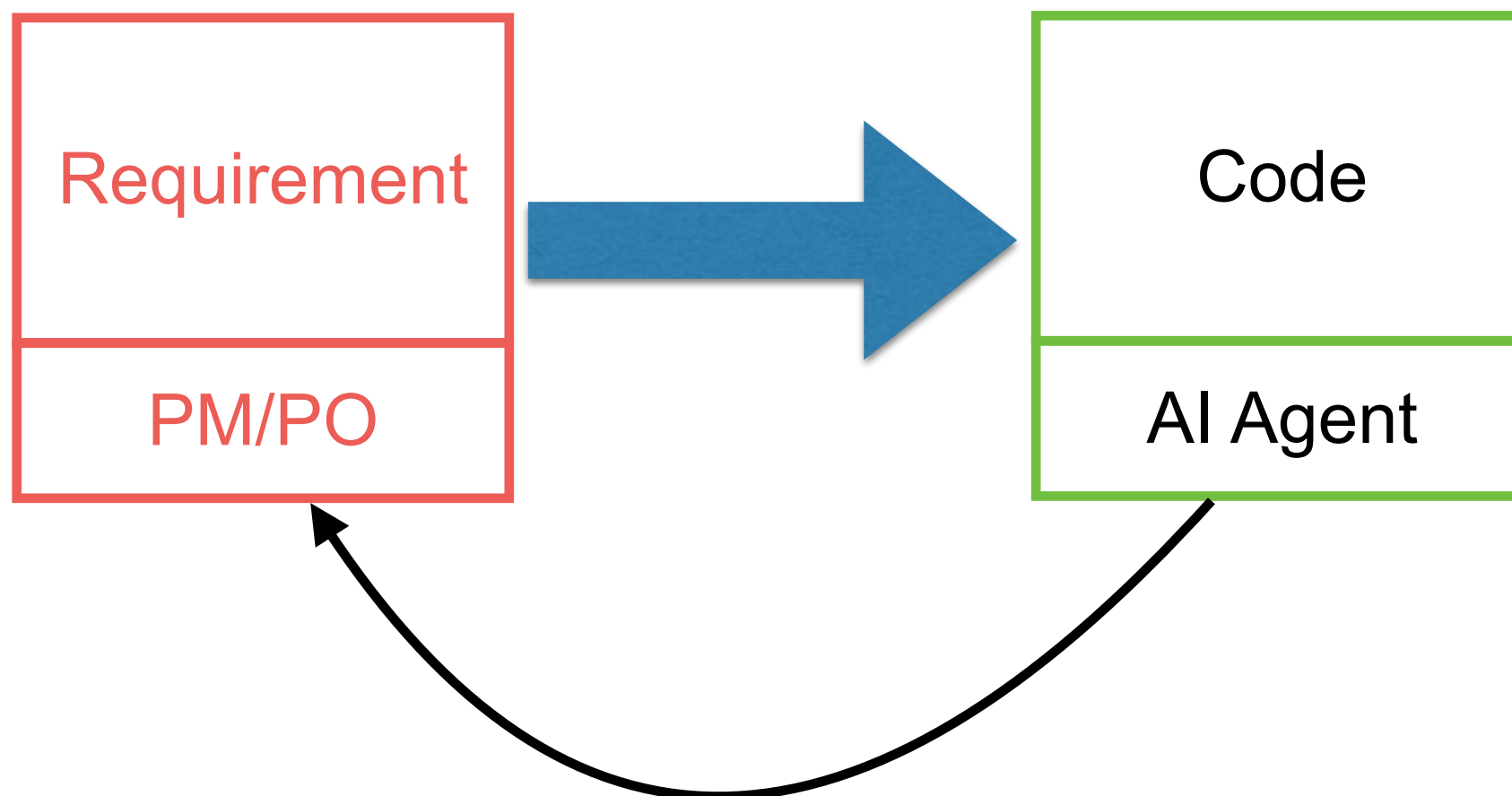


Invest in Real Bottleneck ?



New Bottleneck ?

Decide what to **build**, not code



Get feedback faster

<https://www.theaiopportunities.com/p/is-ai-slowng-down-the-new-bottleneck>



DORA



State of AI-assisted Software Development

Google Cloud

2025

<https://cloud.google.com/blog/products/ai-machine-learning/announcing-the-2025-dora-report>



AI and Productivity ?

AI Adoption

90% of software development professionals are now using AI at work

Productivity

80% report that AI has increased their productivity

Reliance

65% are relying on AI at least a “moderate amount” at work

Code Quality

59% of survey respondents say that AI has positively impacted their code quality

<https://blog.google/innovation-and-ai/technology/developers-tools/dora-report-2025/>

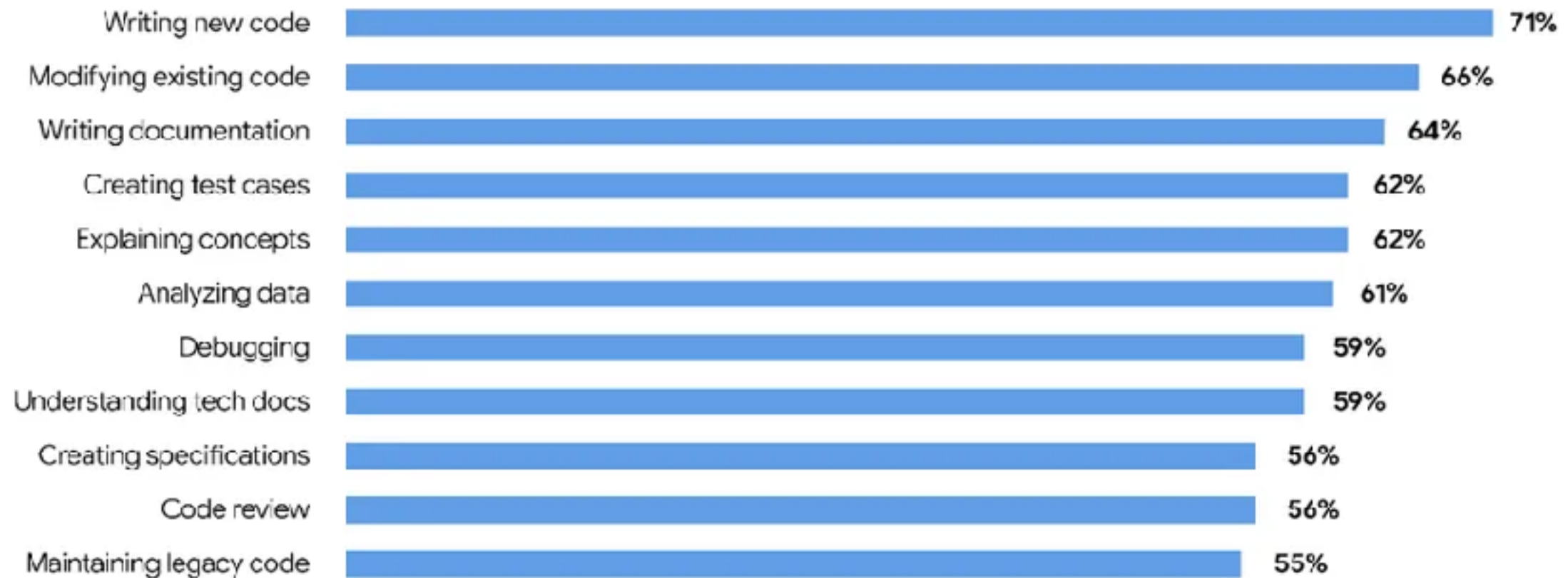


Productivity ?

Reliance on AI by task

DORA
2025

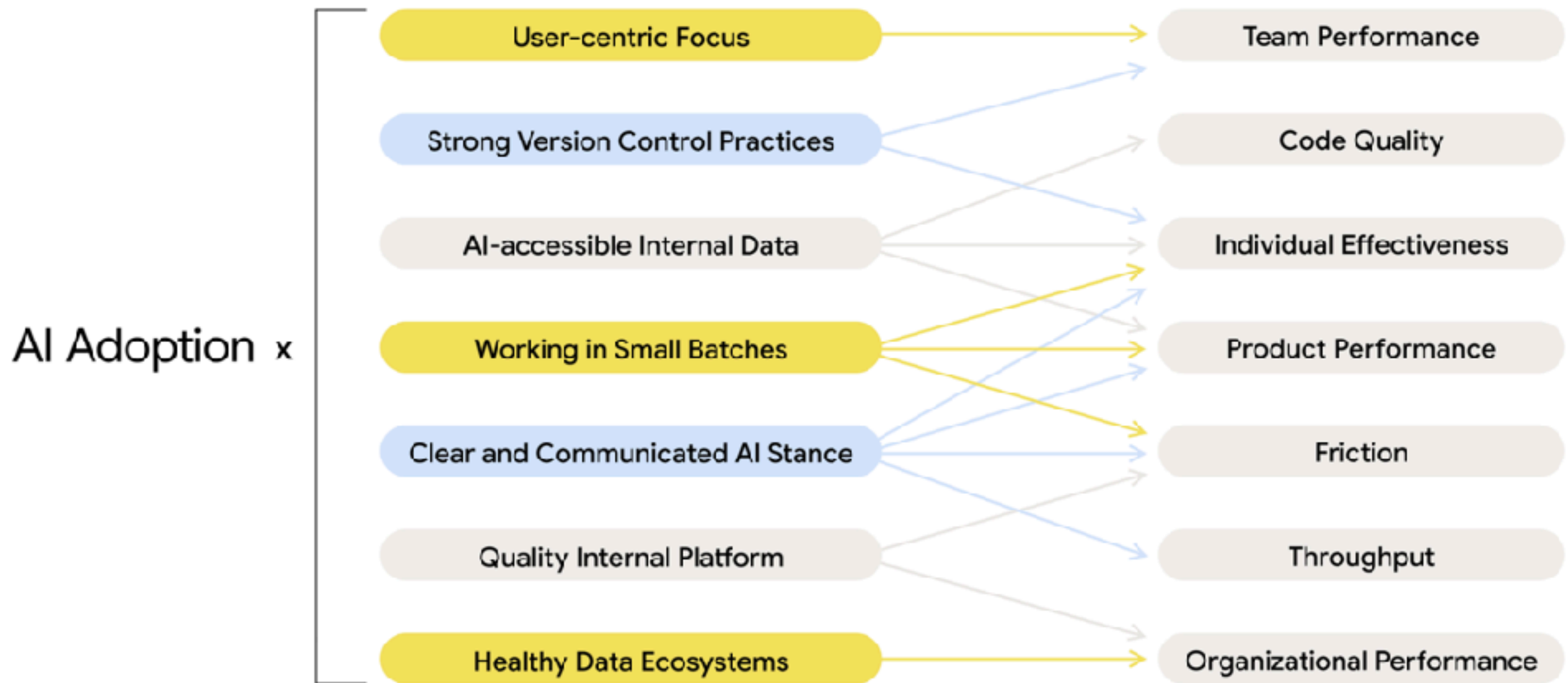
Percentage of task performers who use AI



<https://blog.google/innovation-and-ai/technology/developers-tools/dora-report-2025/>



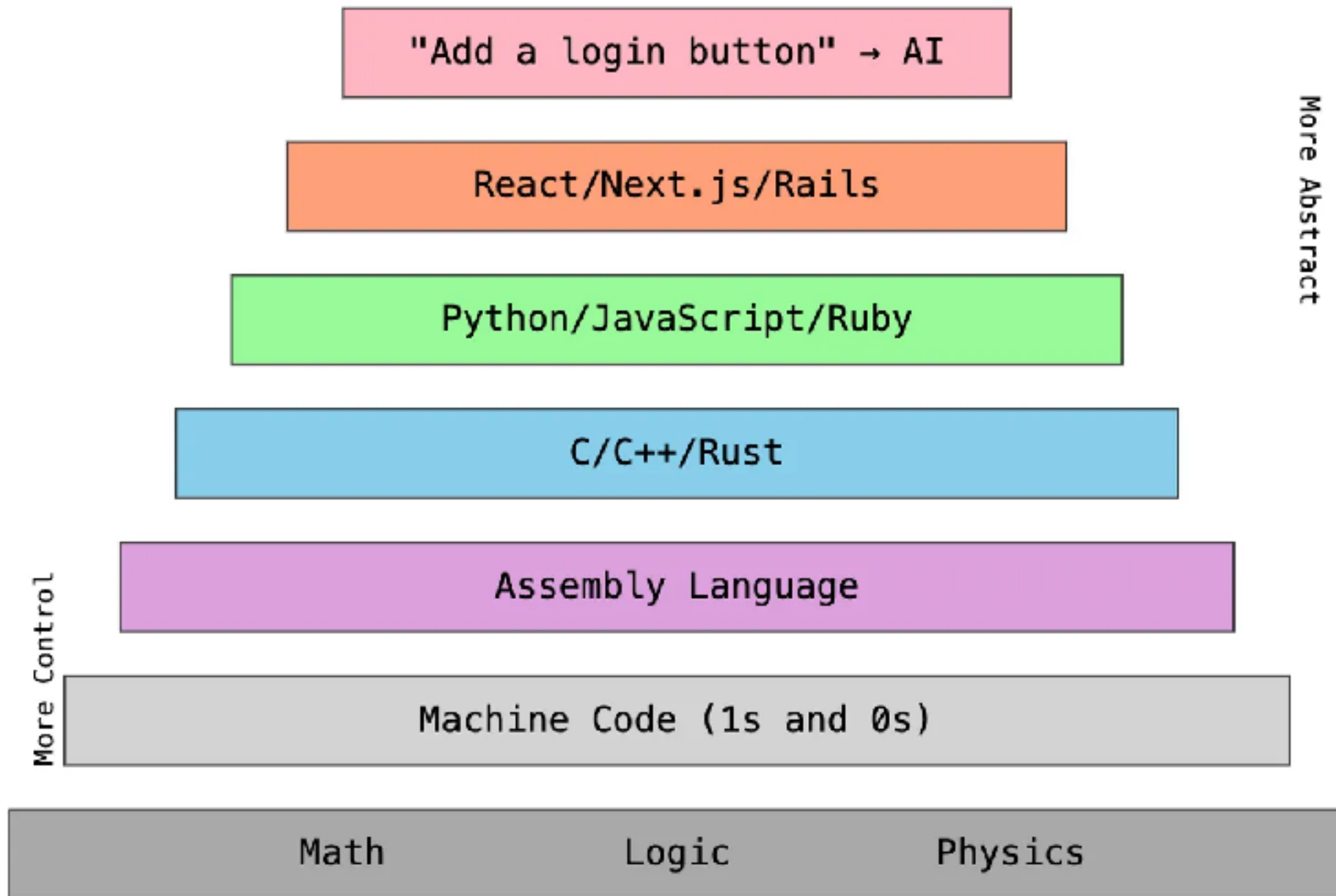
DORA AI Capabilities Model



<https://cloud.google.com/blog/products/ai-machine-learning/announcing-the-2025-dora-report>



The Modern Tower of Abstraction



<https://cline.bot/blog/from-assembly-to-ai-why-vibe-coding-is-just-another-chapter-in-our-abstraction-story>





Andrej Karpathy ✓

@karpathy



There's a new kind of coding I call "vibe coding", where you fully give in to the vibes, embrace exponentials, and forget that the code even exists. It's possible because the LLMs (e.g. Cursor Composer w Sonnet) are getting too good. Also I just talk to Composer with SuperWhisper so I barely even touch the keyboard. I ask for the dumbest things like "decrease the padding on the sidebar by half" because I'm too lazy to find it. I "Accept All" always, I don't read the diffs anymore. When I get error messages I just copy paste them in with no comment, usually that fixes it. The code grows beyond my usual comprehension, I'd have to really read through it for a while. Sometimes the LLMs can't fix a bug so I just work around it or ask for random changes until it goes away. It's not too bad for throwaway weekend projects, but still quite amusing. I'm building a project or webapp, but it's not really coding - I just see stuff, say stuff, run stuff, and copy paste stuff, and it mostly works.

6:17 AM · Feb 3, 2025 · 5.3M Views

<https://x.com/karpathy/status/1886192184808149383>



Vibe Coding

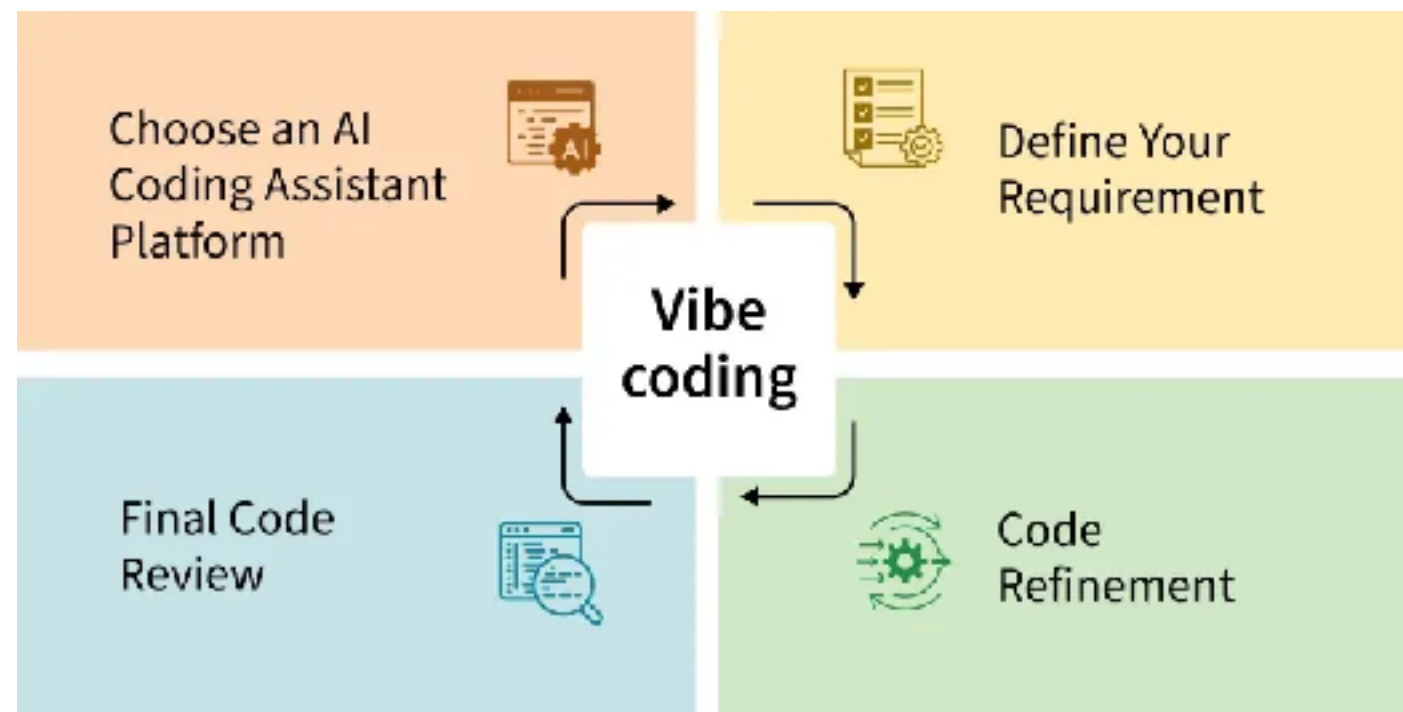
Software development practice that uses artificial intelligence (AI) to generate functional code from **natural language prompts**, accelerating development, and making app building more accessible, especially for those with limited programming experience

<https://cloud.google.com/discover/what-is-vibe-coding>



Types of Vibe Coding ?

Pure vibe coding
Responsible AI-assist development



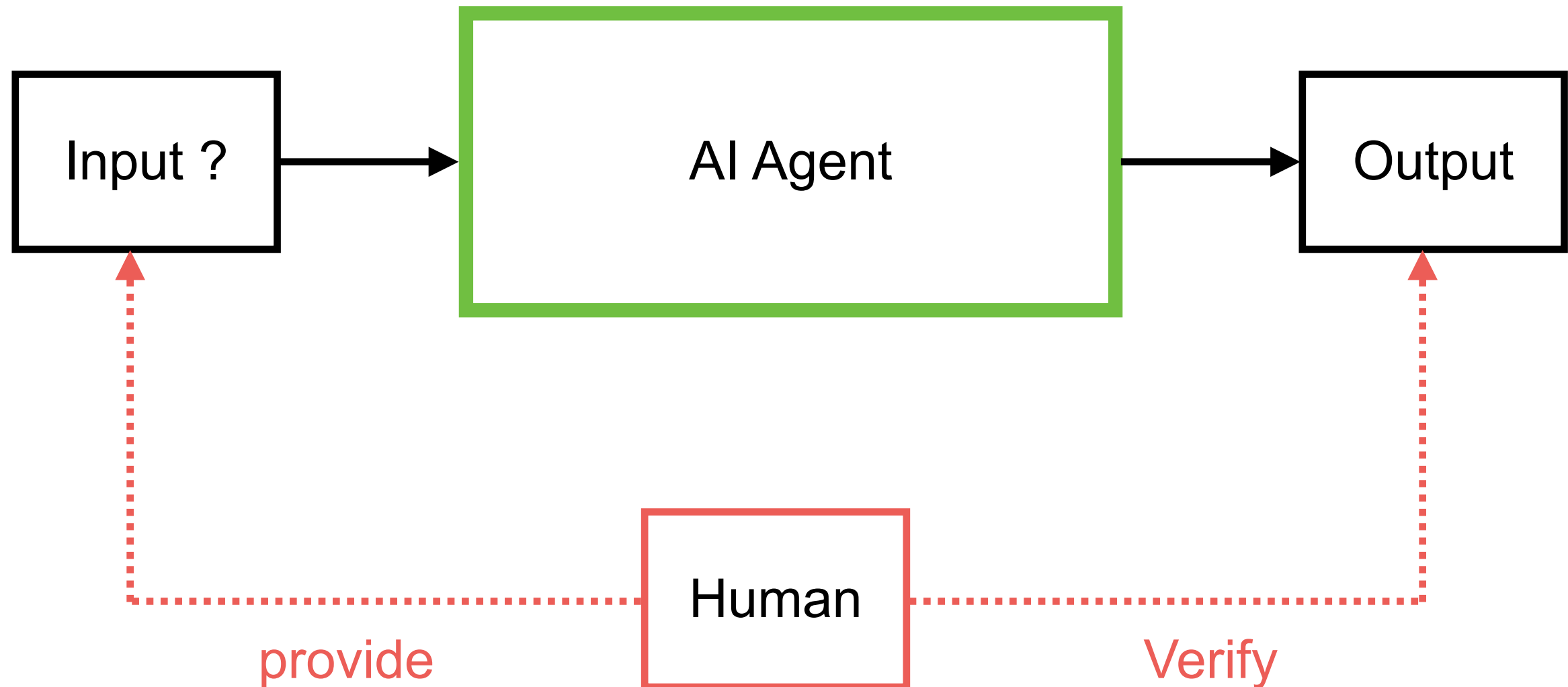
<https://cloud.google.com/discover/what-is-vibe-coding>



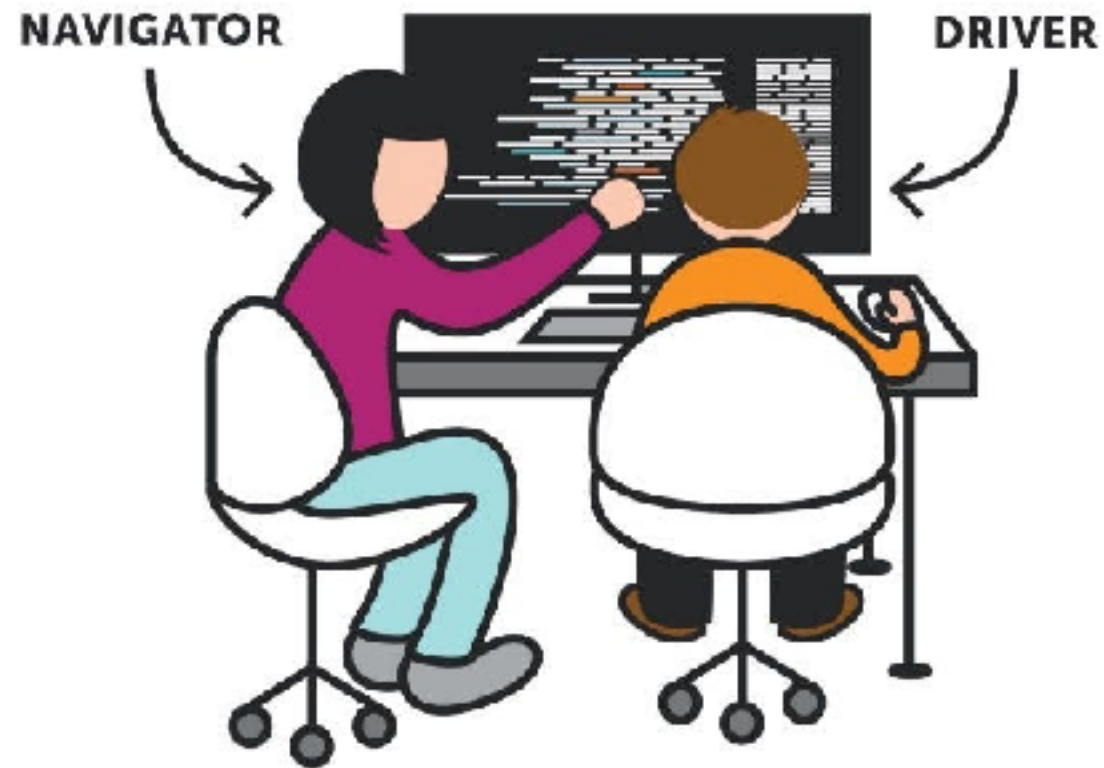
Working with AI Agent !!



Coding with AI Agent



PAIR PROGRAMMING



Reviews, Tests and Understands
with experience

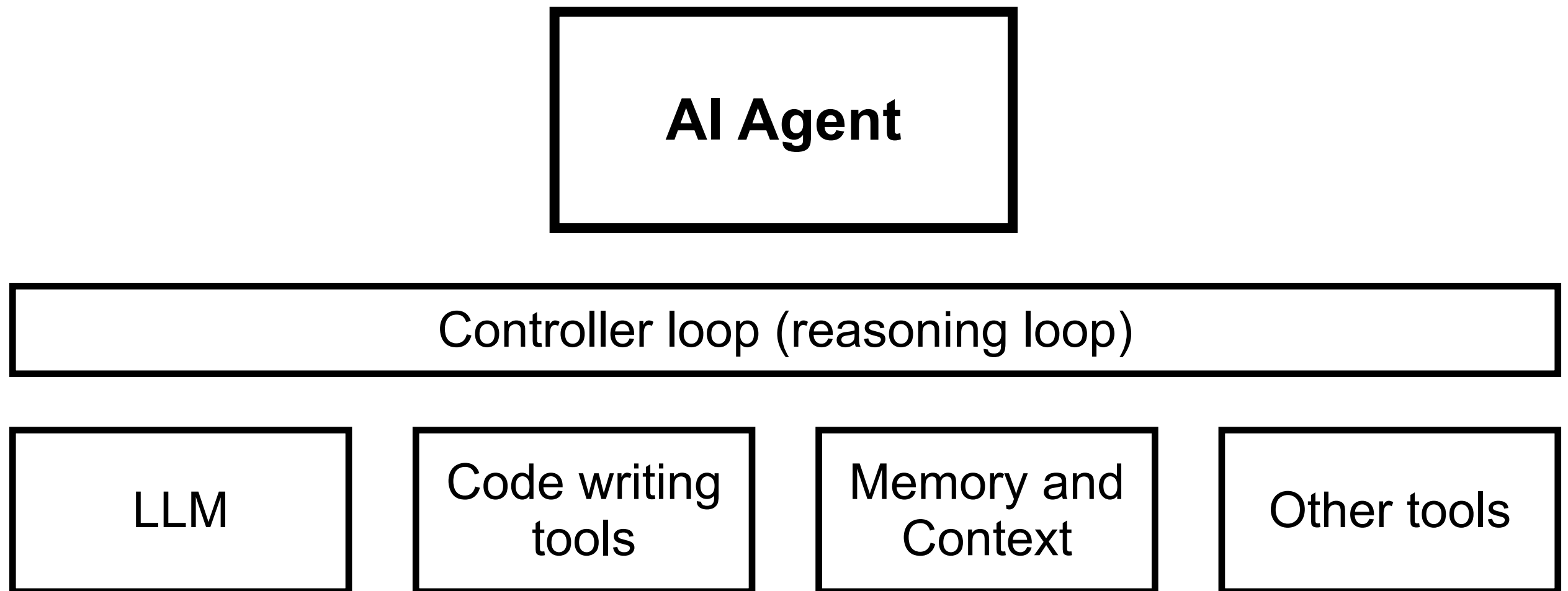


AI draft

Human craft



AI Agent for Coding



Need more secure !!

AI Agent

Controller loop (reasoning loop)

LLM

Code writing
tools

Memory and
Context

Other tools



Security with Claude Code

<https://code.claude.com/docs/en/security>



Security approach ?

Permission-based architecture

Build-in protections

Use responsibility (review before approval)



Permission-based architecture

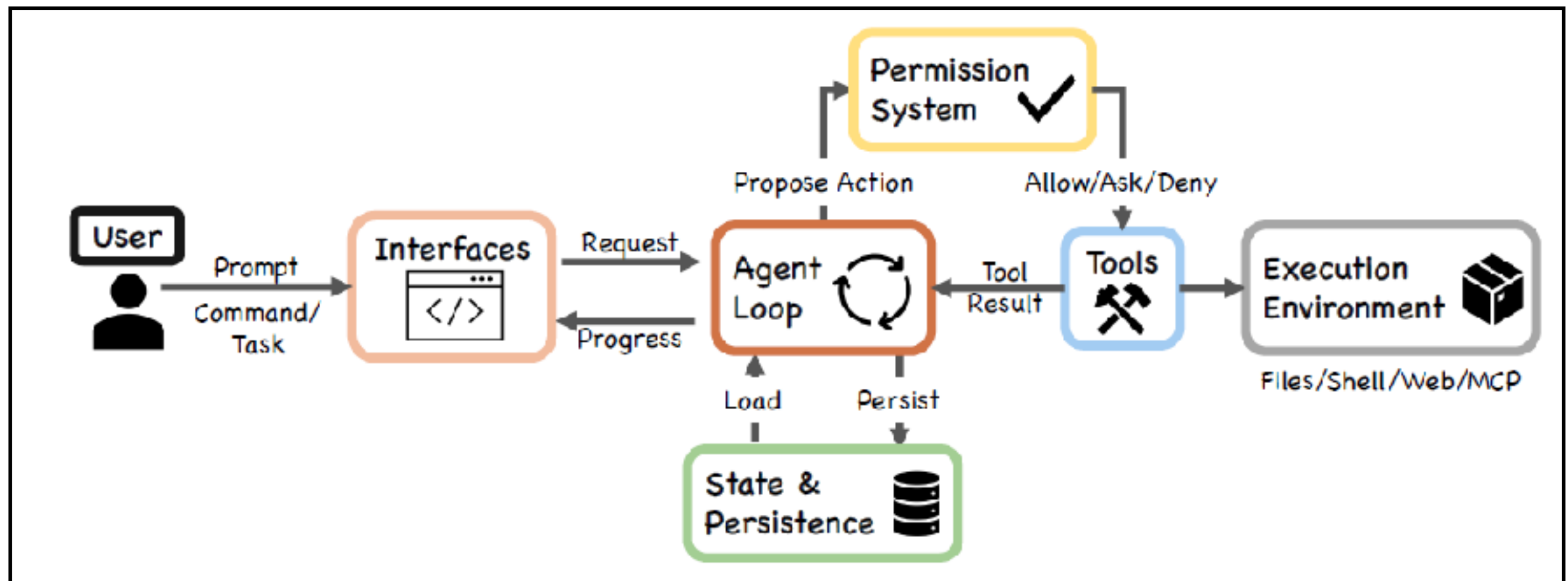
Read-only by default
When edit files, run command
(need human approve)
Human-in-the-loop (HITL)

<https://code.claude.com/docs/en/permissions>



Permission-based architecture

Claude code supports fine-grained permissions



<https://code.claude.com/docs/en/permissions>



Build-in protections

Sandbox

Working directory

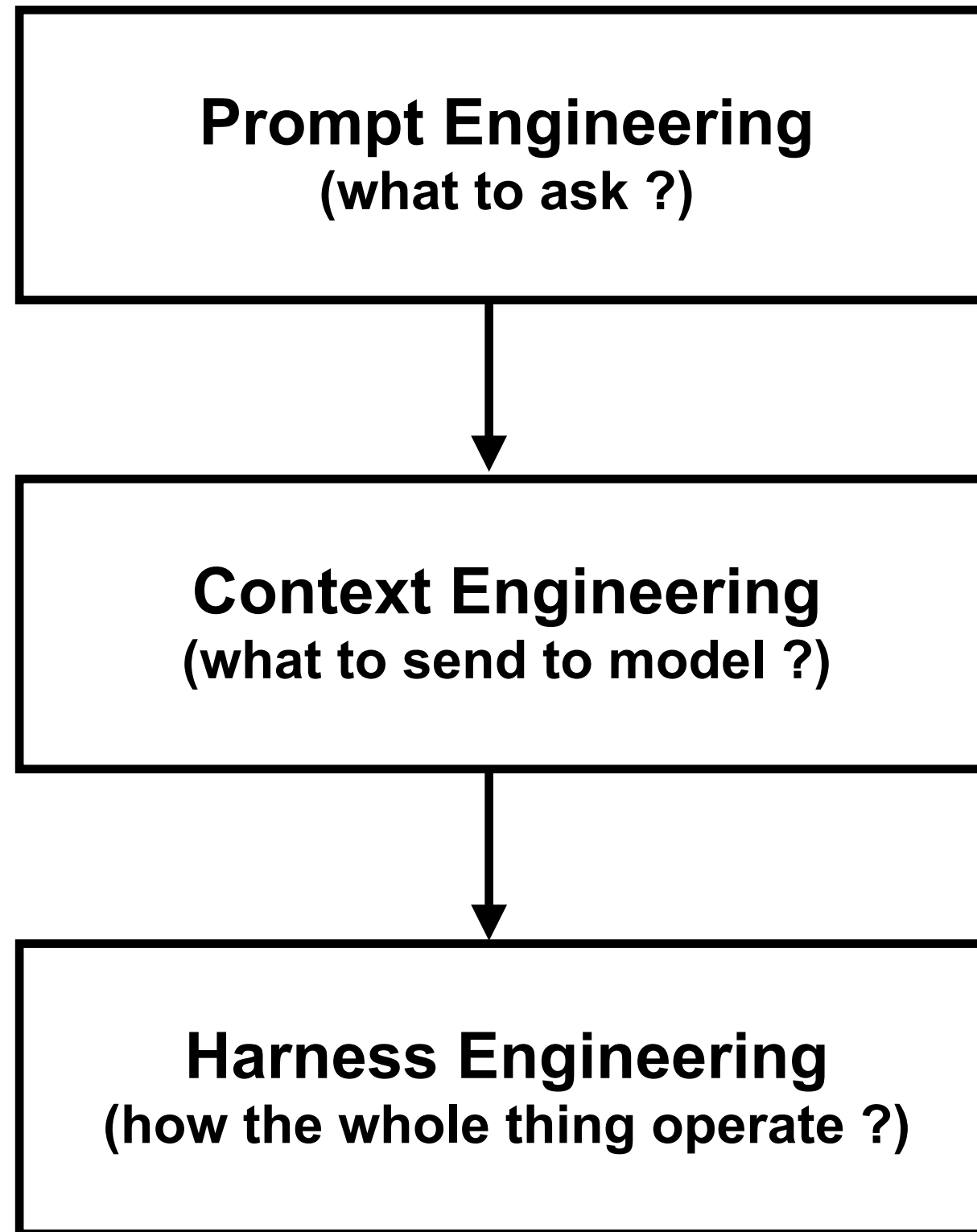
Permission policy

<https://code.claude.com/docs/en/sandbox-environments>



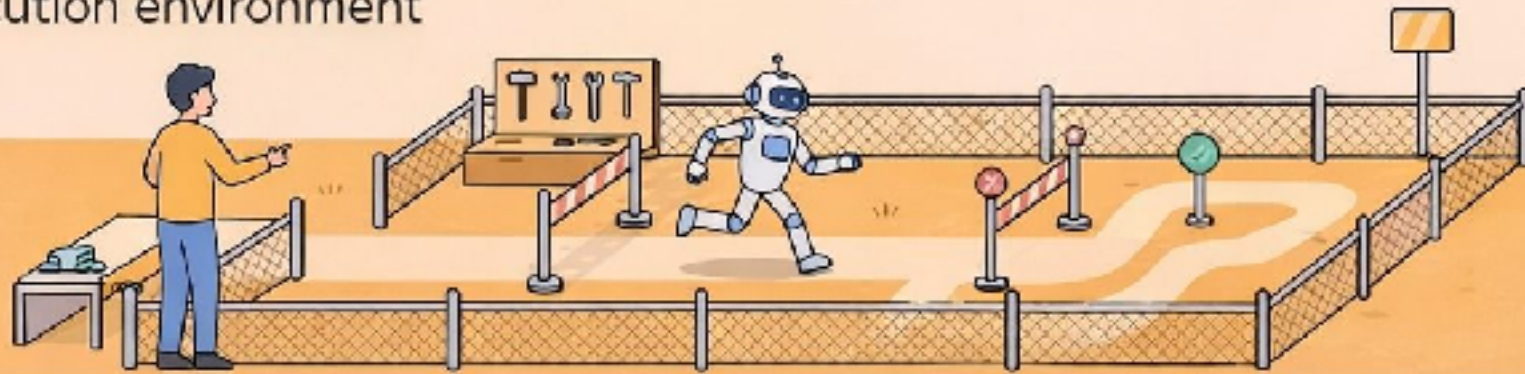
How to work with AI Agent ?





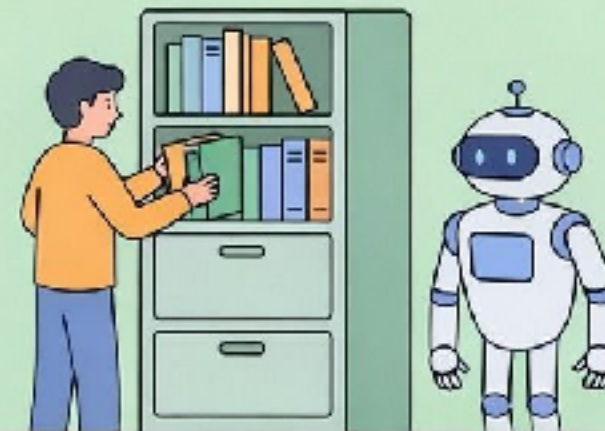
Harness Engineering — Build the World

Design the execution environment



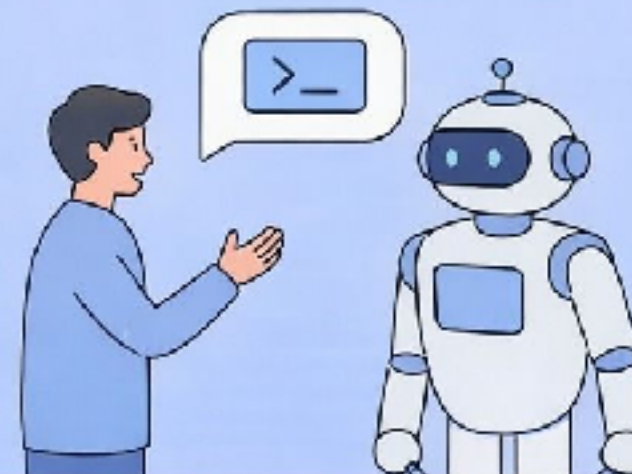
Context Engineering — What to See

Manage visible information



Prompt Engineering — What to Say

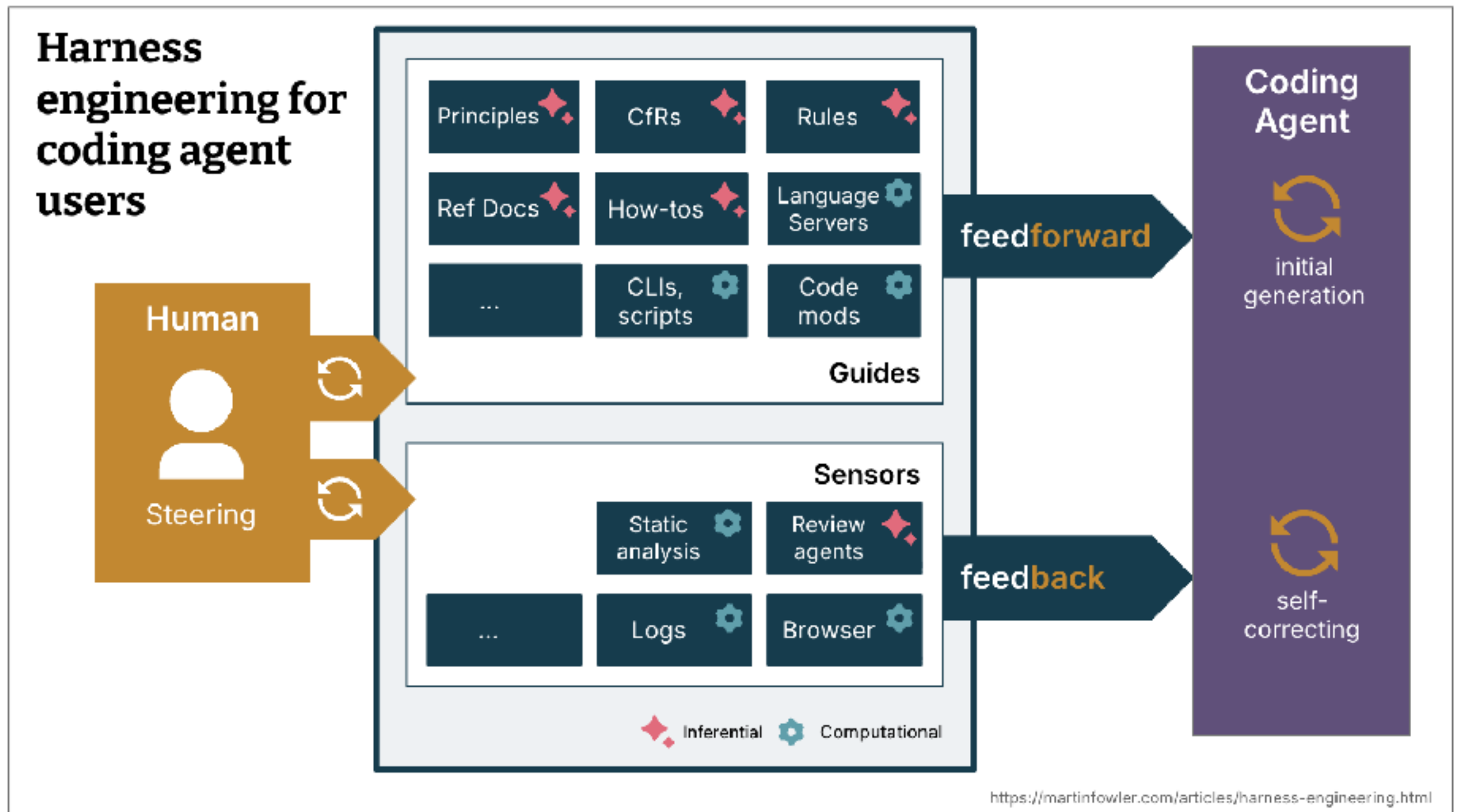
Optimize single-turn interactions



<https://milvus.io/blog/harness-engineering-ai-agents.md>



Harness Engineering



<https://martinfowler.com/articles/harness-engineering.html>



AI Agent = LLM + Harness

Harness = AI Agent - LLM



Workflows with AI Agent ?



Workflows ?

Application level

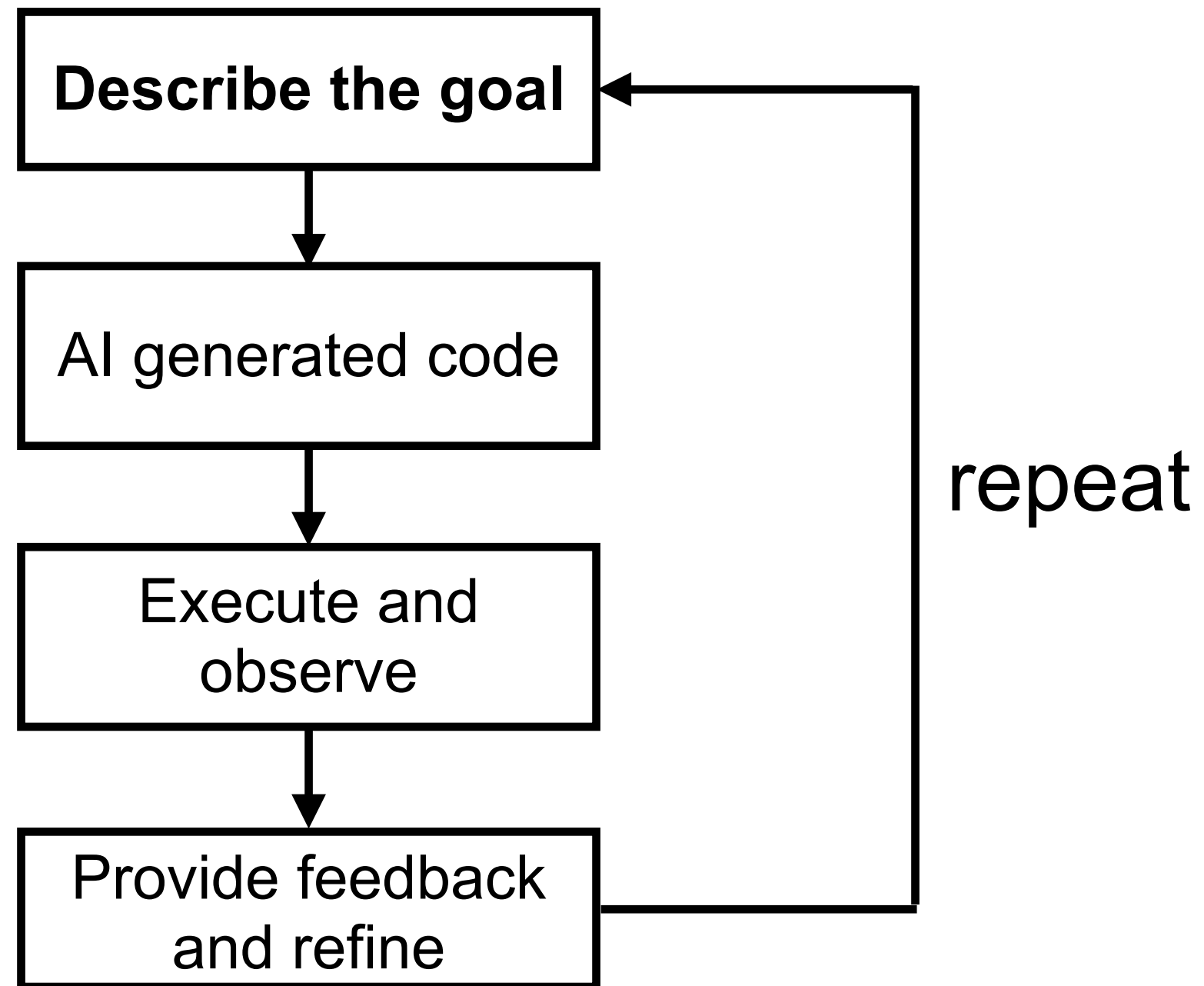
Code level

<https://cloud.google.com/discover/what-is-vibe-coding>



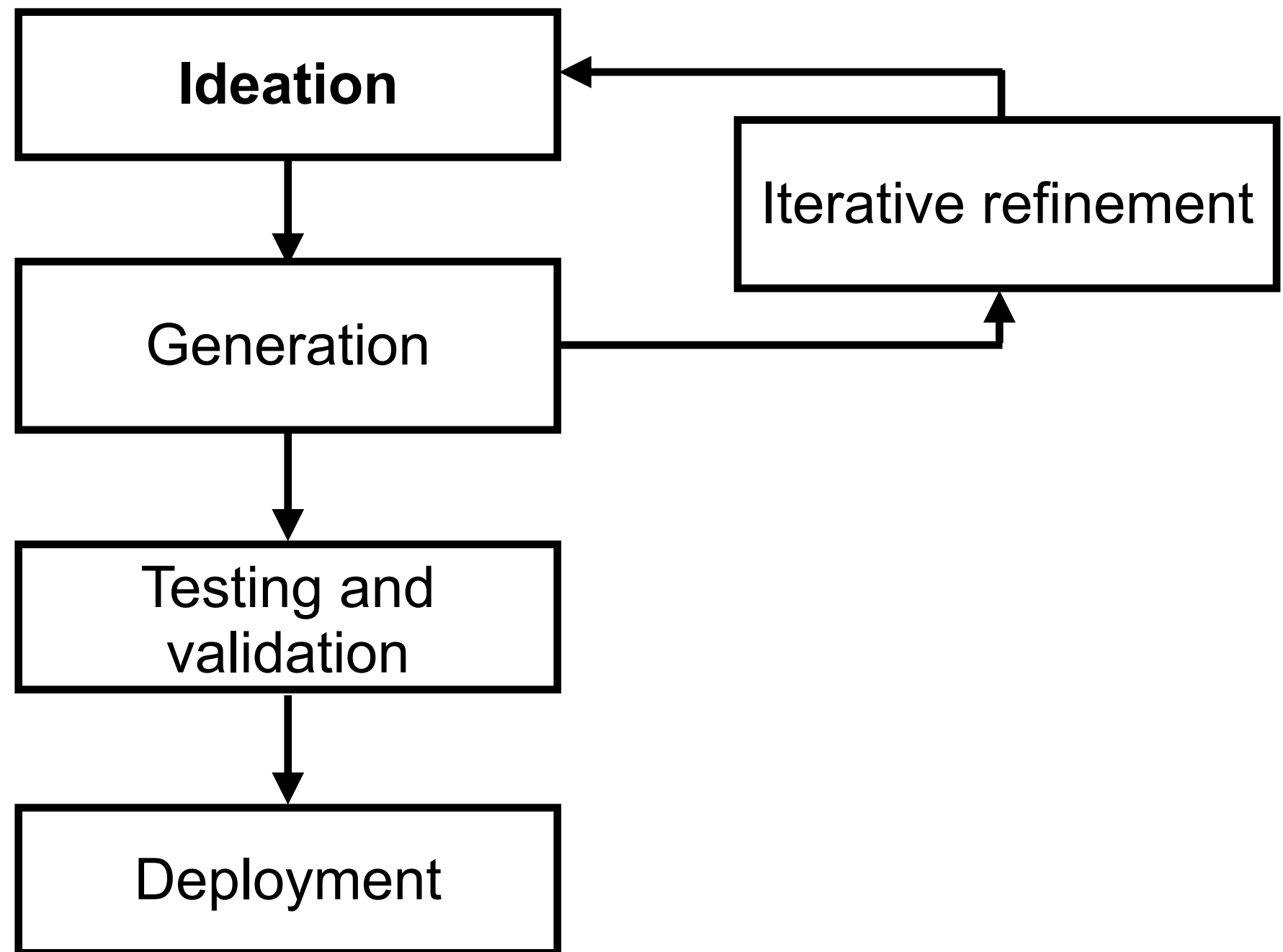
Code level workflow

Conversational loop to create a specific code



Application level workflow

Talking about high-level idea from concept to deployment

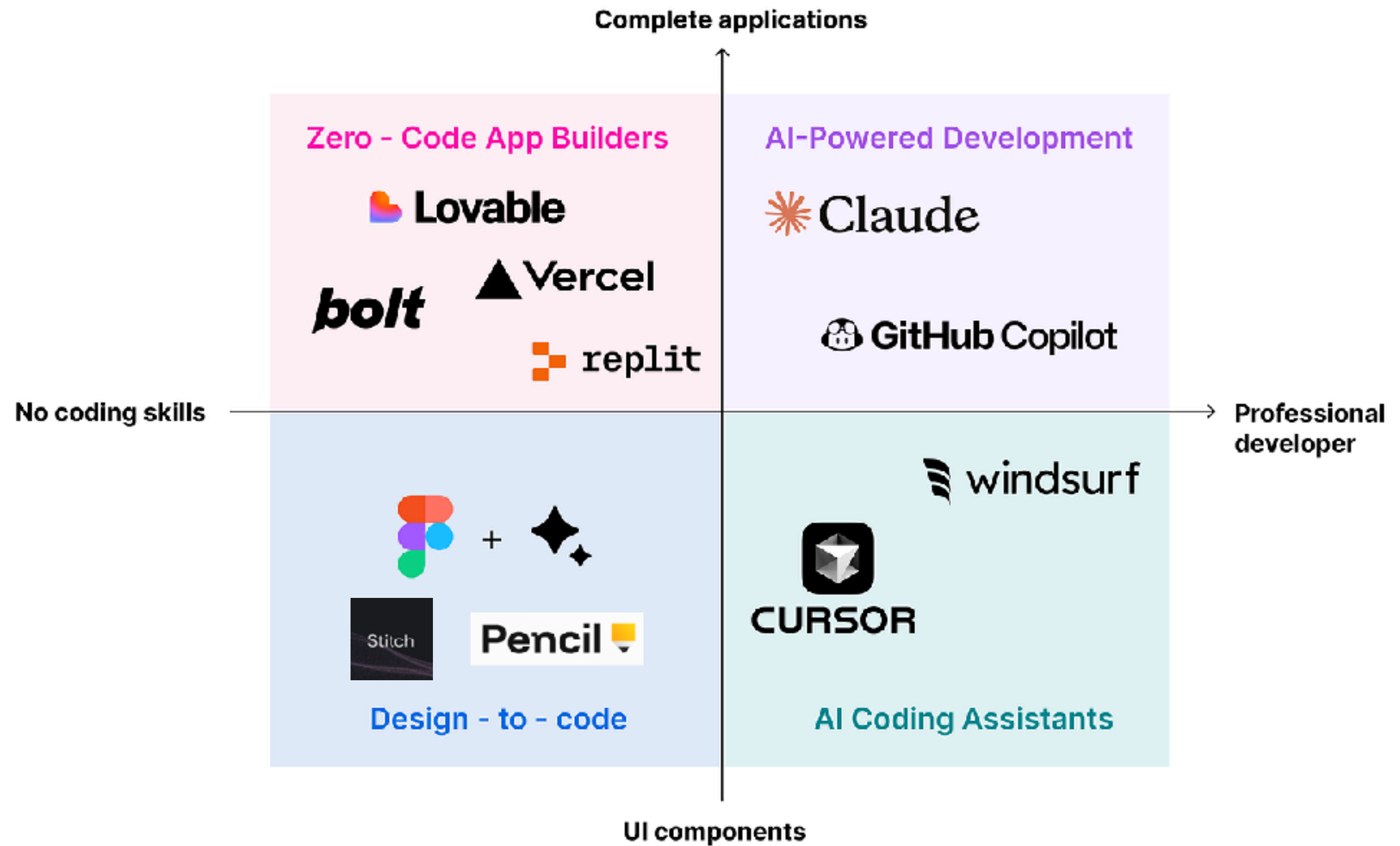


Tools ?

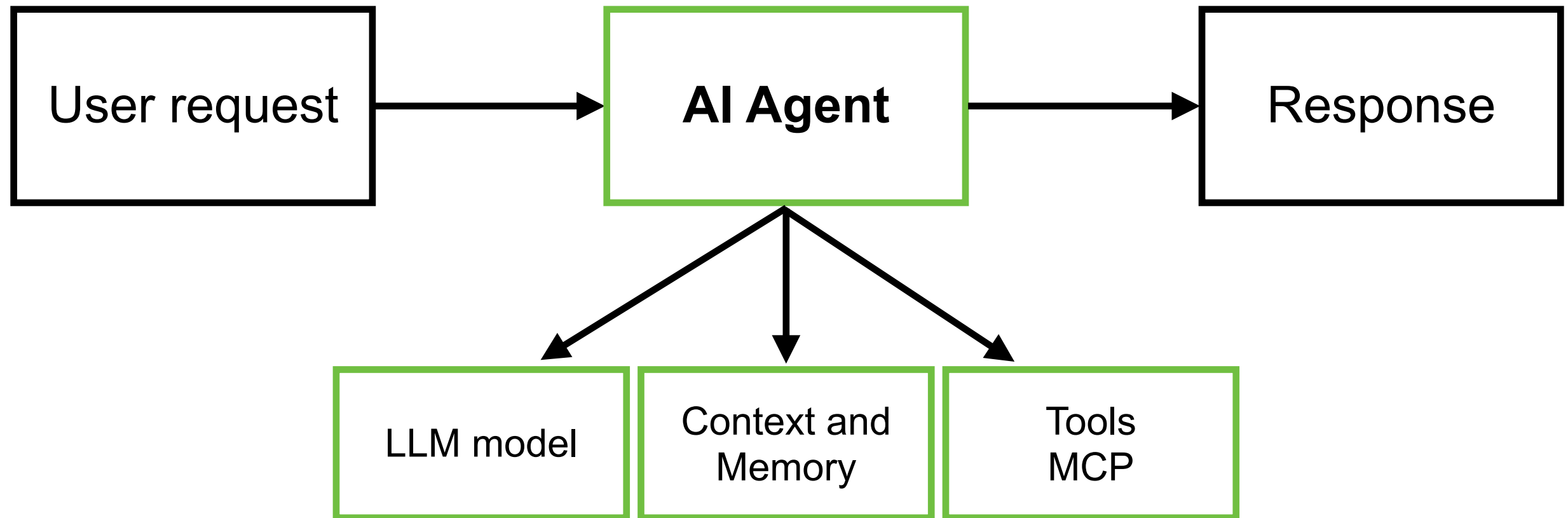


AI Agent for Coding





AI Agent for Coding



How to manage context or knowledge for Agent ?



Token is new oil !!
Token is expensive !!



How to count token ?

GPT-5.x & O1/3

GPT-4 & GPT-3.5 (legacy)

GPT-3 (legacy)

สวัสดีประเทศไทย

Clear

Show example

Tokens

5

Characters

15

สวัสดีประเทศไทย

<https://platform.openai.com/tokenizer>



LLM Model :: GPT 5.5

**GPT-5.5** Default Compare Try in Playground

REASONING

Highest

SPEED

Fast

PRICE
\$5 - \$30
Input - Output

INPUT
   
Text, image

OUTPUT
   
Text

GPT-5.5 is our newest frontier model for the most complex professional work. Learn more in our [latest model guide](#). Reasoning effort supports: none, low, medium (default), high and xhigh.

- ✦ 1,050,000 context window
- ➡ 128,000 max output tokens
- 📅 Dec 01, 2025 knowledge cutoff
- 🗨 Reasoning token support

Pricing

Pricing is based on the number of tokens used, or other metrics based on the model type. For tool-specific models, like search and computer use, there's a fee per tool call. See details in the [pricing page](#).

Text tokens

Per 1M tokens - Batch API price ☐

Input

\$5.00

Cached input

\$0.50

Output

\$30.00

<https://developers.openai.com/api/docs/models/gpt-5.5>



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Development

66

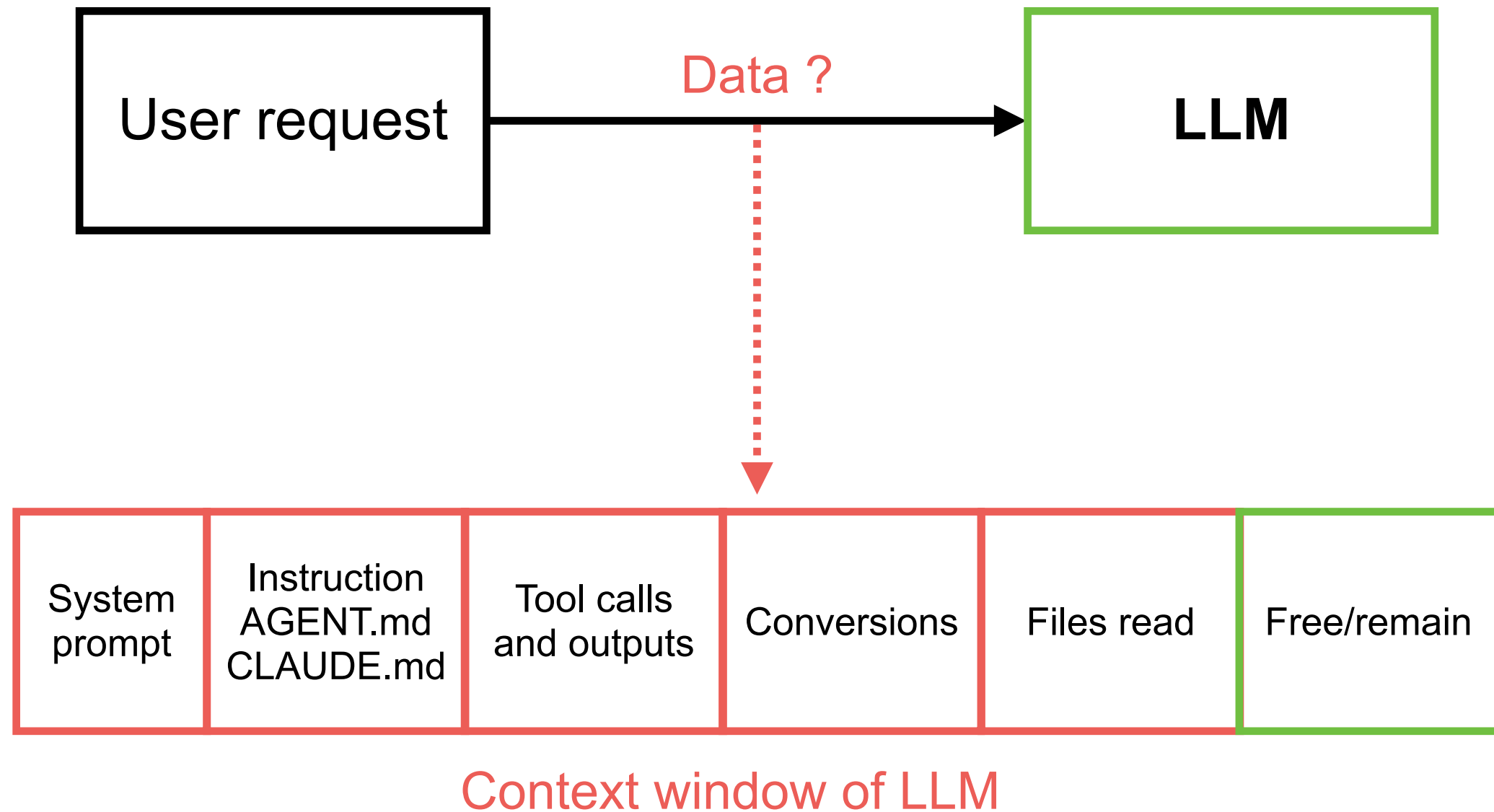
LLM Model :: Anthropic

Model	Base Input Tokens	5m Cache Writes	1h Cache Writes	Cache Hits & Refreshes	Output Tokens
Claude Opus 4.7	\$5 / MTok	\$6.25 / MTok	\$10 / MTok	\$0.50 / MTok	\$25 / MTok
Claude Opus 4.6	\$5 / MTok	\$6.25 / MTok	\$10 / MTok	\$0.50 / MTok	\$25 / MTok
Claude Opus 4.5	\$5 / MTok	\$6.25 / MTok	\$10 / MTok	\$0.50 / MTok	\$25 / MTok
Claude Opus 4.1	\$15 / MTok	\$18.75 / MTok	\$30 / MTok	\$1.50 / MTok	\$75 / MTok
Claude Opus 4	\$15 / MTok	\$18.75 / MTok	\$30 / MTok	\$1.50 / MTok	\$75 / MTok
Claude Sonnet 4.6	\$3 / MTok	\$3.75 / MTok	\$6 / MTok	\$0.30 / MTok	\$15 / MTok
Claude Sonnet 4.5	\$3 / MTok	\$3.75 / MTok	\$6 / MTok	\$0.30 / MTok	\$15 / MTok
Claude Sonnet 4	\$3 / MTok	\$3.75 / MTok	\$6 / MTok	\$0.30 / MTok	\$15 / MTok

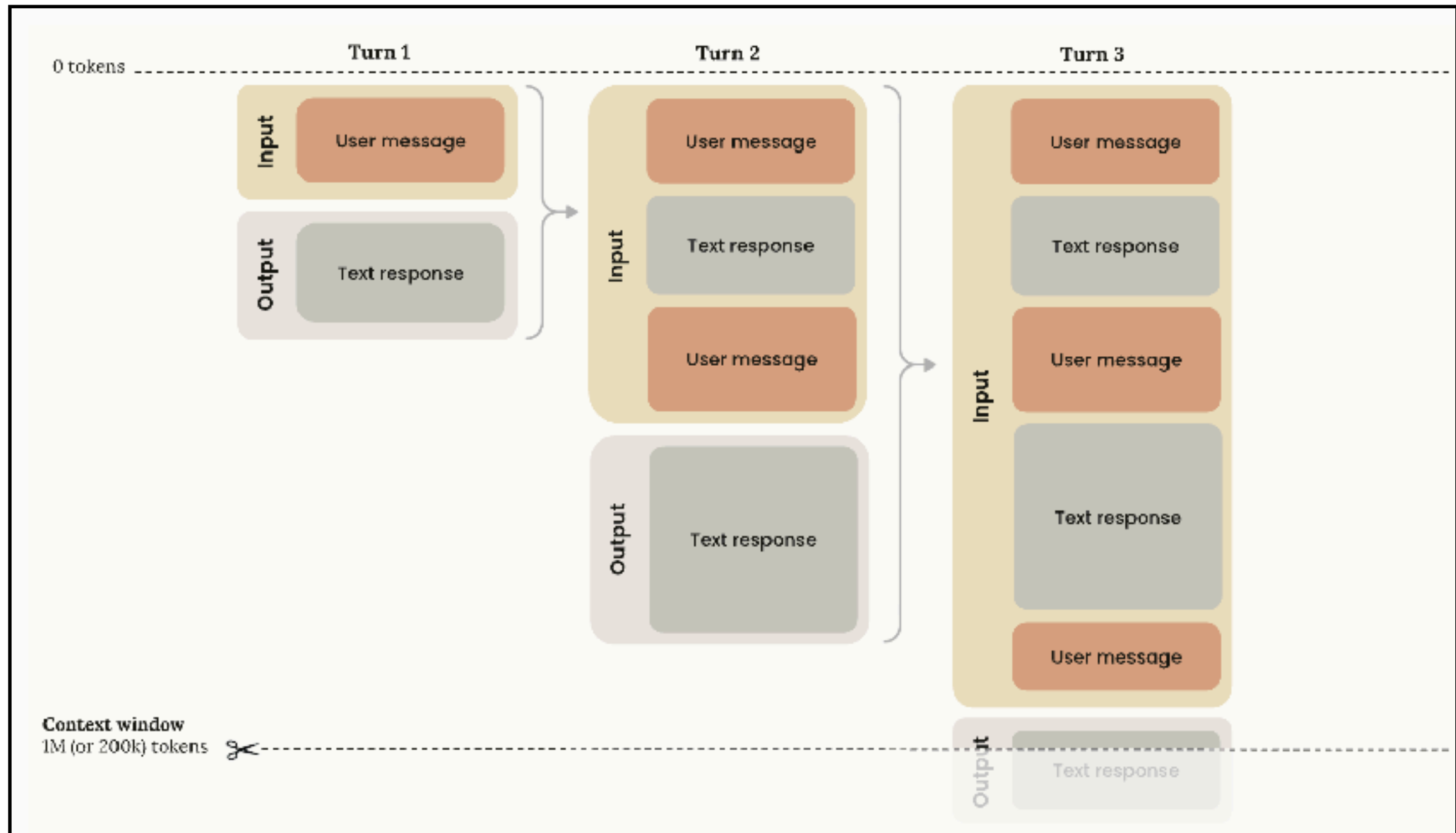
<https://platform.claude.com/docs/en/about-claude/pricing>



Context Window ?



Context Window with Claude



<https://platform.claude.com/docs/en/build-with-claude/context-windows>



Types of Knowledge/Context ?

Shared context

Static Knowledge

Prompt engineering
Global Instructions, Rules
AGENTS.md
CLAUDE.md
DESIGN.md

Task context

Dynamic Knowledge

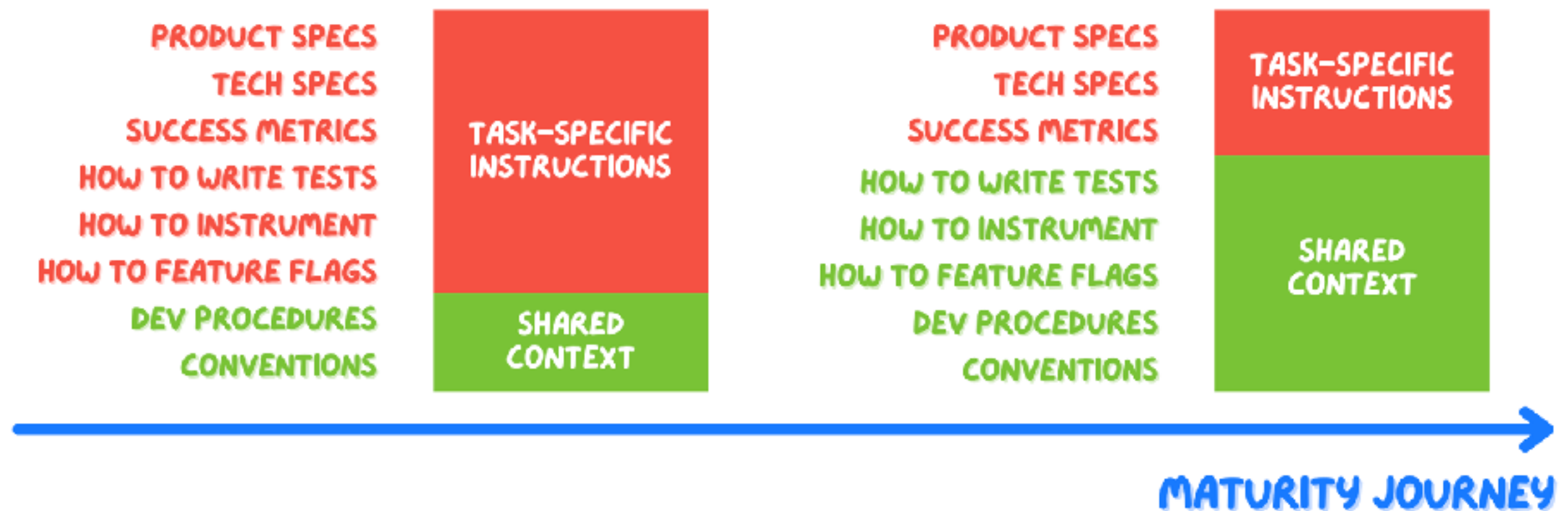
Memory
Graph/Vector database
MCP tools
Specific Instructions
Agent SKILLS

Sub-agent and Modes

Context engineering ?



Types of Knowledge/Context ?

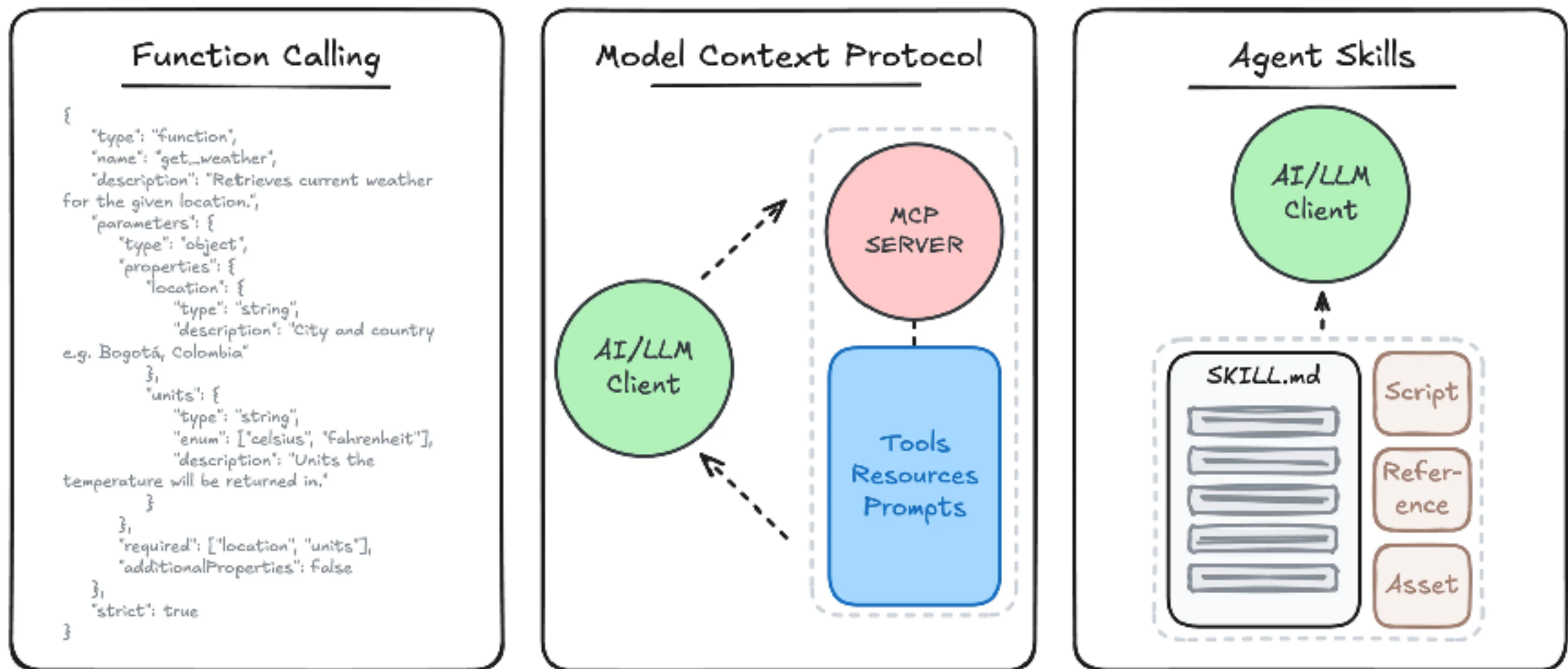


<https://refactoring.fm/p/context-capabilities-and-tech-hubs>



Agent Skills

Standard way to give AI agent new capabilities and expertise



<https://agentskills.io/home>



Vocabulary before start !!



Vocabulary before start !!

Rules

Prompts

MCP servers

Instructions

Commands

Subagents

Modes

Hooks

Agent Skills



Rules and instructions

Static context

Rules are files that contain essential context
Codebase, best practices, common pitfalls
Business requirements

Prevent LLM models from hallucinating
and to ensure specific standards



Commands and Prompts

Static context

Text instructions sent to the agent

Repeat workflow that can be triggered on demand

Allow teams to share standardized workflows
And run complex tasks conditionally

git add -> generate commit message -> git commit



Subagents and Modes

Specific persona or task to specialized agent
Agent, Plan, Ask, Edit mode
Work AI as a team

Make the agent's behavior more deterministic, reliable
and discoverable



Agent Skills

Dynamic context

Create workflow + resources + script

Skills do not bloat the initial context window

Only loaded when the agent determines (relevant with task)

Provide an open standard for sharing capabilities with a team without hitting context limits

<https://agentskills.io/>



Hooks

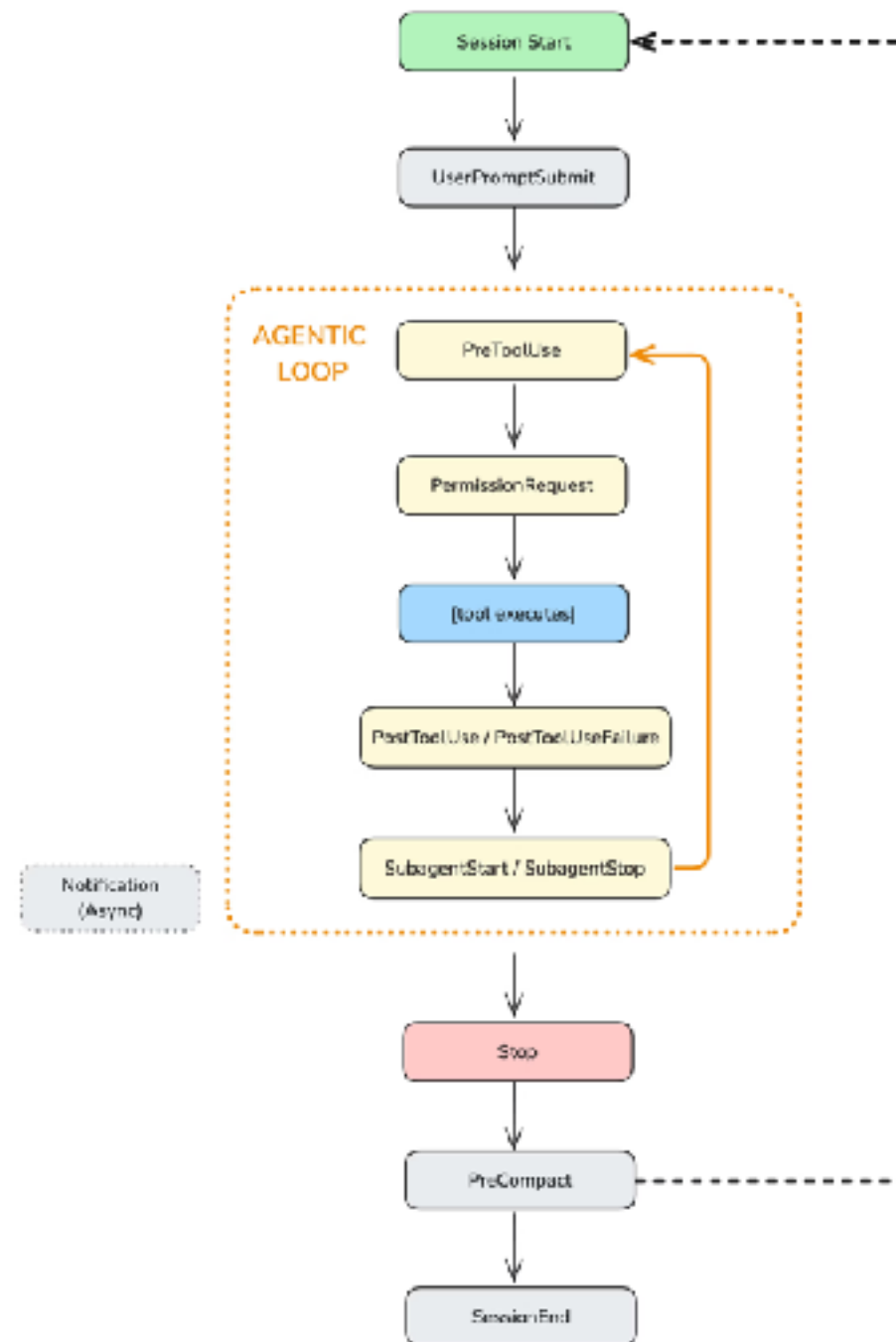
Customize and extent behavior of agent
Triggers for actions before/after conversation

Setup environment, run test, cleanup resources,
Notification, automated pre-commit check ...

<https://code.claude.com/docs/en/hooks>



Hooks in Agent's Lifecycle

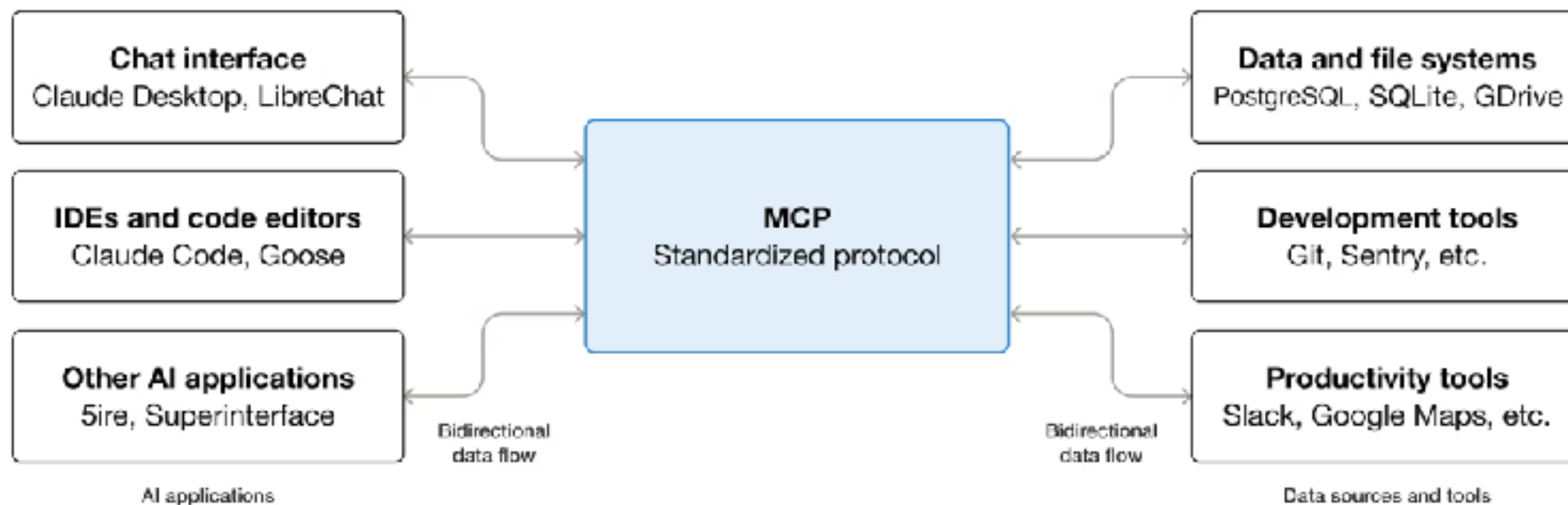


<https://code.claude.com/docs/en/hooks>



MCP (Model Context Protocol)

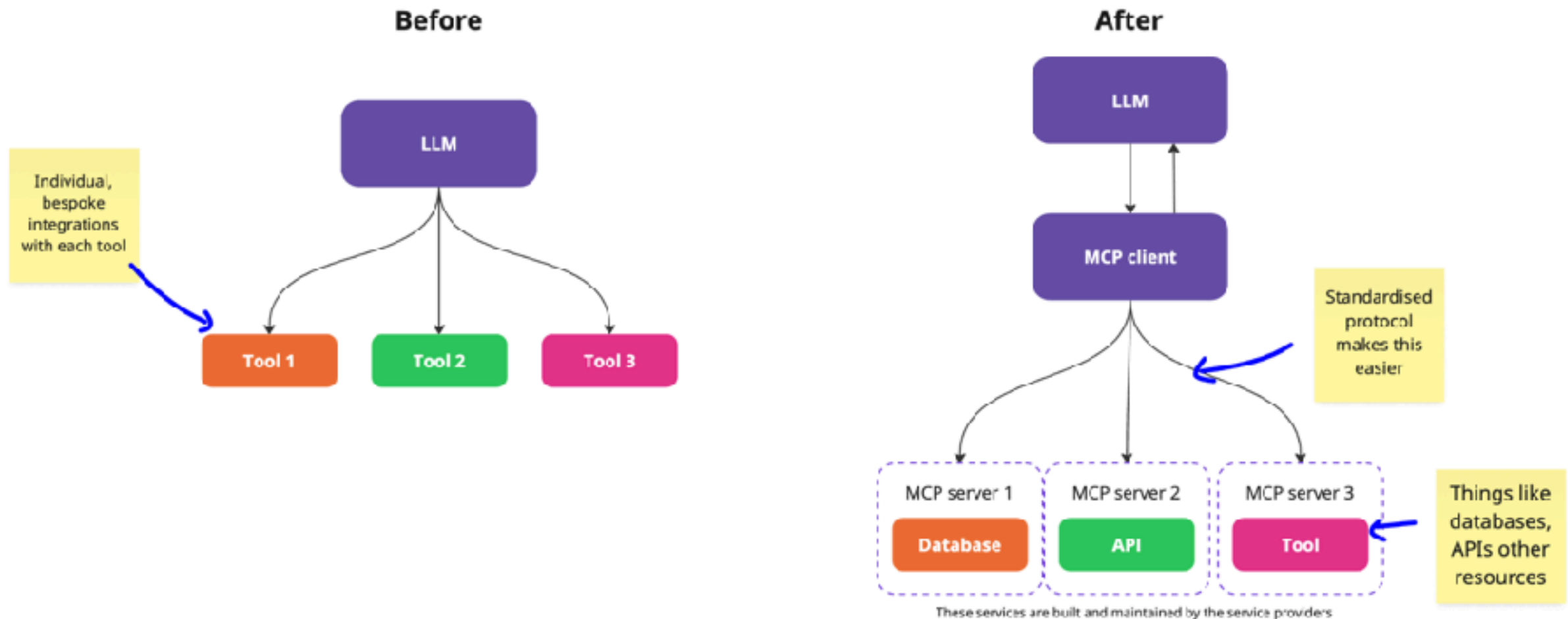
Open standard for connecting AI app to external system



<https://modelcontextprotocol.io/>



MCP (Model Context Protocol)



<https://departmentofproduct.substack.com/p/mcp-explained-a-simple-guide-for>

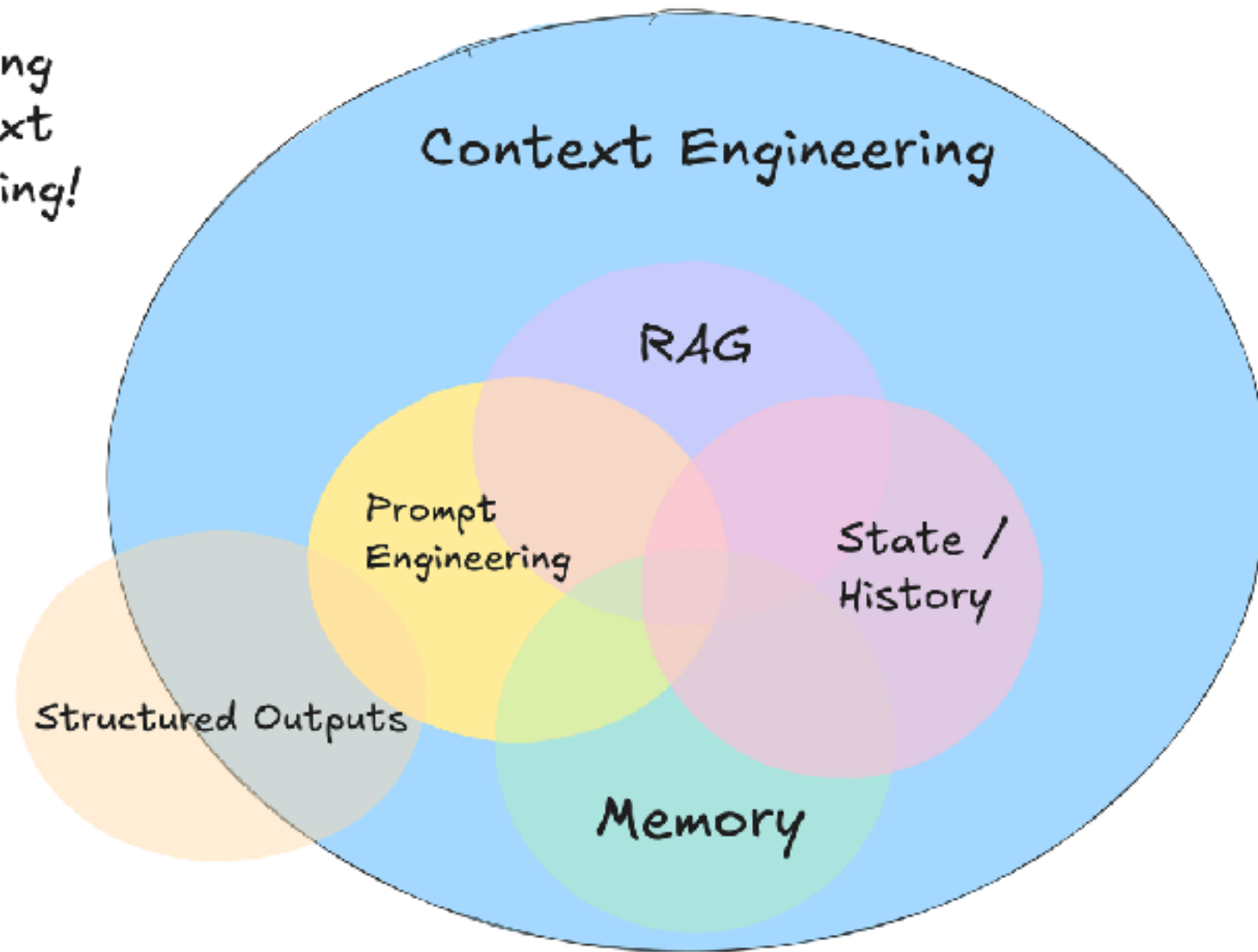


How to manage context ?



Context Engineering

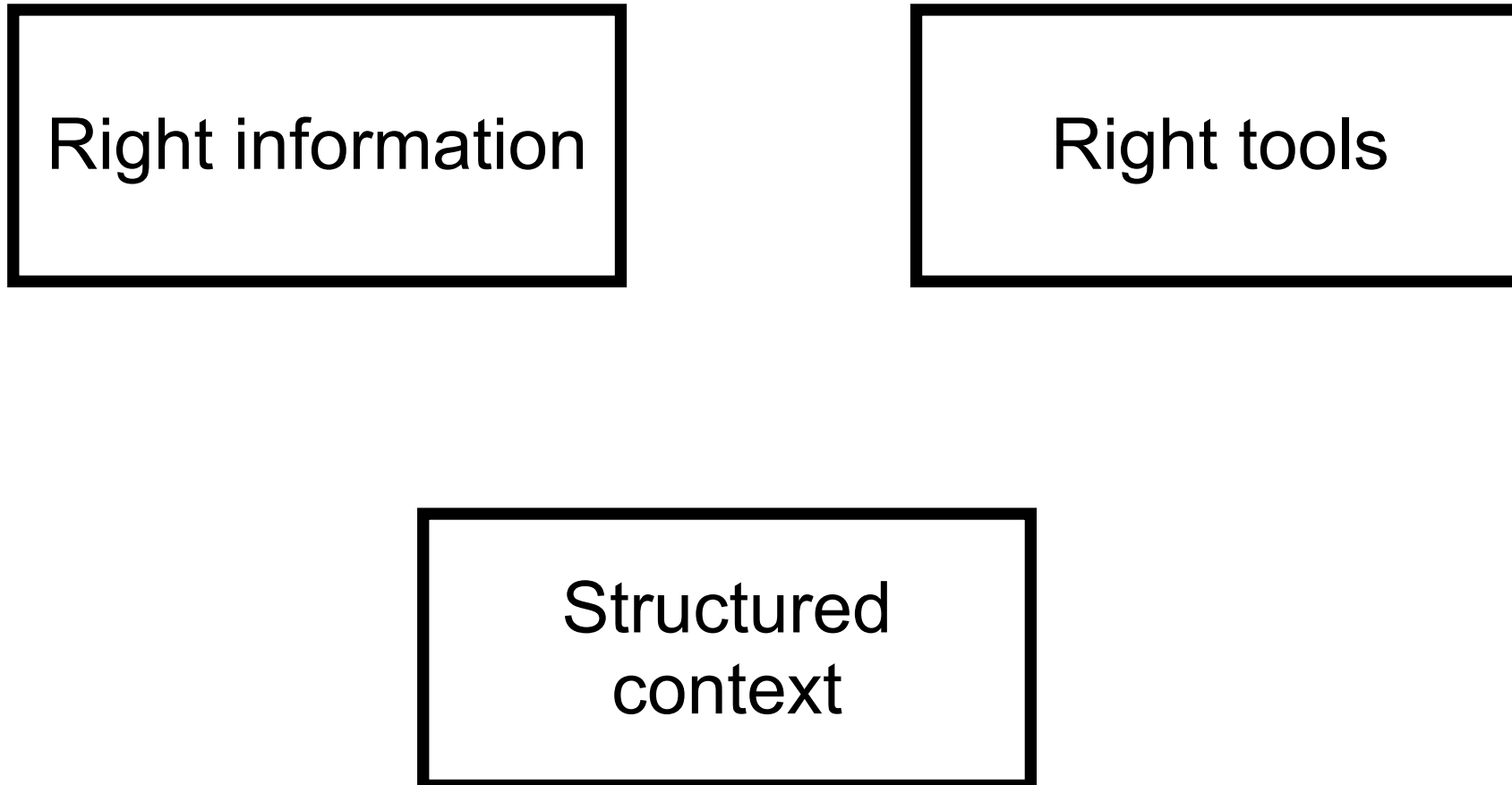
Everything
is Context
Engineering!



<https://www.promptingguide.ai/guides/context-engineering-guide>



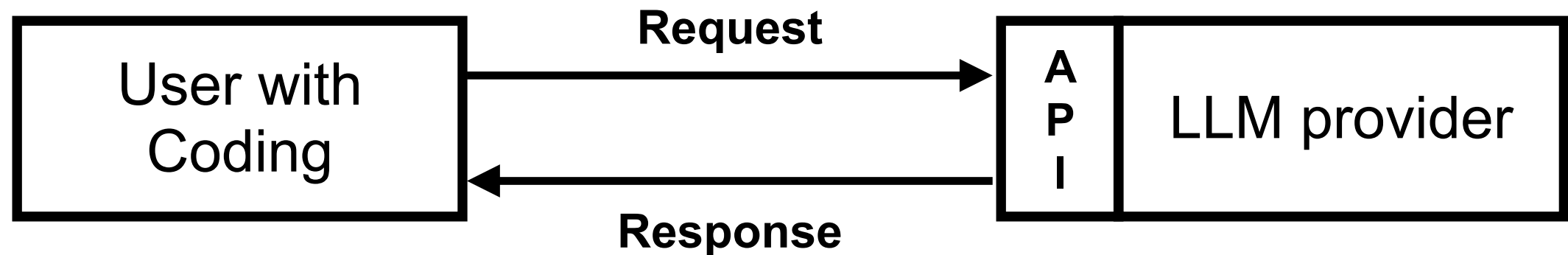
Context Engineering



Context window size ?

Amount of text a model can process and remember
at once time

Measure with **token**



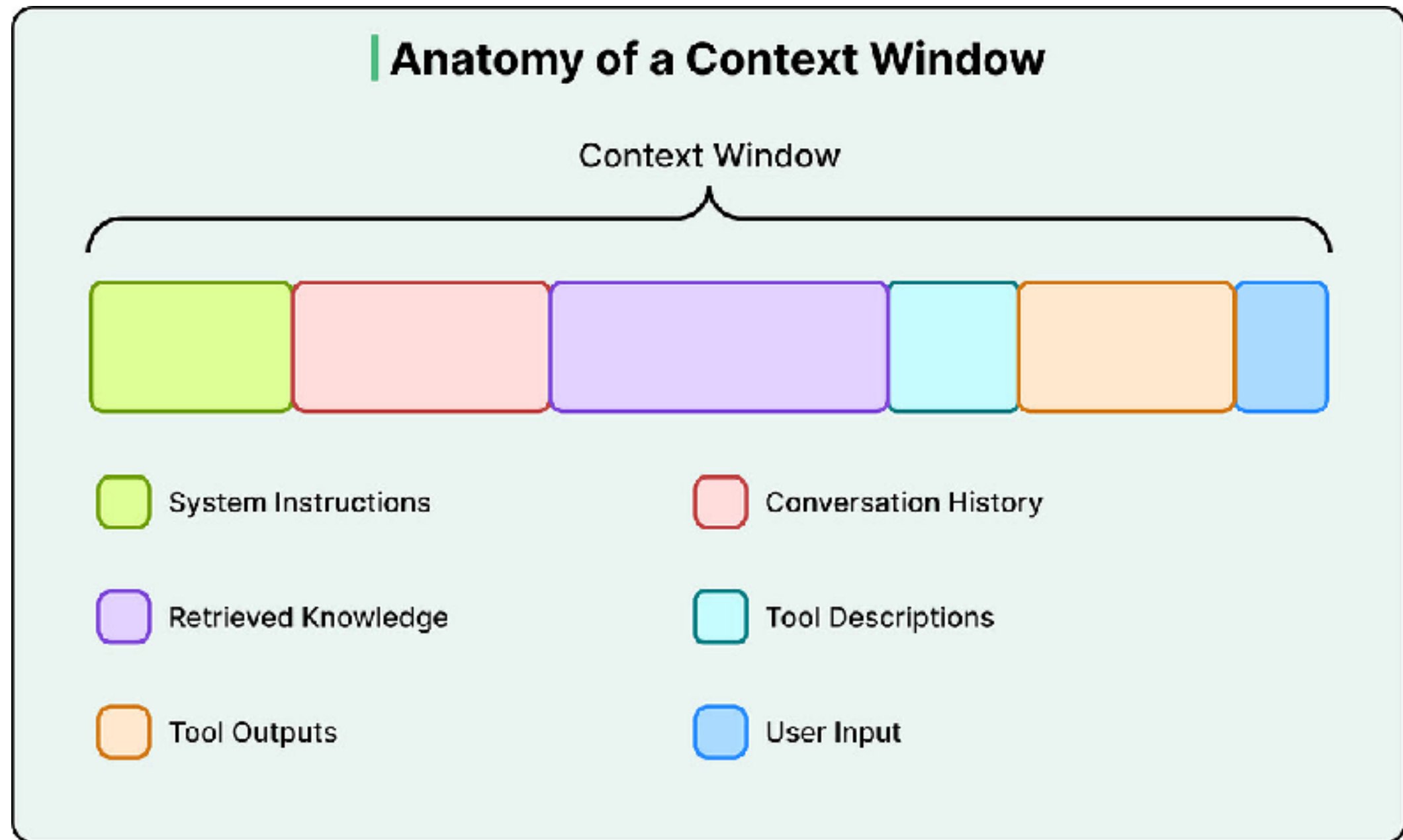
Context window size of models

Different models may have different window size

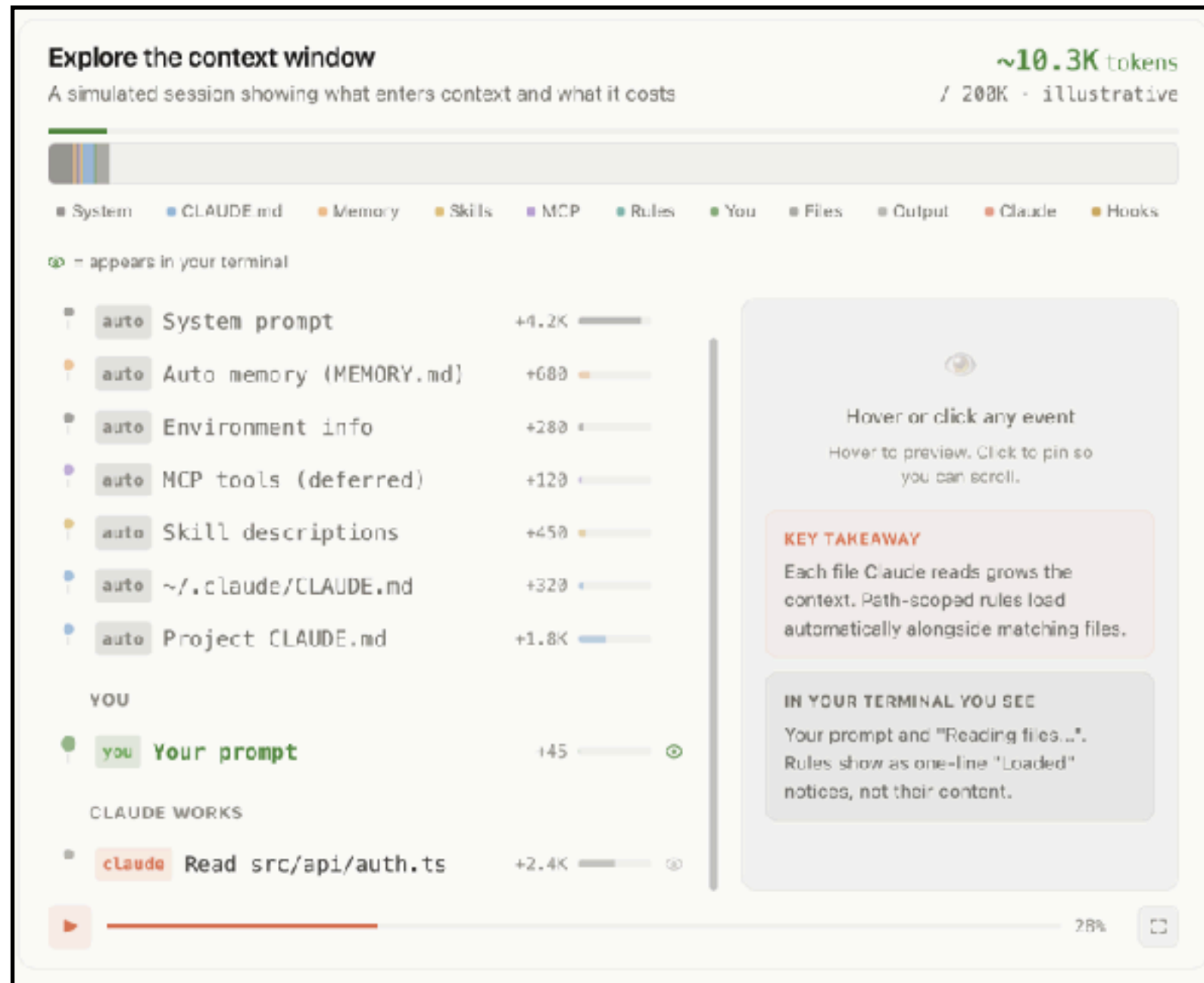
Model name	Context window size
GPT 4.1	1M
Gemini 1.5, 2.5	1M
Llama 4	1M
Claude Opus 4.8	1M
Claude Haiku 3.5	200,000
DeepSeek R1	1M



Anatomy of Context Window



Explore Context Window



<https://code.claude.com/docs/en/context-window#what-the-timeline-shows>

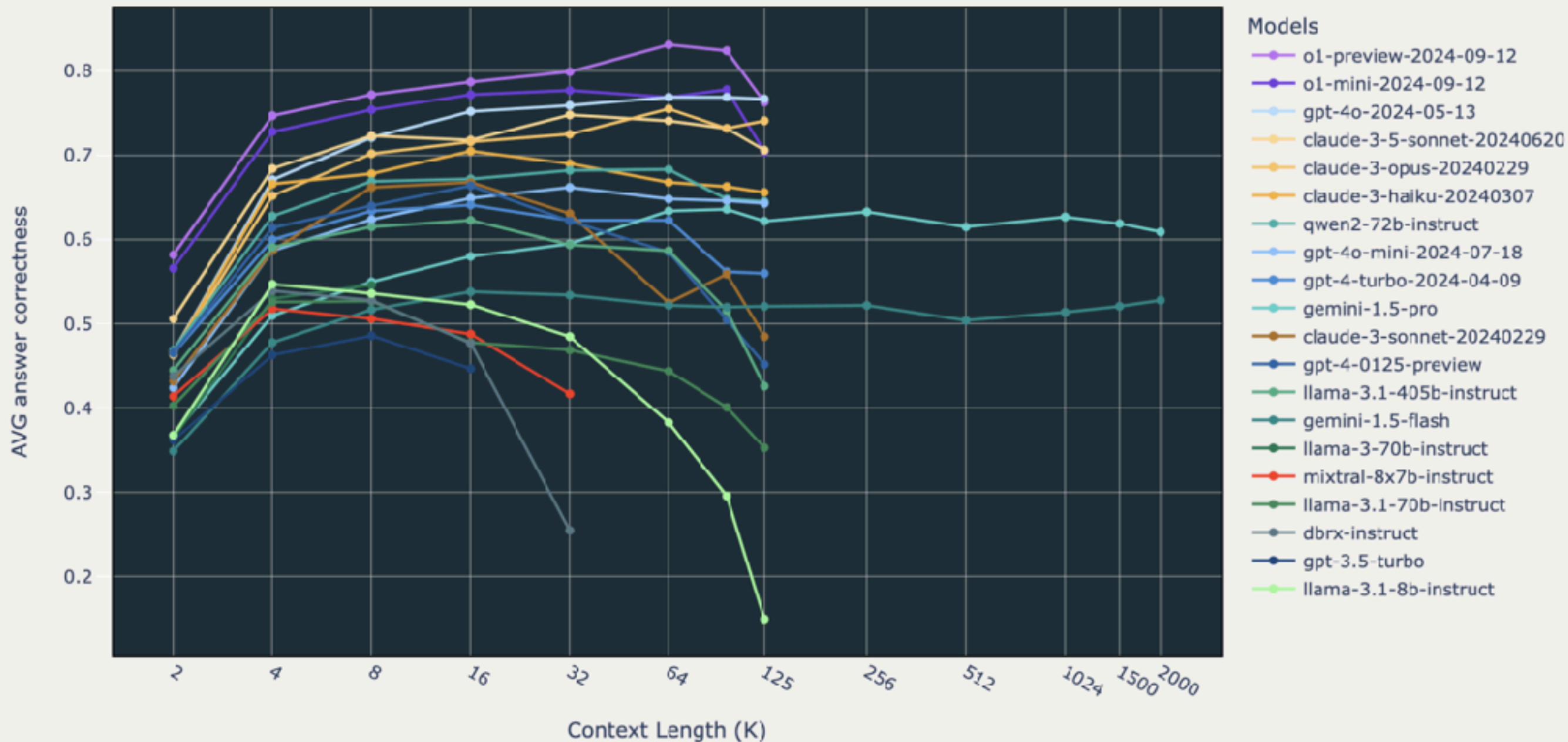


Long context == Better result ?



Long context LLM is better !!

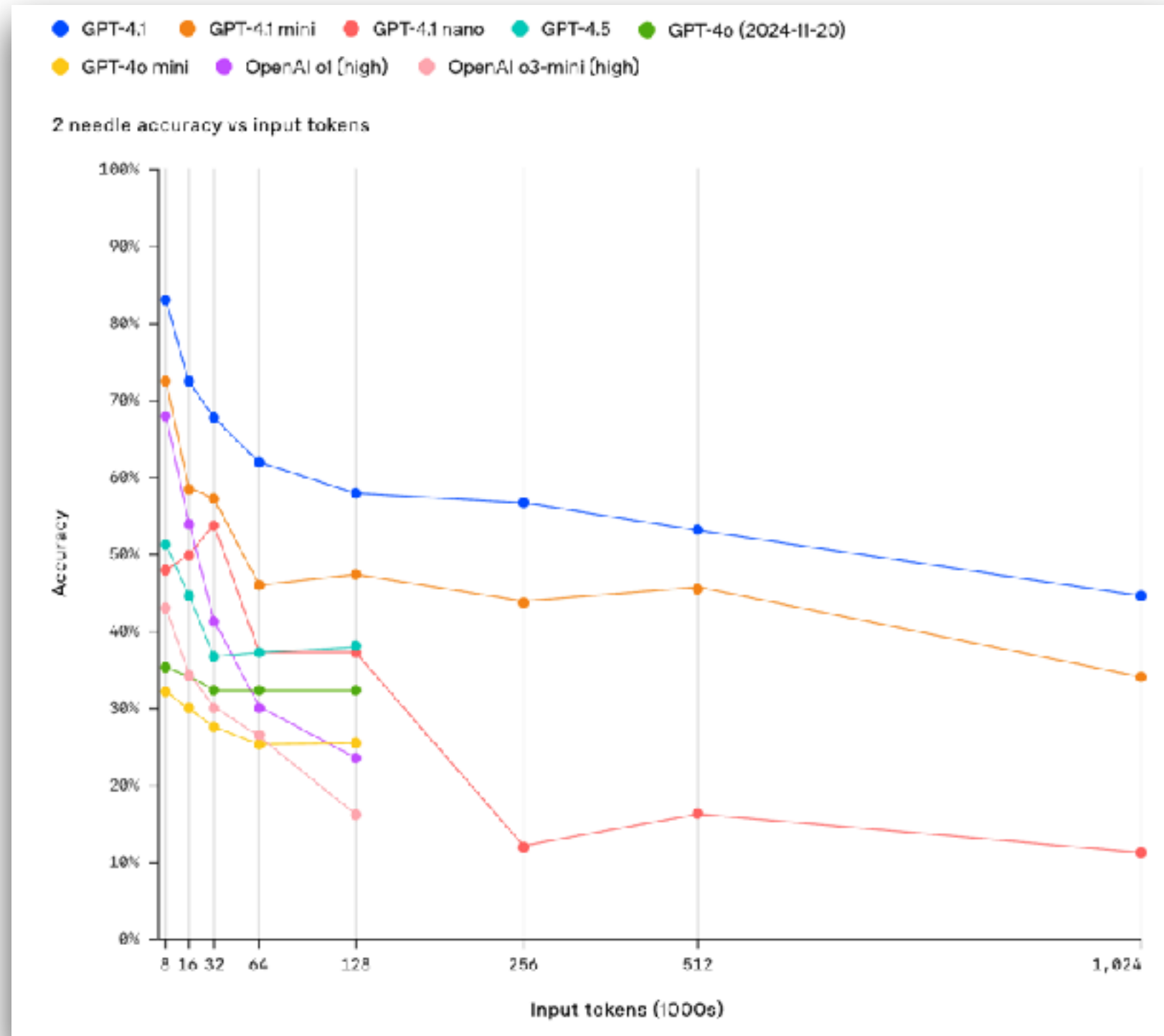
Long Context RAG Performance of LLMs



<https://arxiv.org/abs/2411.03538>



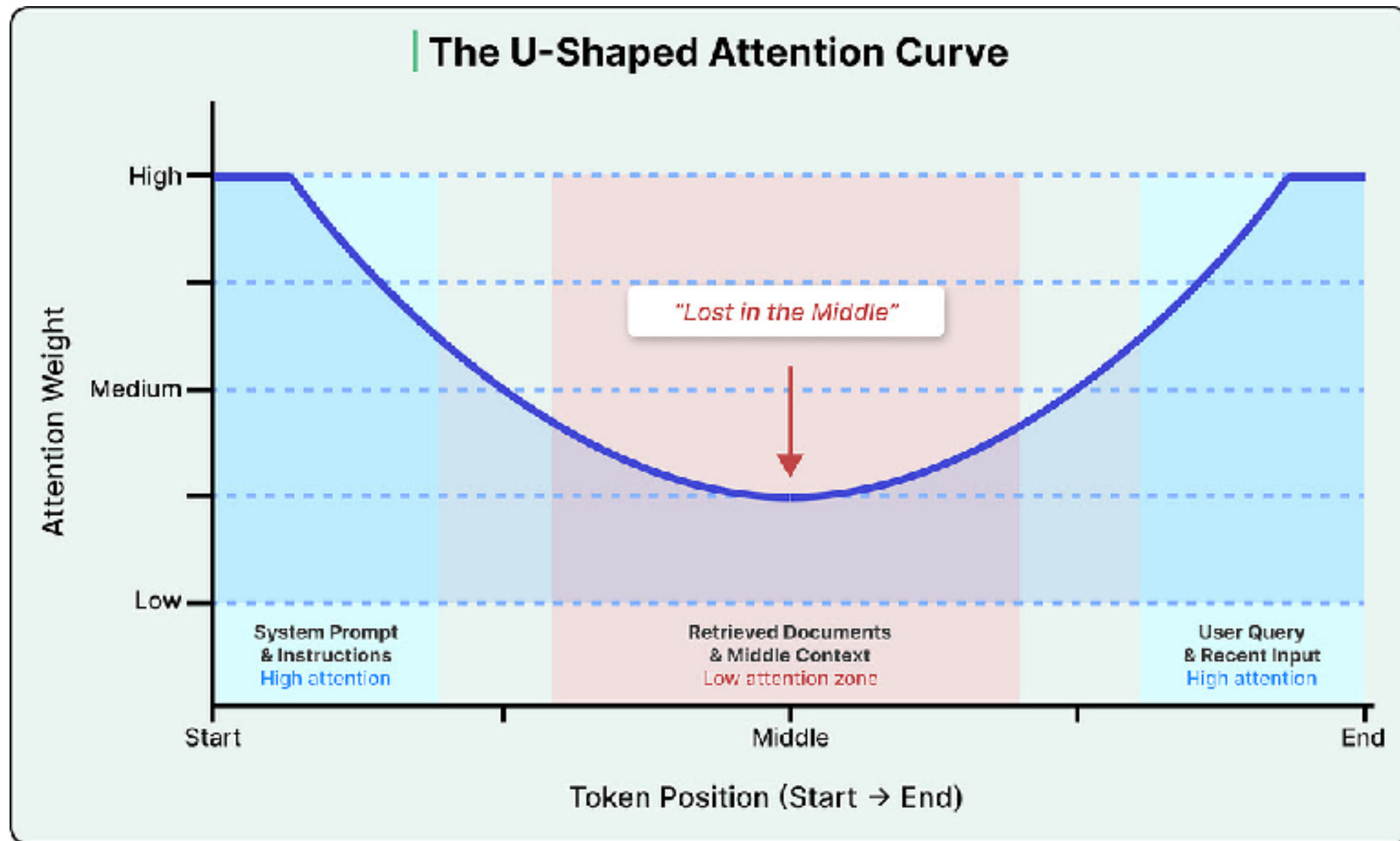
Long context LLM is better !!



<https://openai.com/index/gpt-4-1/>



Loss in the middle of LLM

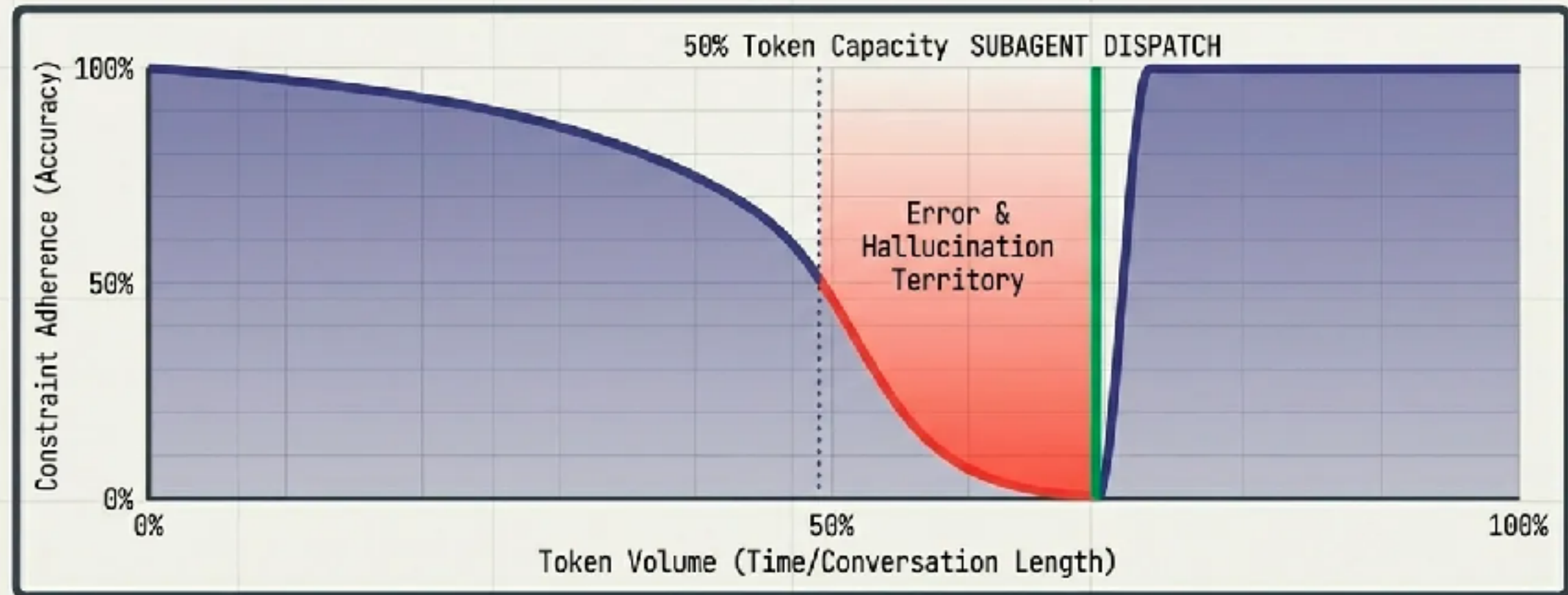


<https://arxiv.org/abs/2307.03172>



Need to provide clean context !!

Gate 4: Beating Context Degradation



Superpowers dispatches fresh subagents for every granular task, providing a clean context window completely free from earlier conversational noise.

<https://pub.towardsai.net/from-vibe-coding-to-viable-coding-how-superpowers-turns-ai-chatbots-into-engineering-partners-59c081f02a26>

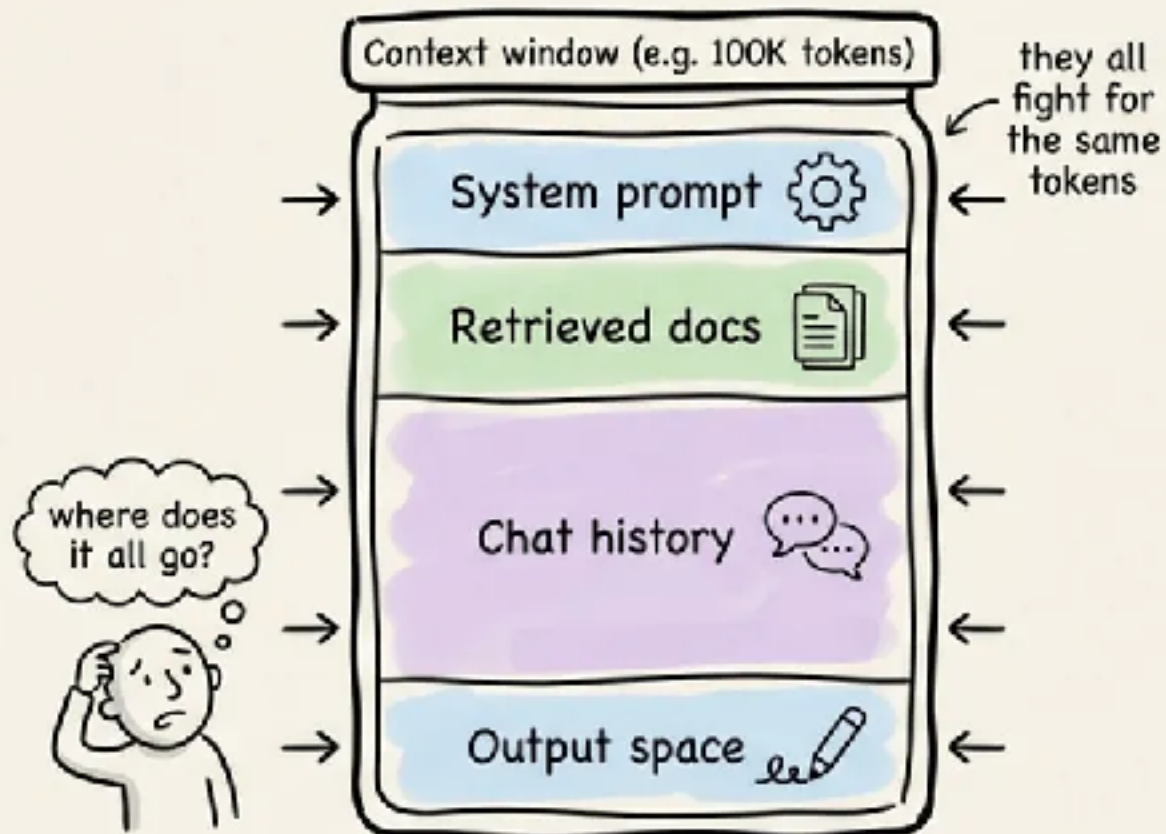


Better context == Better result ?

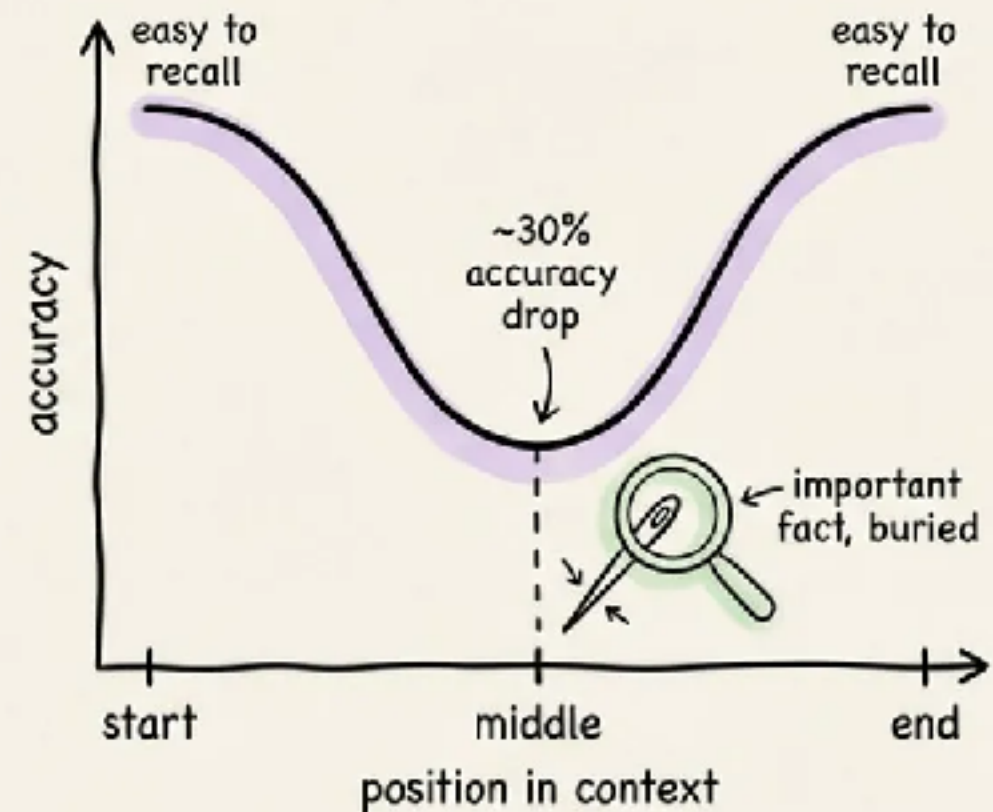


More tokens isn't more memory.

Context = a shared budget



Lost in the middle



Bigger windows don't beat structured memory.

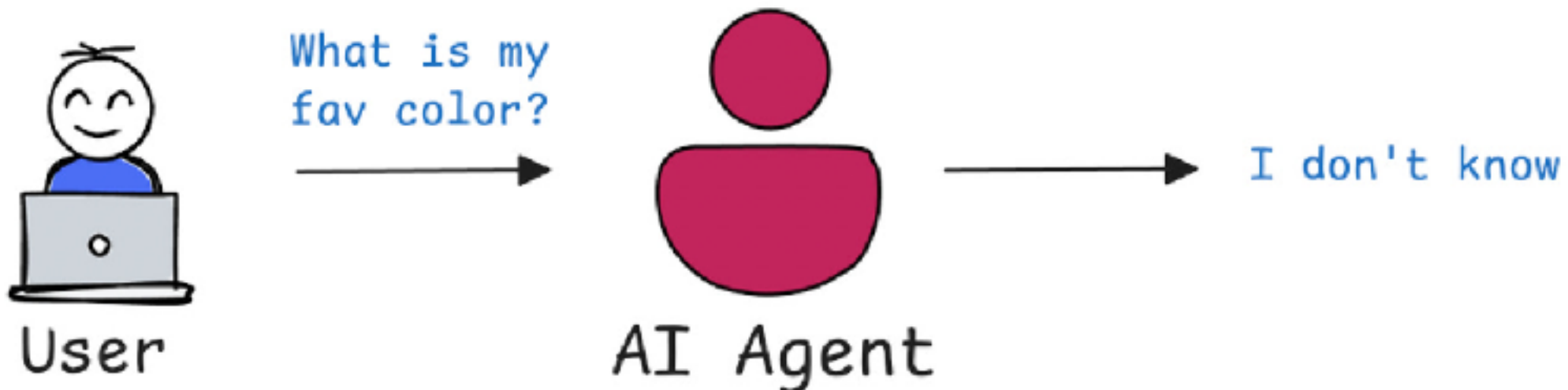
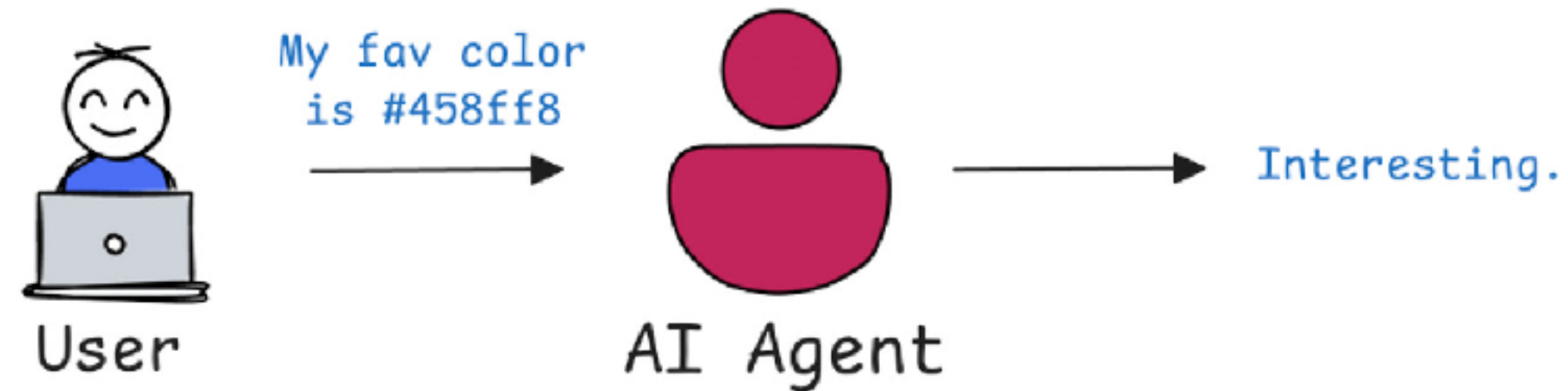
<https://blog.dailydoseofds.com/p/evolution-of-agent-landscape-from>



Agent Memory Memory Bank



Interaction without memory

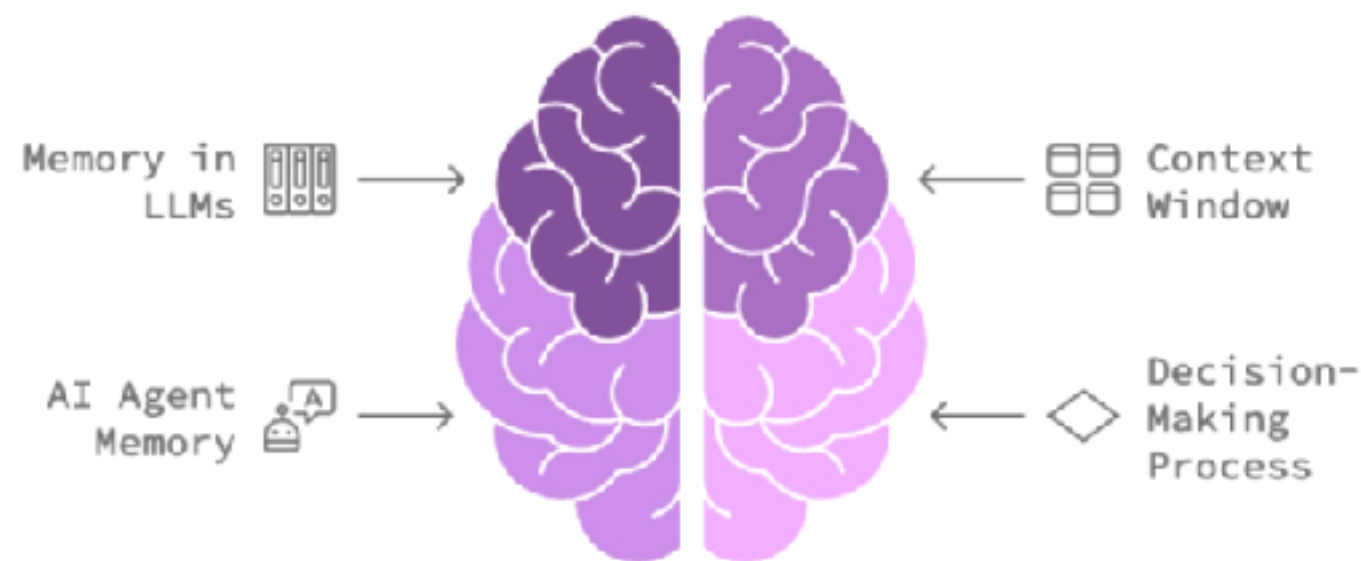


Why AI Agent memory important ?

Enable personalized and context-aware interactions

Improve decision-making and adaptability

Enhance LLM's performance and response



<https://www.falkordb.com/blog/ai-agents-memory-systems/>



Types of Memory

To make intelligent decisions
Agents combine different types of memory

Short-term



Current goals and steps
File system (markdown, JSON)
In-memory database

Episodic



Logs of past interactions
Success and failure

Semantic

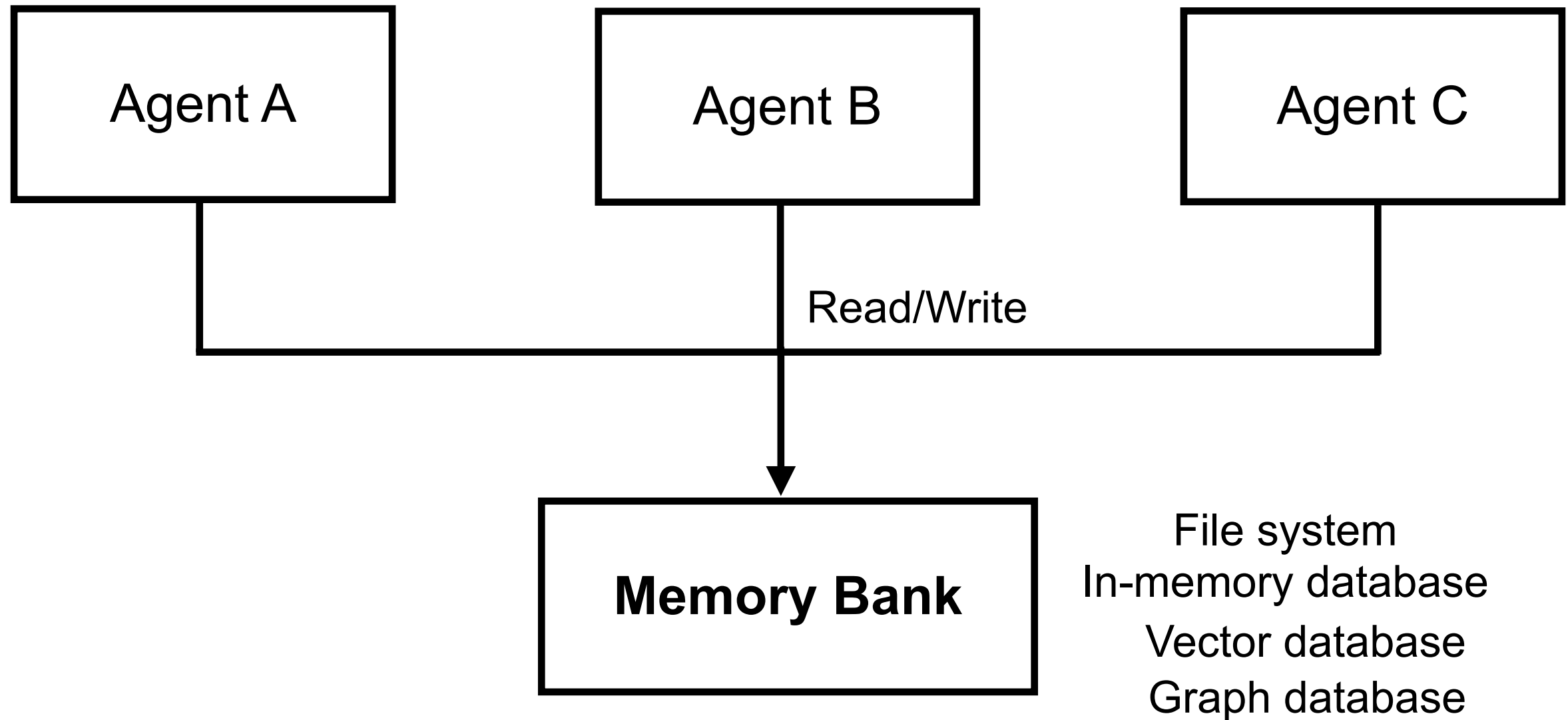


Long-term knowledge
Fact and strategies
RAG
Vector and graph database

<https://bhavishyapandit9.substack.com/p/how-memory-works-in-agentic-ai-a>

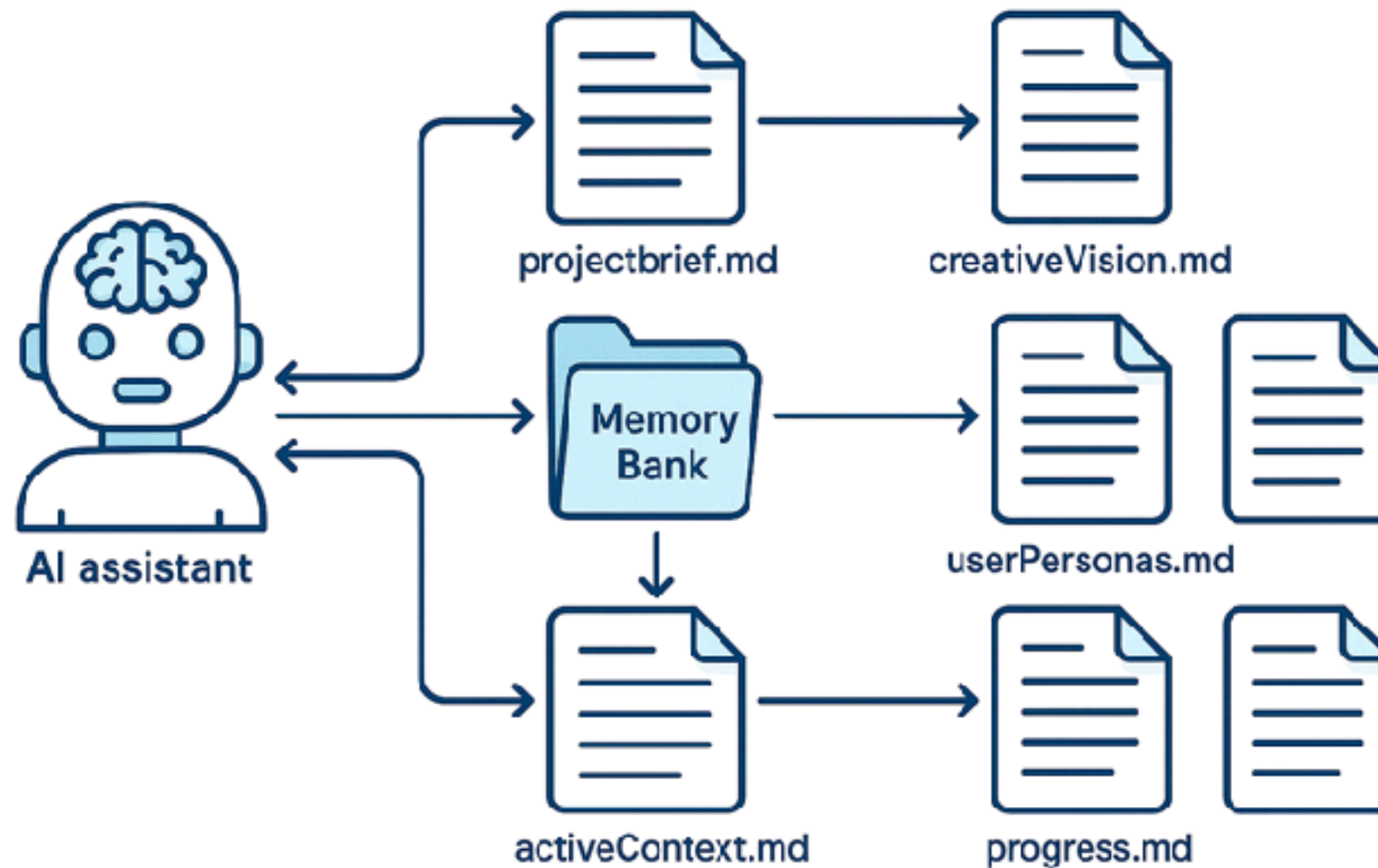


Shared memory between Agent



Cline Memory Bank

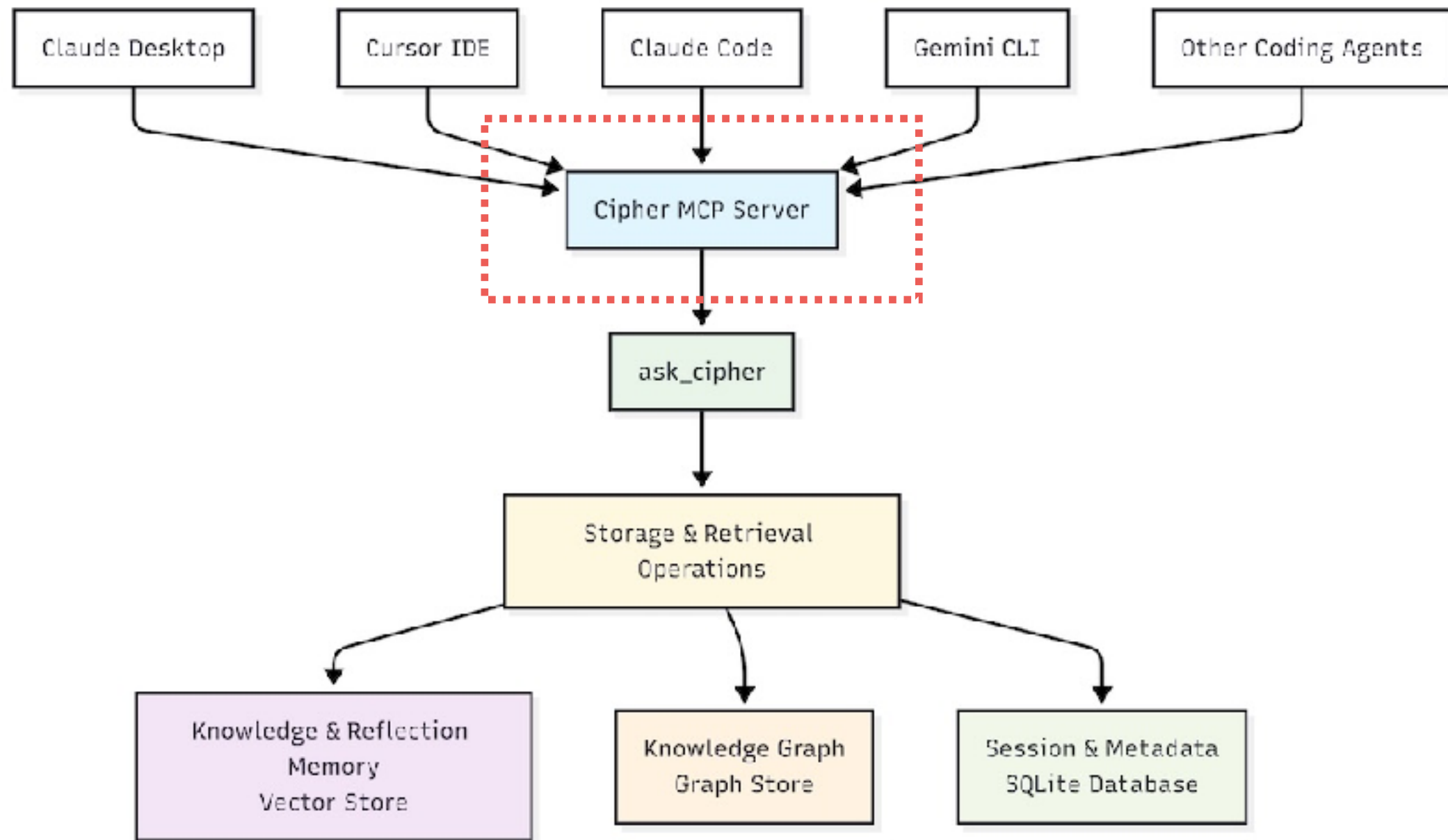
Working with file system



<https://docs.cline.bot/prompting/cline-memory-bank>



Memory Bank via MCP



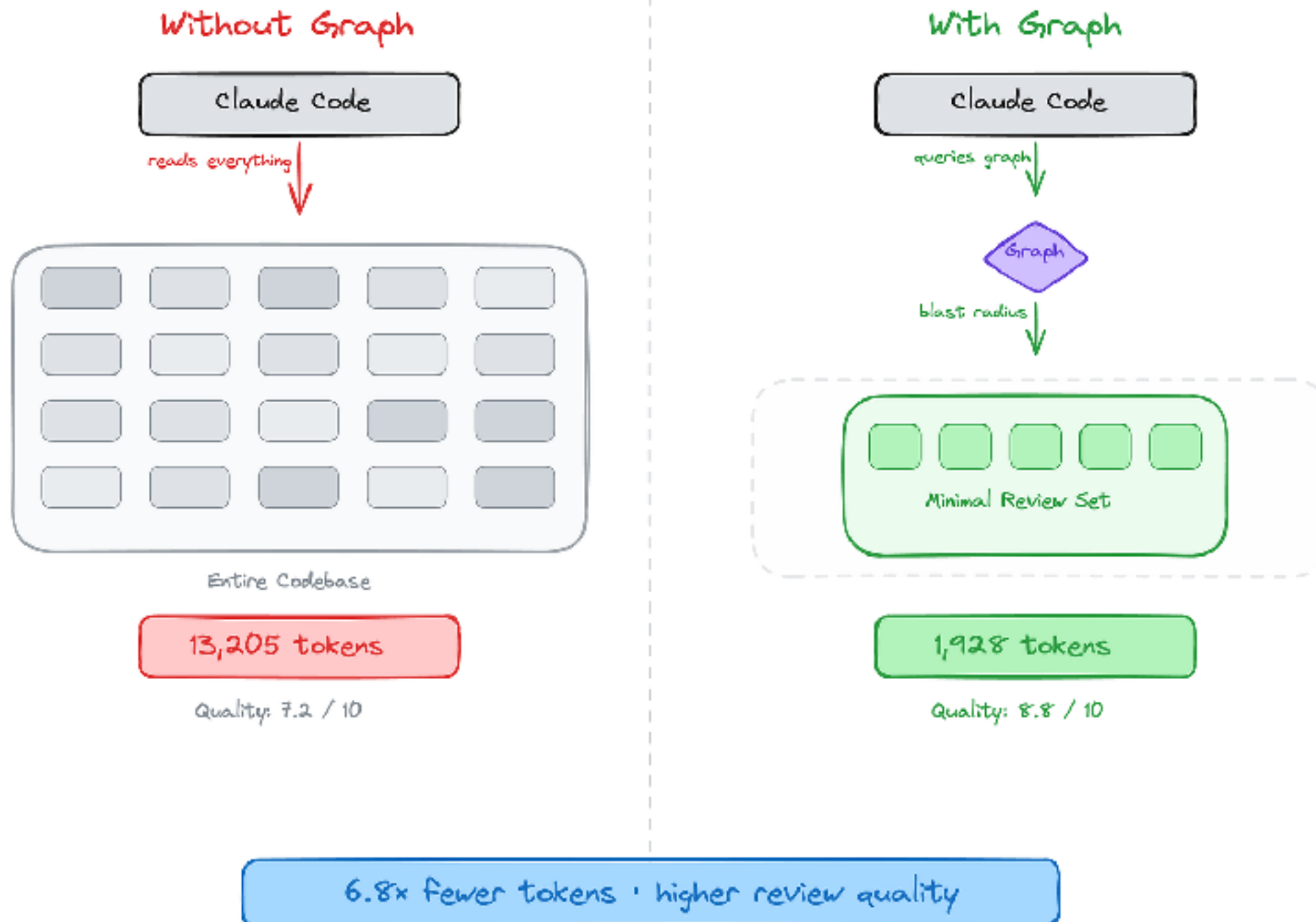
<https://www.somkiat.cc/share-memory-for-ai-ide/>



How to reduce Token usage ?



The Token Problem



<https://code-review-graph.com/>



Claude Code ?

Manage context with slash command

/context

/compact

/clear

/rewide

<https://code.claude.com/docs/en/context-window>



Context Engineering best practices ?

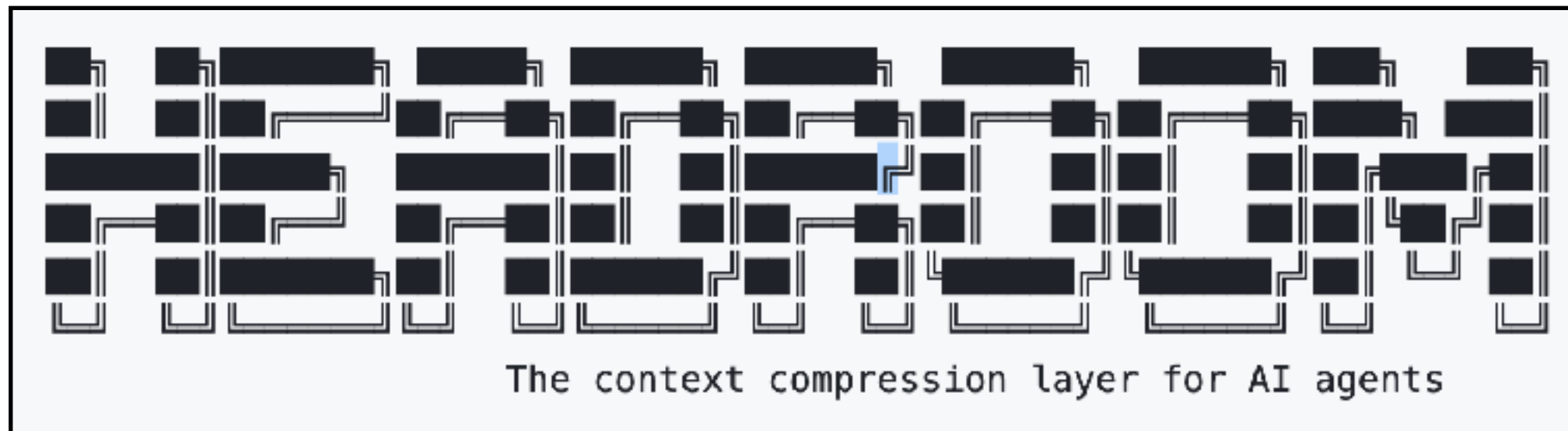
Keep CLAUDE.md lean (200 lines)
Be specific with file loading (with @file)
Leverage **Subagents**
Start fresh for new task

<https://claude.com/blog/using-claude-code-session-management-and-1m-context>



More tools ?







Supercharge your Agent with Semantic Code Intelligence
and save 💰 in the process!

► FEWER TOKENS ► FEWER TOOL CALLS ► 100% LOCAL

```
$ cargo install tokensave
```

also via `brew install aovestdipaperino/tap/tokensave` or `scoop bucket add tokensave`
`https://github.com/aovestdipaperino/scoop-aovestdipaperino && scoop install tokensave`

<https://tokensave.dev/>



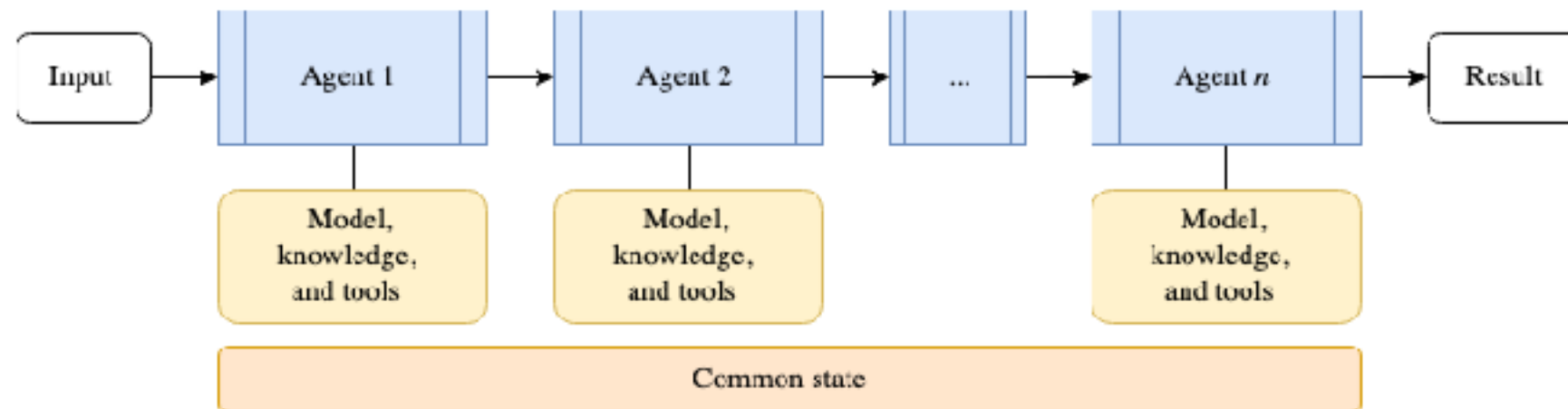
Sub-agent and Multi-agent !!

Reduce size of knowledge !!



Monolith to Multi-agent

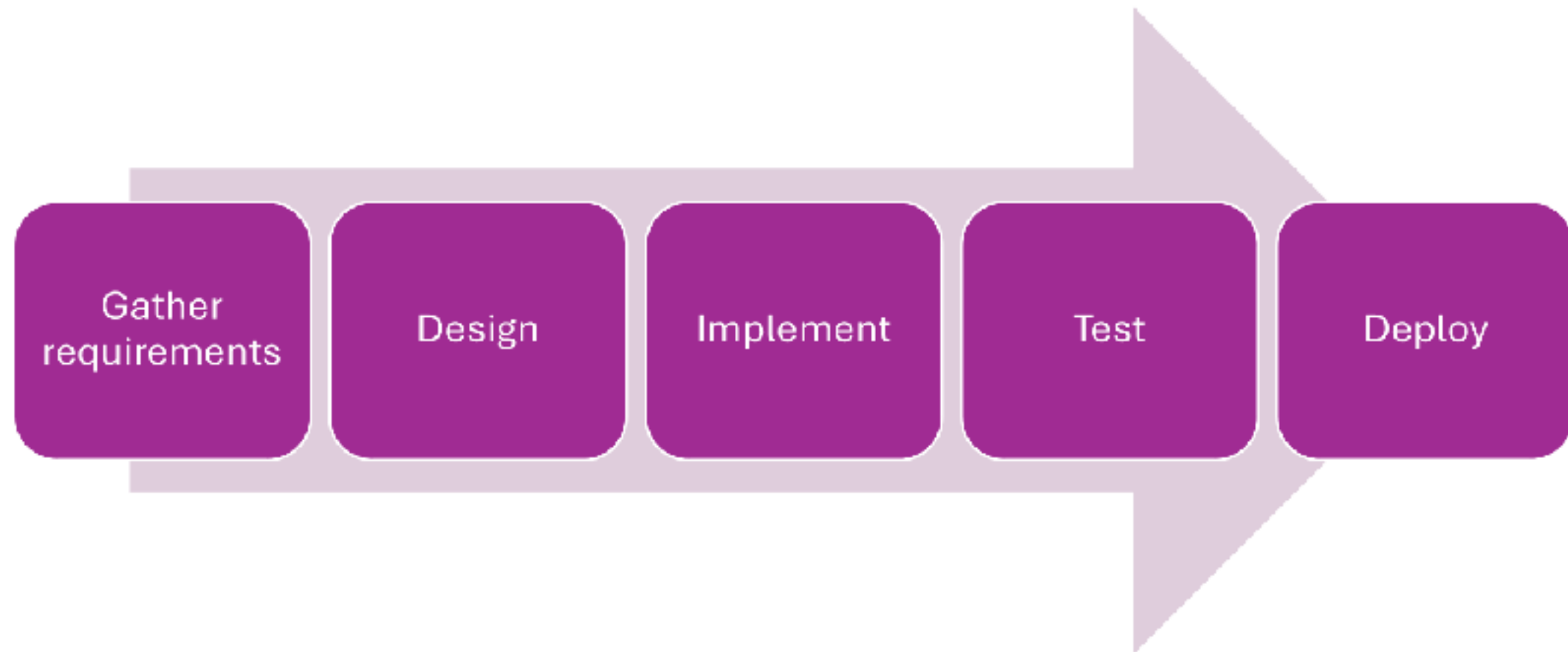
Specialization
Scalability
Maintainability
Optimization



<https://learn.microsoft.com/en-us/azure/architecture/ai-ml/guide/ai-agent-design-patterns>



Design your Agents ?



<https://devblogs.microsoft.com/dotnet/introducing-microsoft-agent-framework-preview/>



Design your Agents ?

Each step may contain subtasks
Different specialists have different steps
Progress isn't always linear

Bugs found during testing may send back to
implementation



Design your Agents ?

Full Stack Agent !!

Frontend

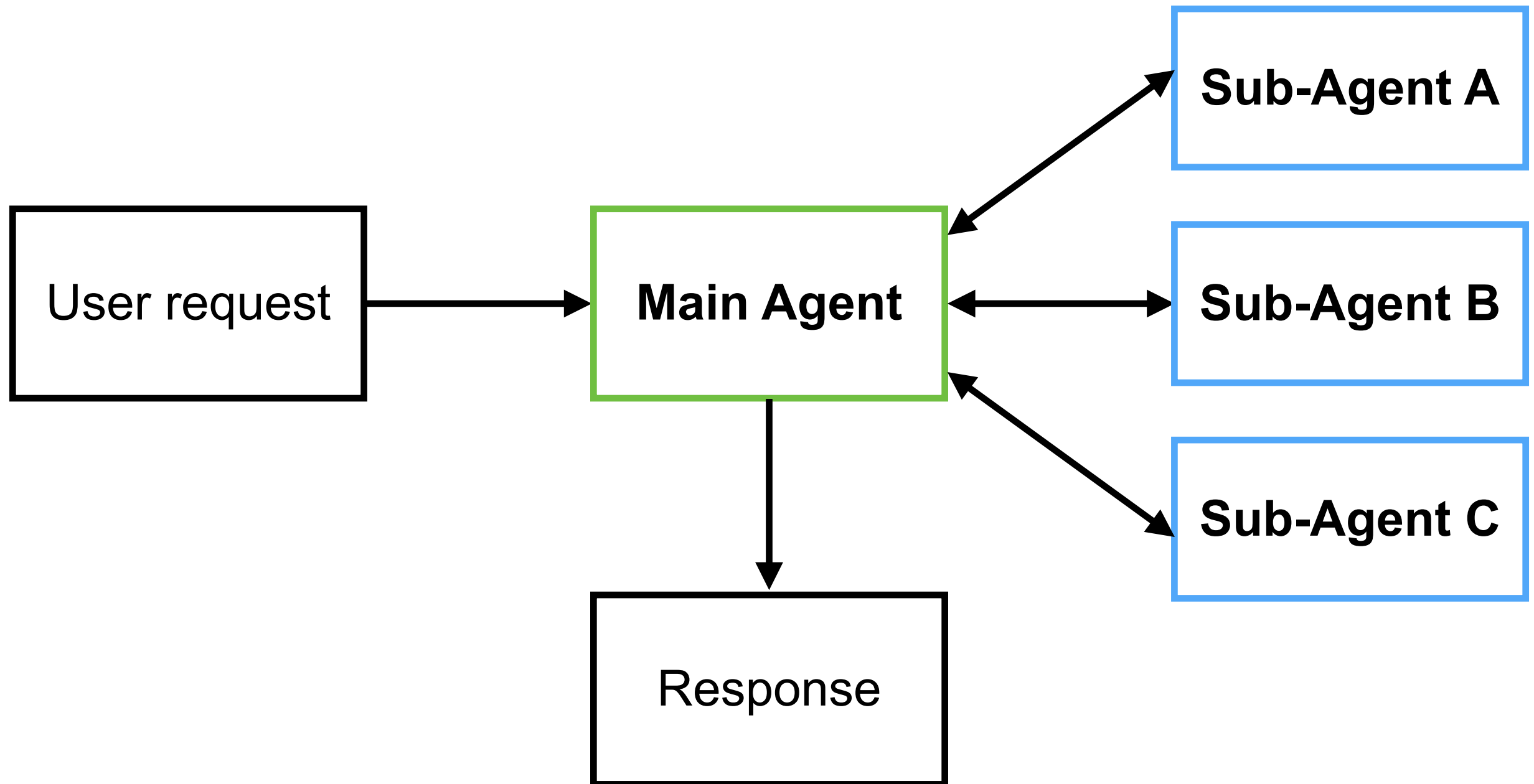
Backend

Database



Multi AI Agents !!

Sub-agent with **centralized** orchestration

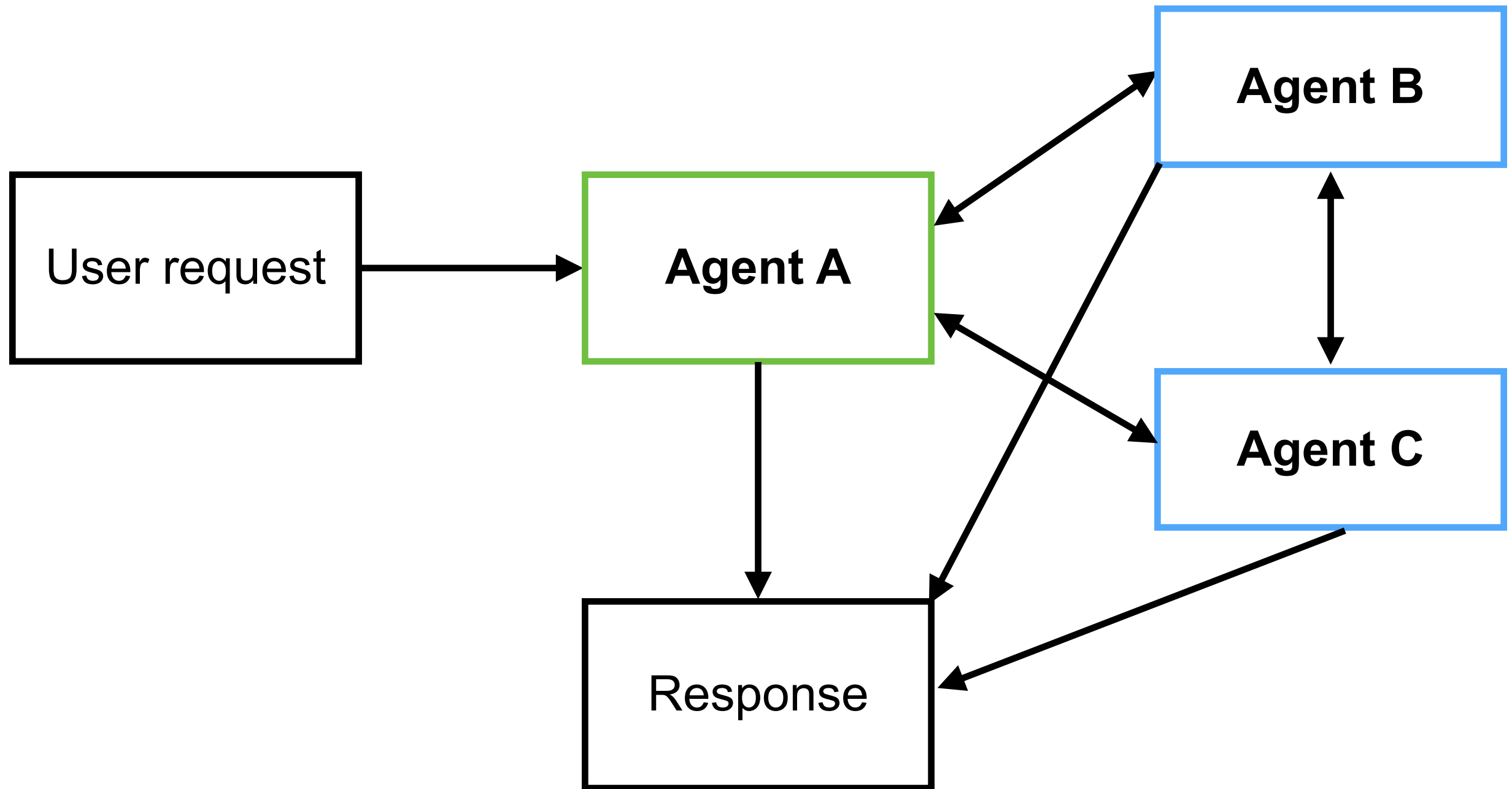


<https://www.blog.langchain.com/choosing-the-right-multi-agent-architecture/>



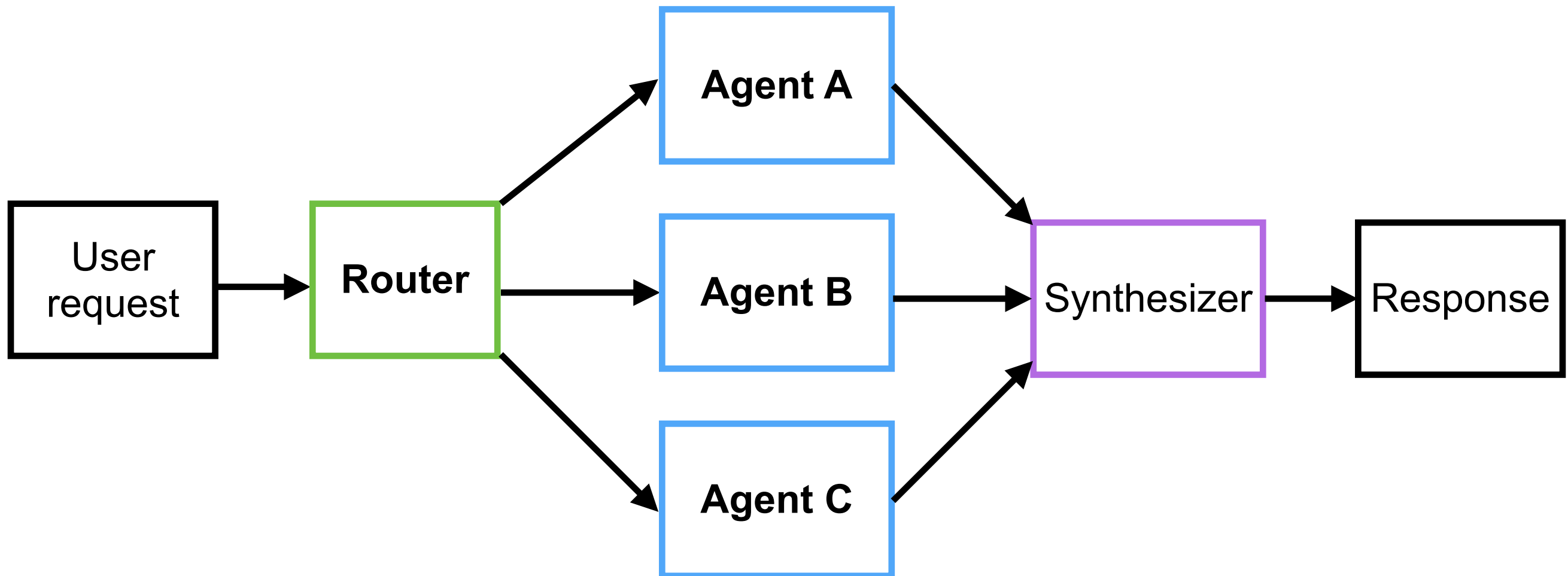
Multi AI Agents !!

State-driven transitions based-on conversation context



Multi AI Agents !!

Router and parallel run



Prompt vs Agent Skills ?



Command

<https://agentskills.io/>



Agent Skills

Moving from a paradigm of
“prompt” to “programming by instruction”

Agent loads specialized prompts and knowledge on-demand

Specialized
task

Reduce
repetition

Compose
capabilities

Efficient
loading

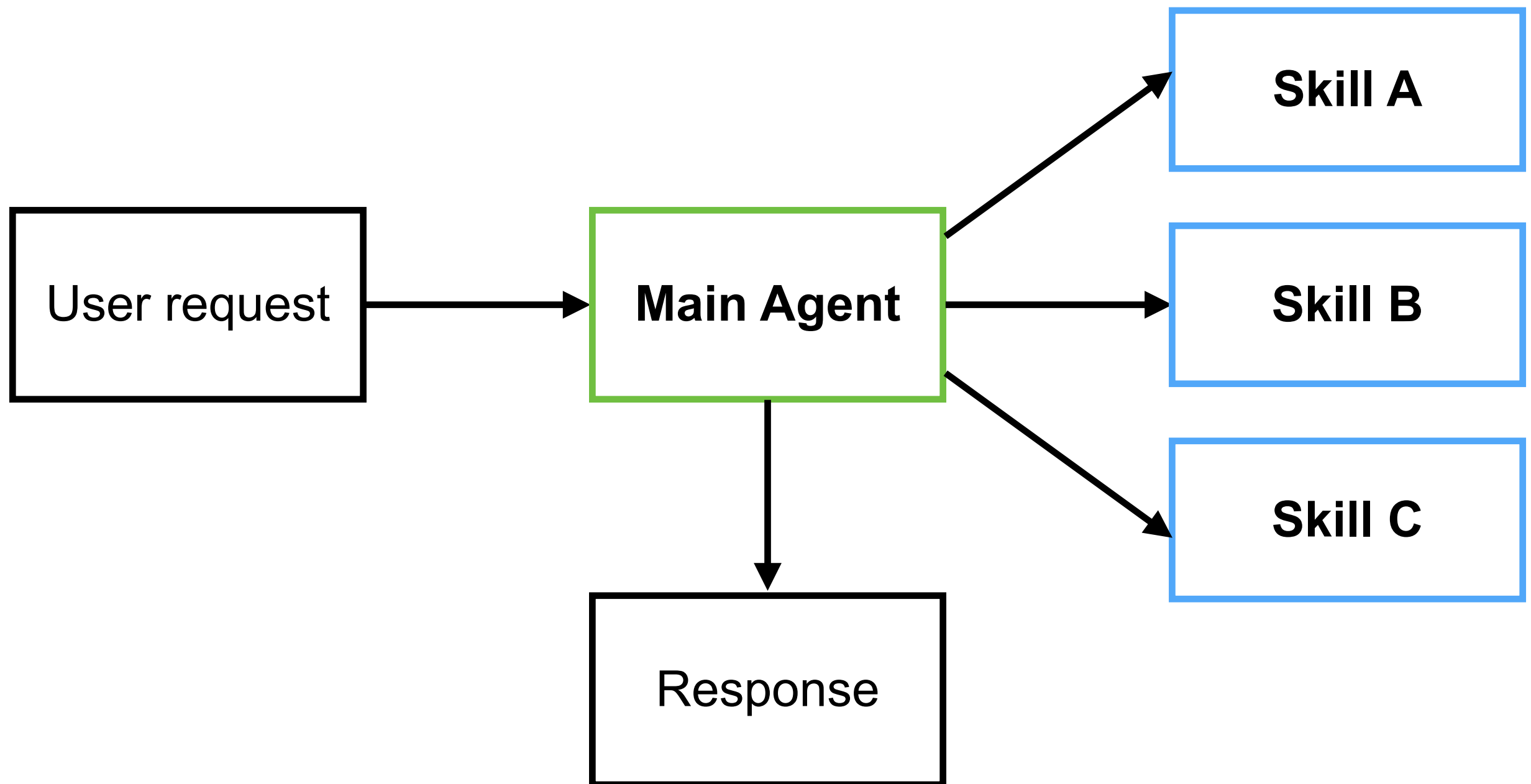
Interoperability

<https://agentskills.io/>



Agent with Skills

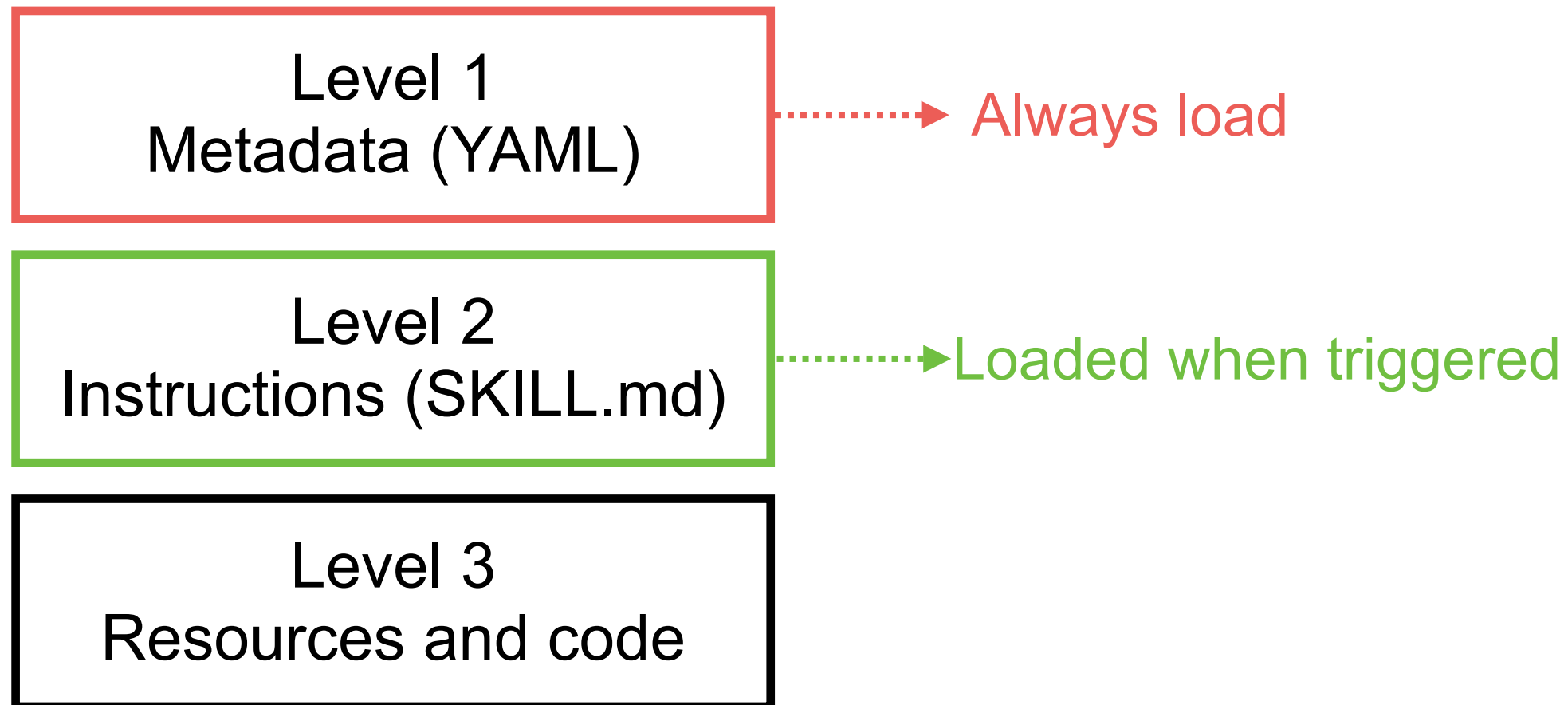
Load skills on-demand from context



<https://www.blog.langchain.com/choosing-the-right-multi-agent-architecture/>



Structure of Agent Skills



<https://agentskills.io/>



Prompt vs Skill

Features	Prompt	Agent Skills
Persistence	Ephemeral (chat-specific)	Persistent (Useable cross conversation)
Activation	Manual by user	Automatic (Triggered by context)
Structure	Unstructured text/ instruction	Structured markdown File system-based
Context load	Full instruction	Load only when needed
Token cost	Higher (Repeated usage)	Lower (Load on-demand)
Use case	One-time, creative quick query	Repeatable workflows Specialized tasks

<https://agentskills.io/>



Awesome Agent Skills

The screenshot shows the Agent-Skills.md website interface. At the top left is the logo and name 'Agent-Skills.md'. At the top right is a 'Submit' button and a share icon. The main heading is 'Find awesome Agent Skills'. Below it is a subtitle: 'Agent-Skills.md is a agent skills marketplace, to find the right agent skills for you.' A search bar contains the text 'c#' with a magnifying glass icon on the left and a close 'x' icon on the right. Below the search bar, three skill cards are displayed:

- csharp-developer**
Use when building C# applications with .NET 8+, ASP.NET Core APIs, or Blazor webapps. Invoke for Entity Framework Core, minimal APIs, async patterns, CQRS with MediatR. Keywords: C#, .NET, ASP.NET...
Tags: c#, .net, aspnet-core, blazor
User: Jeffallan (24 stars, 5 forks)
- async-expert**
Expert in asynchronous programming patterns across languages (Python asyncio, JavaScript/TypeScript promises, C# async/await, Rust futures). Use for concurrent programming, event loops, async patterns...
Tags: asynchronous-programming, concurrency, performance-optimization, error-handling
User: martinholovsky (9 stars, 2 forks)
- xUnit Test Framework**
Execute and generate xUnit tests for C#/.NET projects with FluentAssertions and Moq support.
Tags: dotnet, xunit, test-framework, moq
User: FortiumPartners (5 stars, 1 fork)

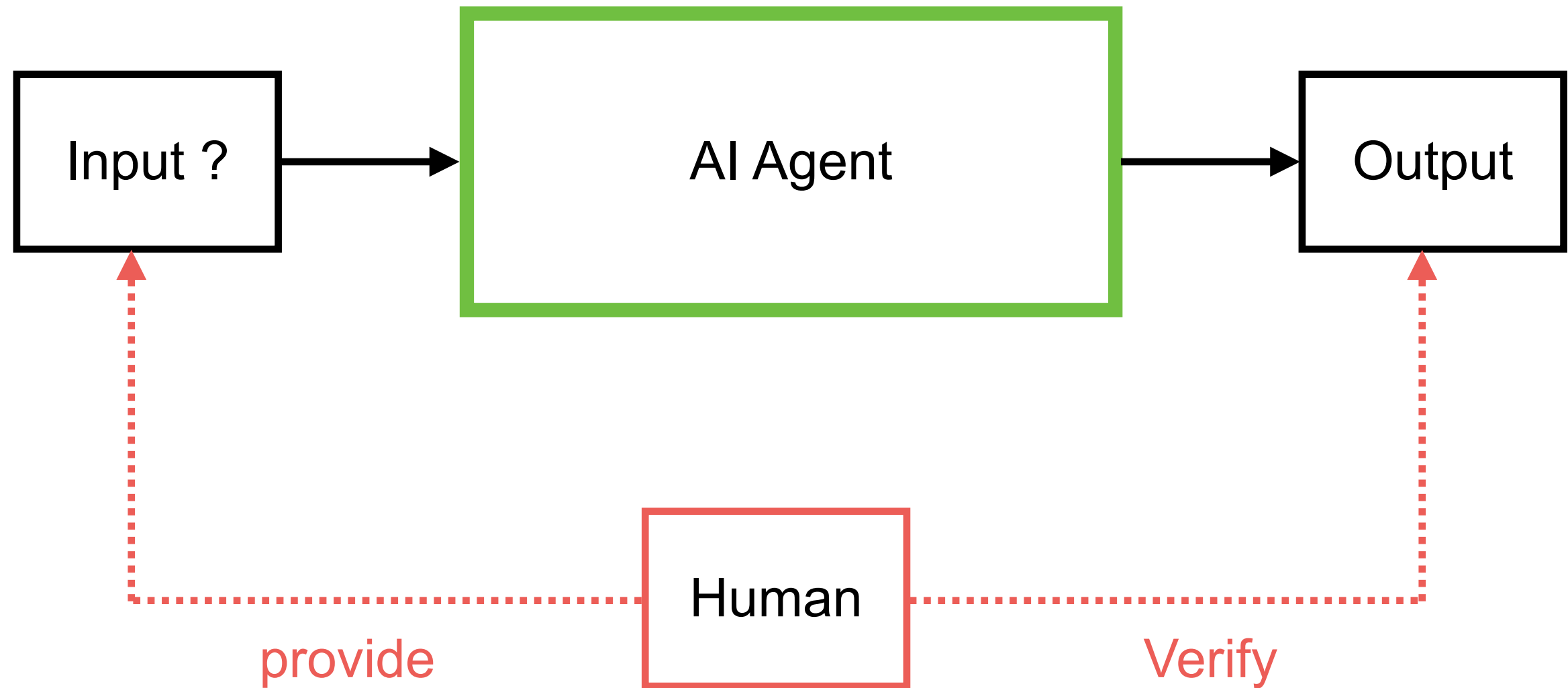
<https://agent-skills.md/>



How to provide inputs ?



How to provide inputs ?



Manage context !! In Software development



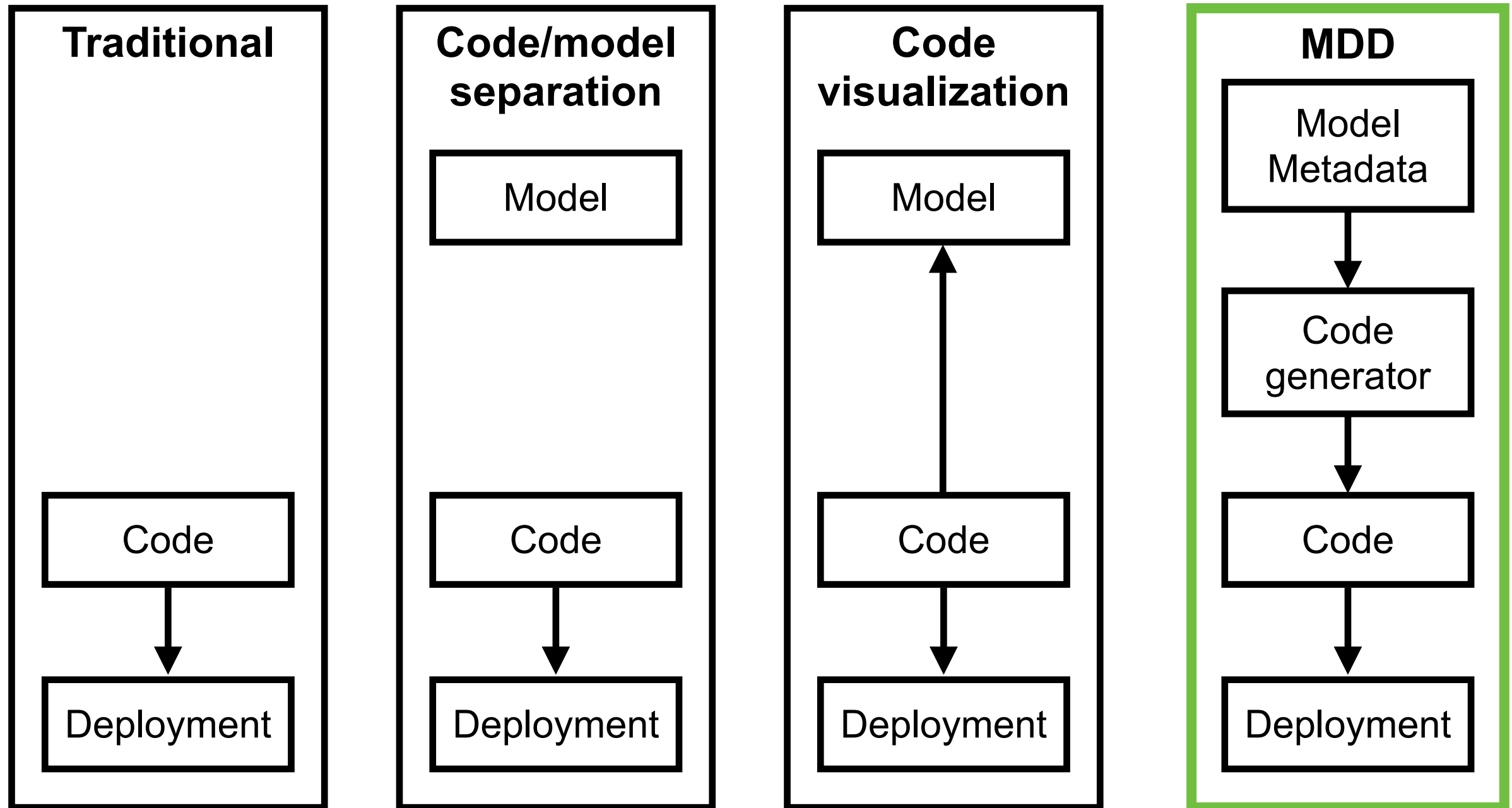
Specification ?
Architecture ?
Modeling ?
Coding convention ?
Project structure ?
Better practice ?
Technology stack ?
Test cases



Model-Driven Development (MDD)



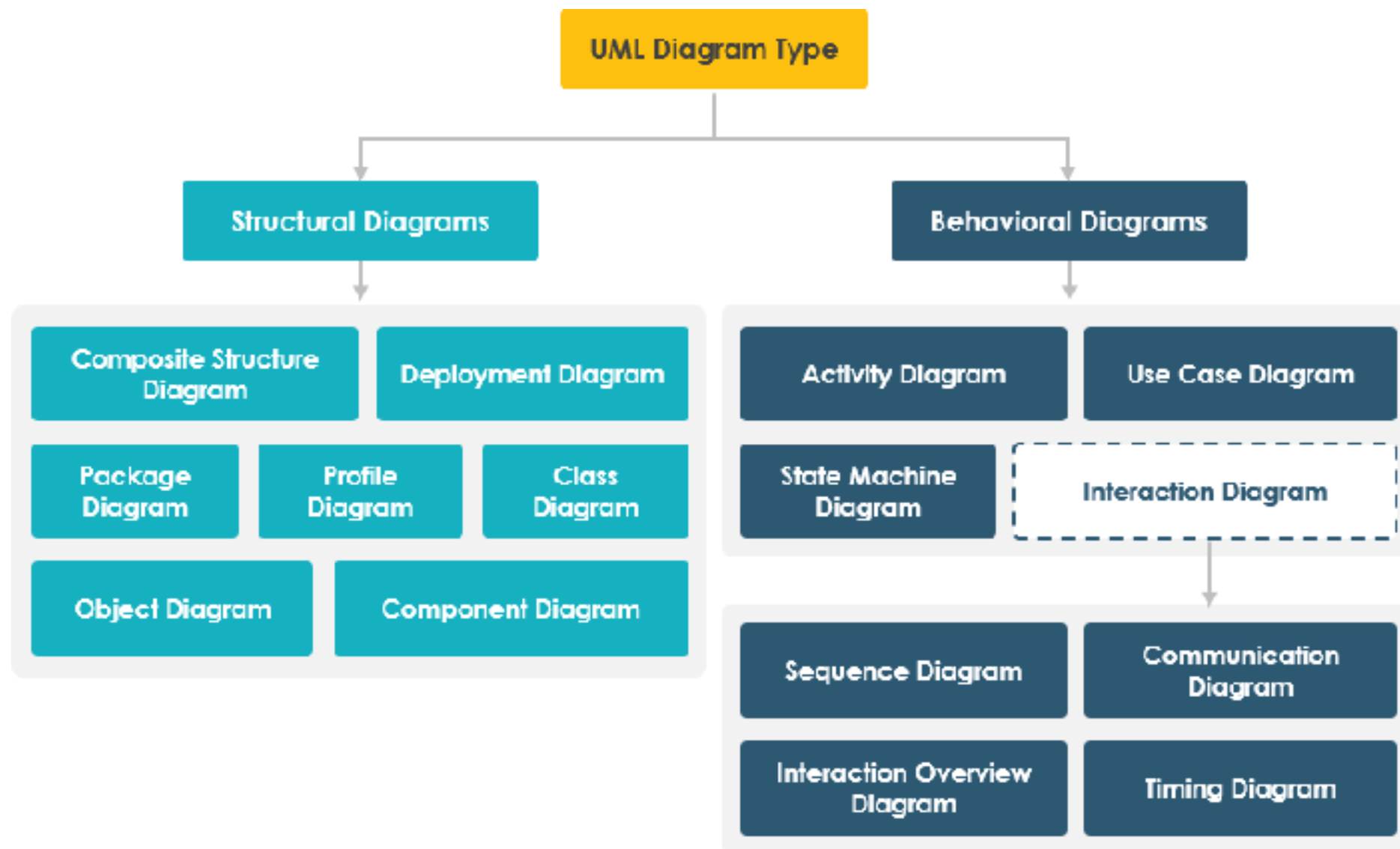
Traditional coding to MDD



<https://arxiv.org/abs/2410.18489>



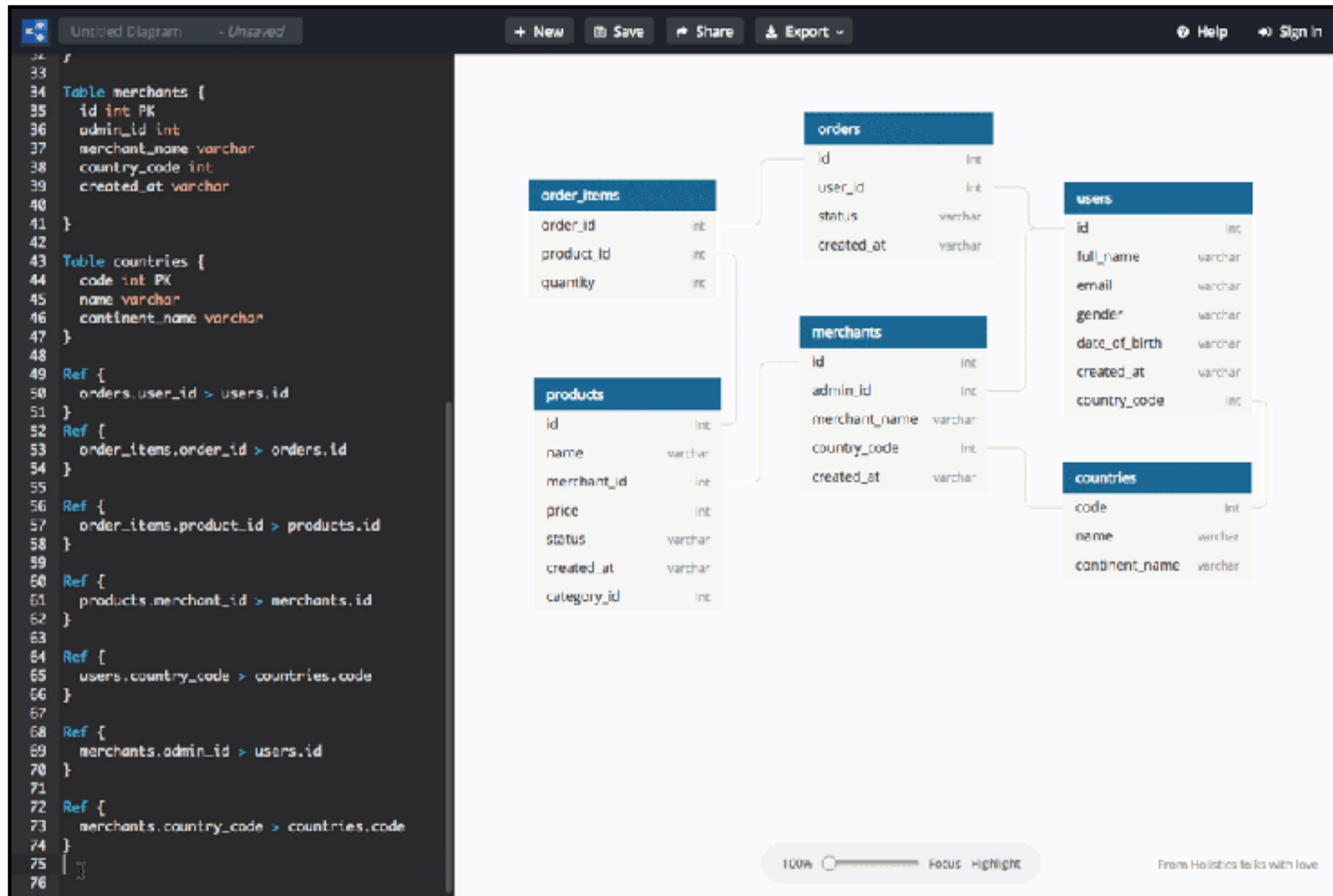
UML Diagrams !!



<https://www.visual-paradigm.com/guide/uml-unified-modeling-language/behavior-vs-structural-diagram/>



ER Diagrams for Database !!



<https://dbml.dbdiagram.io/home/>



Mermaid Diagram

Diagram as a Code (AI-friendly)

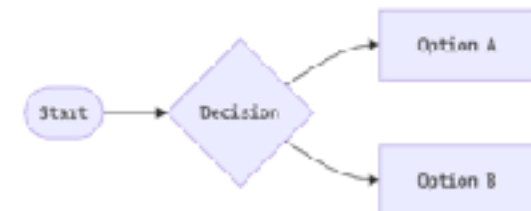
Mermaid Diagramming and charting tool

JavaScript based diagramming and charting tool that renders Markdown-inspired text definitions to create and modify diagrams dynamically.

Try Editor

Get started

```
1 flowchart LR
2   A(["Start"])
3   A --> B["Decision"]
4   B --> C["Option A"]
5   B --> D["Option B"]
```

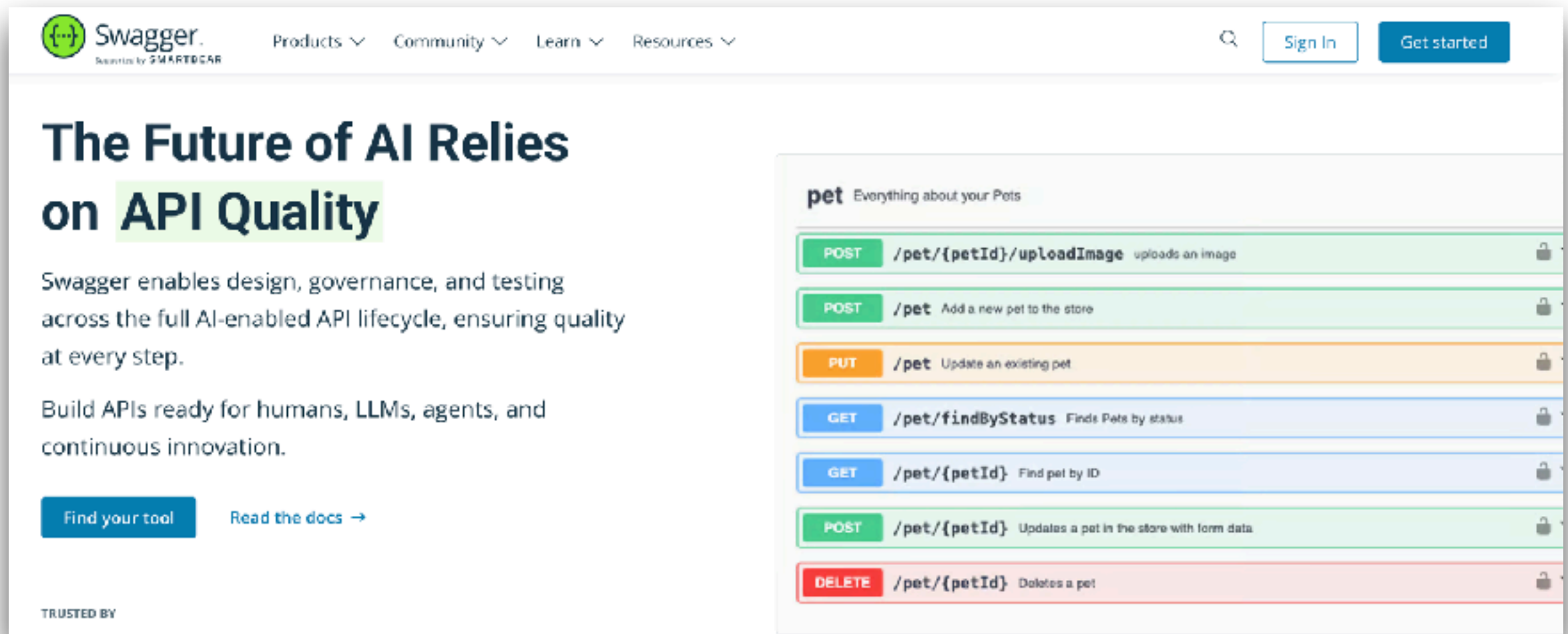


<https://mermaid.js.org/>



Swagger or OpenAPI

Standardized specification for describe REST APIs



The screenshot displays the Swagger.io homepage. On the left, a hero section titled "The Future of AI Relies on API Quality" features a green highlight on "API Quality". Below the title, text states: "Swagger enables design, governance, and testing across the full AI-enabled API lifecycle, ensuring quality at every step." and "Build APIs ready for humans, LLMs, agents, and continuous innovation." Two buttons are present: "Find your tool" and "Read the docs →". At the bottom left, it says "TRUSTED BY".

On the right, a preview of a REST API specification for a "pet" API is shown. The header reads "pet Everything about your Pets". The API endpoints are listed in a table with color-coded rows: POST (green), PUT (orange), GET (blue), and DELETE (red). Each row includes the HTTP method, the endpoint path, a brief description, and a lock icon.

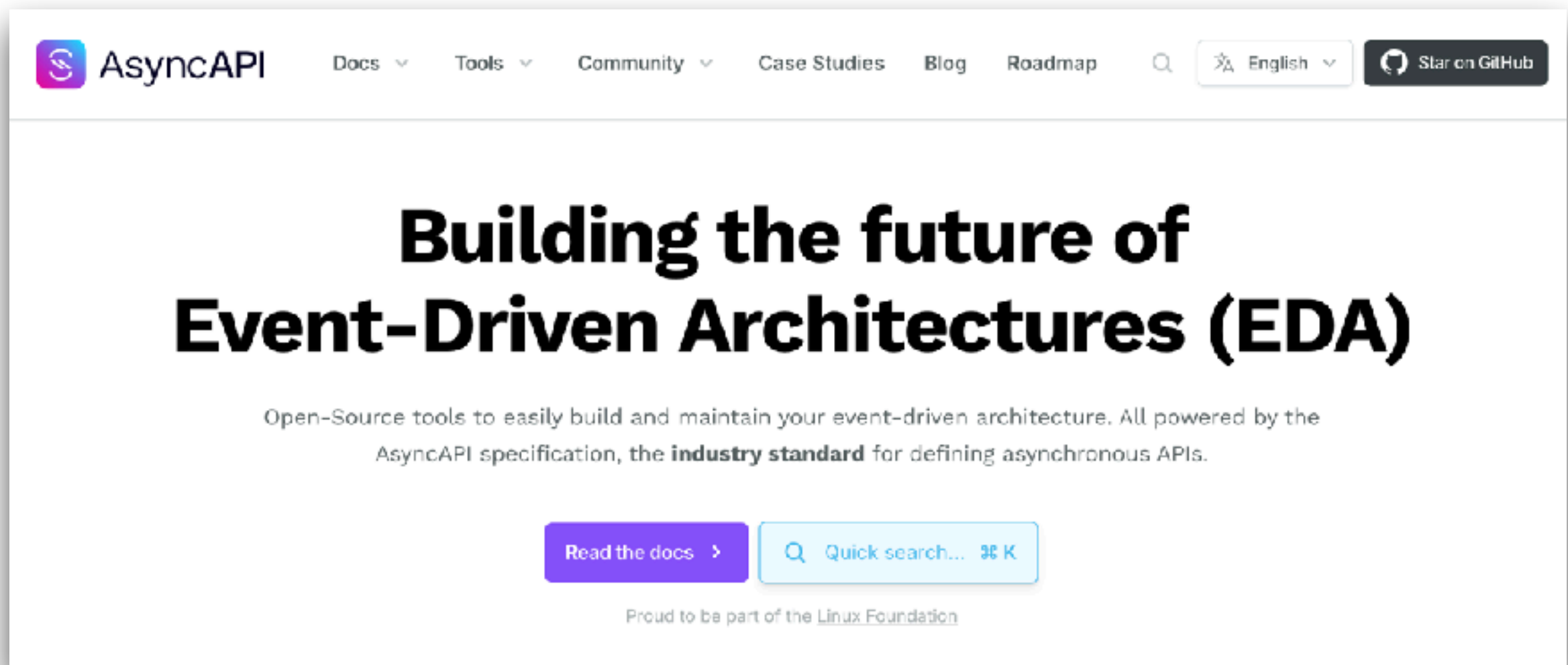
Method	Endpoint	Description
POST	/pet/{petId}/uploadImage	uploads an image
POST	/pet	Add a new pet to the store
PUT	/pet	Update an existing pet
GET	/pet/findByStatus	Finds Pets by status
GET	/pet/{petId}	Find pet by ID
POST	/pet/{petId}	Updates a pet in the store with form data
DELETE	/pet/{petId}	Deletes a pet

<https://swagger.io/>



AsyncAPI

Standardized specification for describe asynchronous and event-driven APIs (Kafka, AMQP, WebSocket)



<https://www.asyncapi.com/>



JSONSchema

Declarative language used to annotate and validate the structure, constraints and data type of JSON documents

Data validation

Documentation

Automated
testing

<https://json-schema.org/>



Test Cases (Human + AI)

Effective regression tests, role in the maintain codebase
More safer to change code !!

Coverage

Property-based
testing

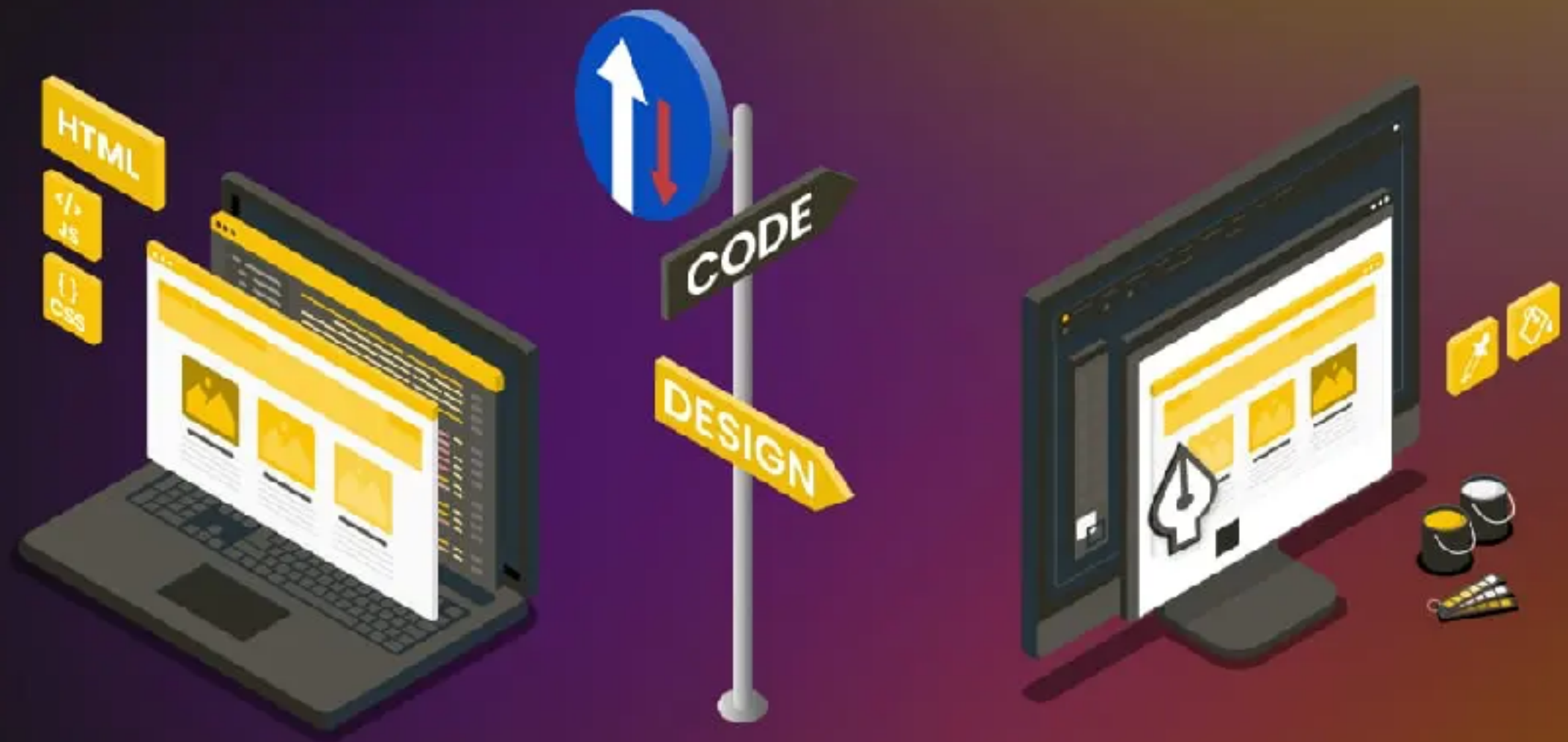
Mutation
testing

Fuzz testing

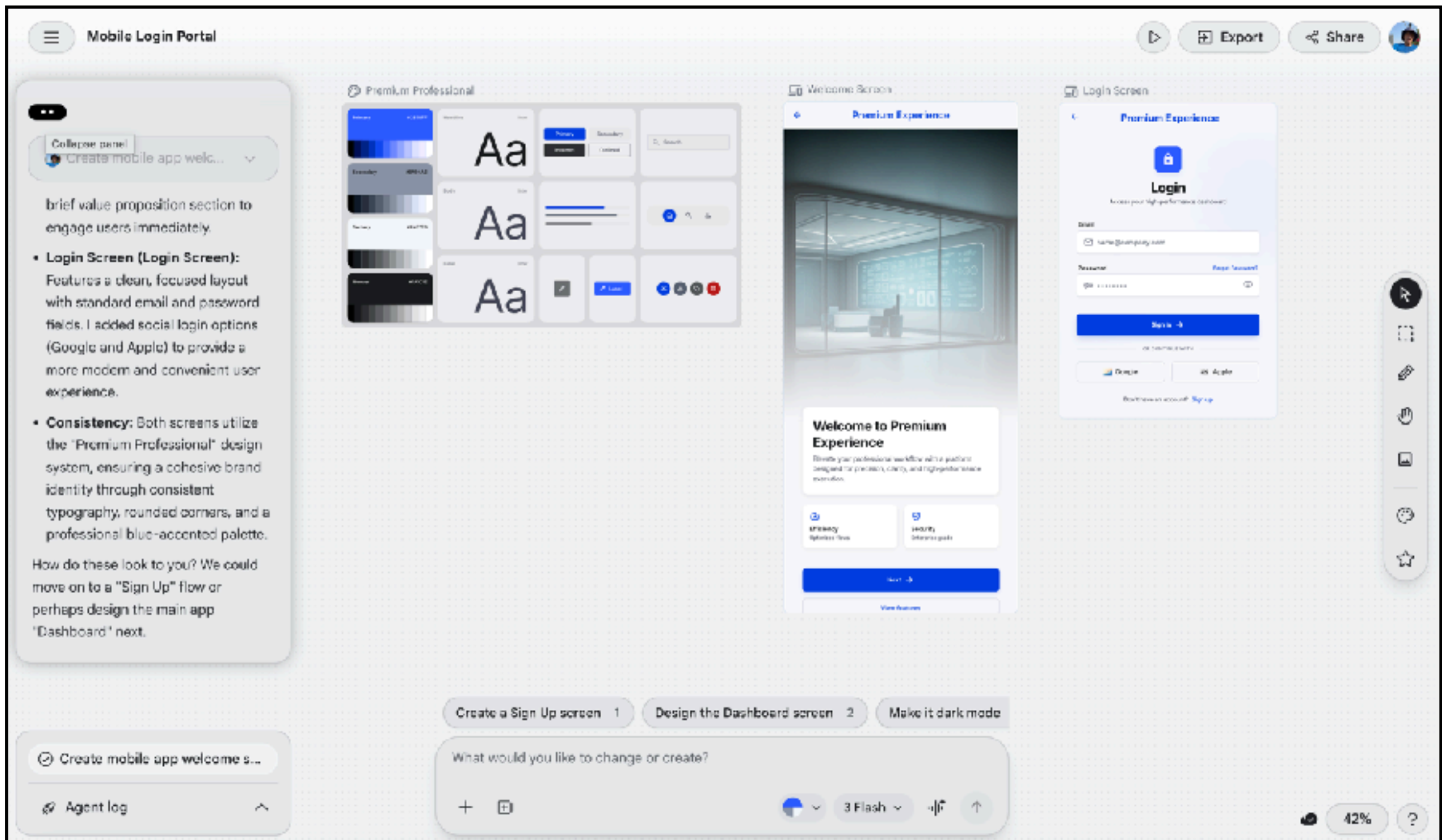
<https://martinfowler.com/articles/sensors-for-coding-agents.html>



Design to Code Code to Design



Google Stitch



<https://stitch.withgoogle.com/>

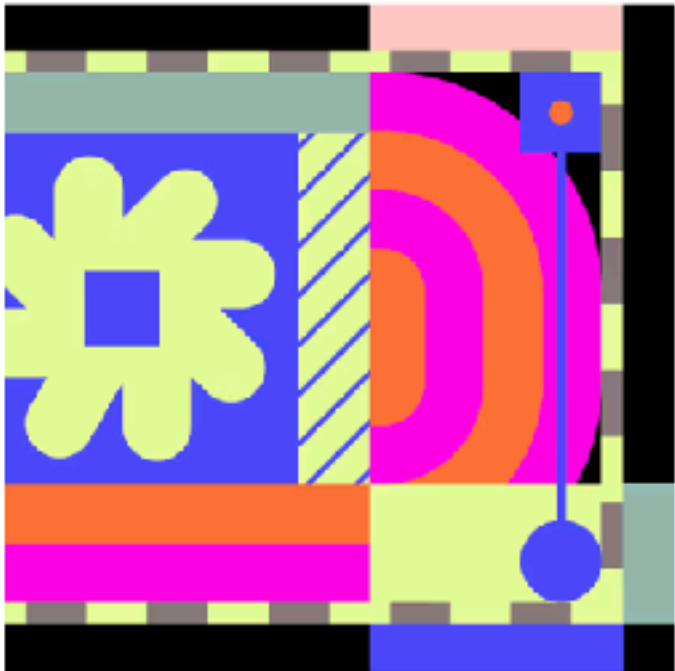


Figma

MCP CLIENT TOOLS

Connect Figma to top MCP clients

However you like to code, the Figma MCP server adds Figma context to your favorite agentic tools.





Claude Code

- Supports Figma Design, Figma Make, and FigJam
- Remote and local servers
- Read and write access

[View install guide](#)



Claude

- Supports Figma Design, Figma Make, and FigJam
- Remote and local servers
- Read and write access

[View install guide](#)



Cursor

- Supports Figma Design, Figma Make, and FigJam
- Remote and local servers
- Read and write access

[Add MCP to Cursor](#)



VS Code

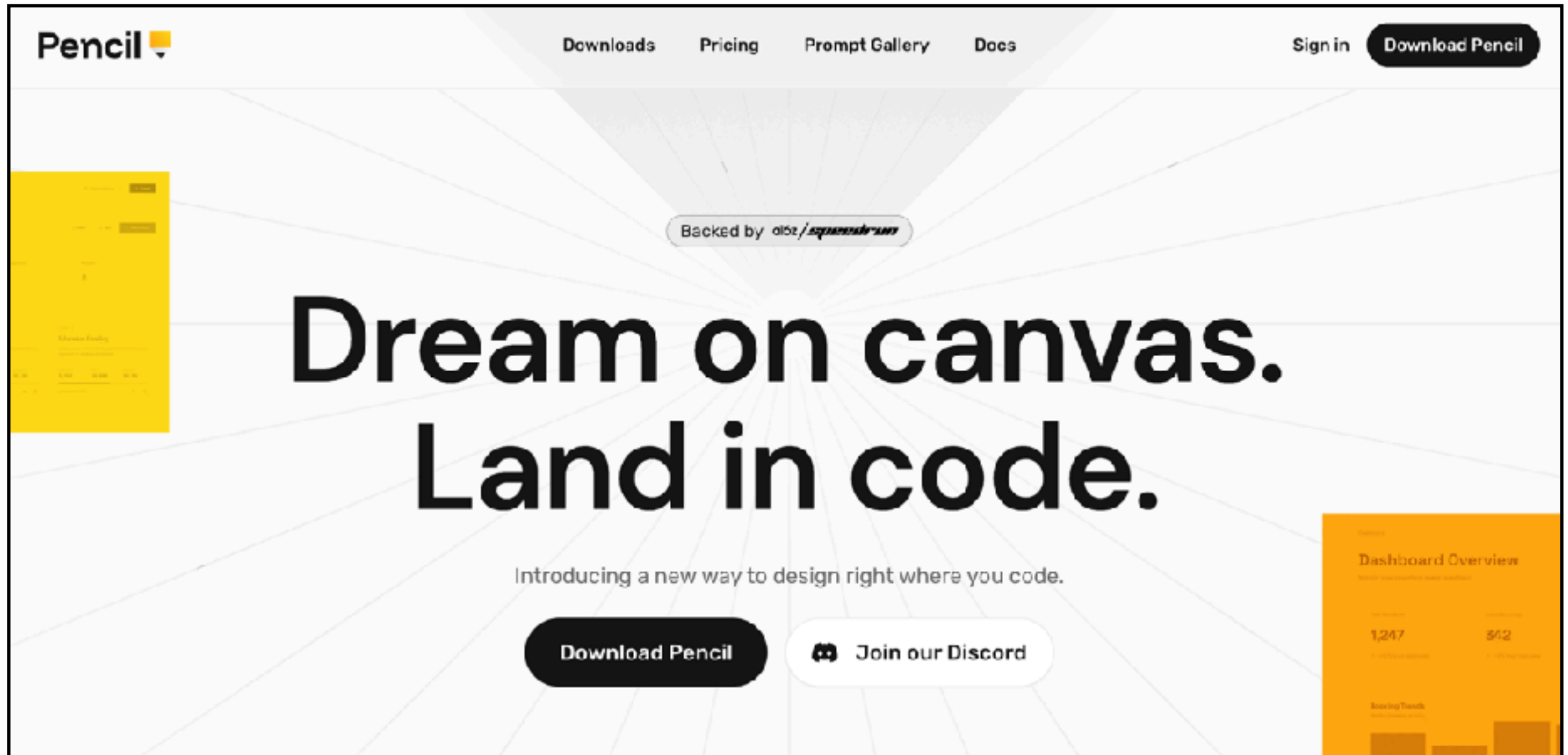
- Supports Figma Design, and FigJam
- Remote and local servers
- Read and write access

[Add MCP to VS Code](#)

<https://www.figma.com/mcp-catalog/>



Pencil



<https://www.pencil.dev/>



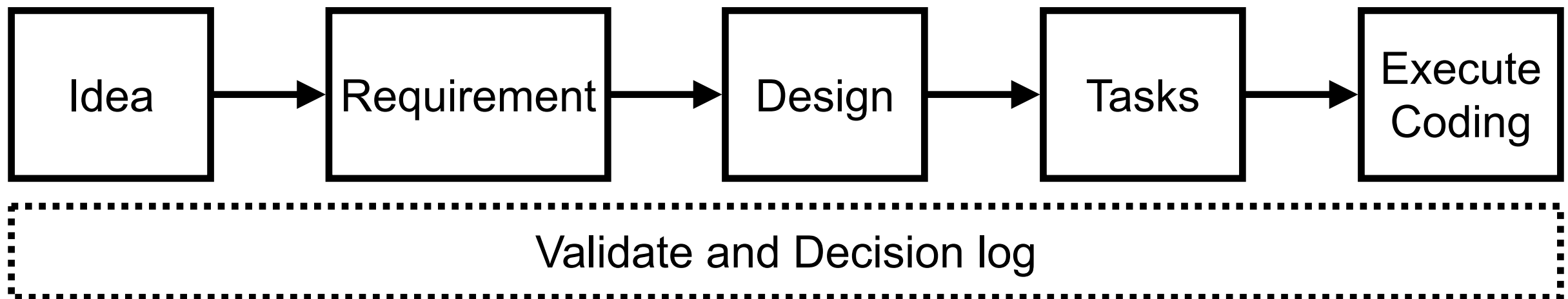
Specification-Driven Development (SDD)



SDD ?

Specification or Documentation-First

Modern software development approach where detailed, structure specifications are created before writing code

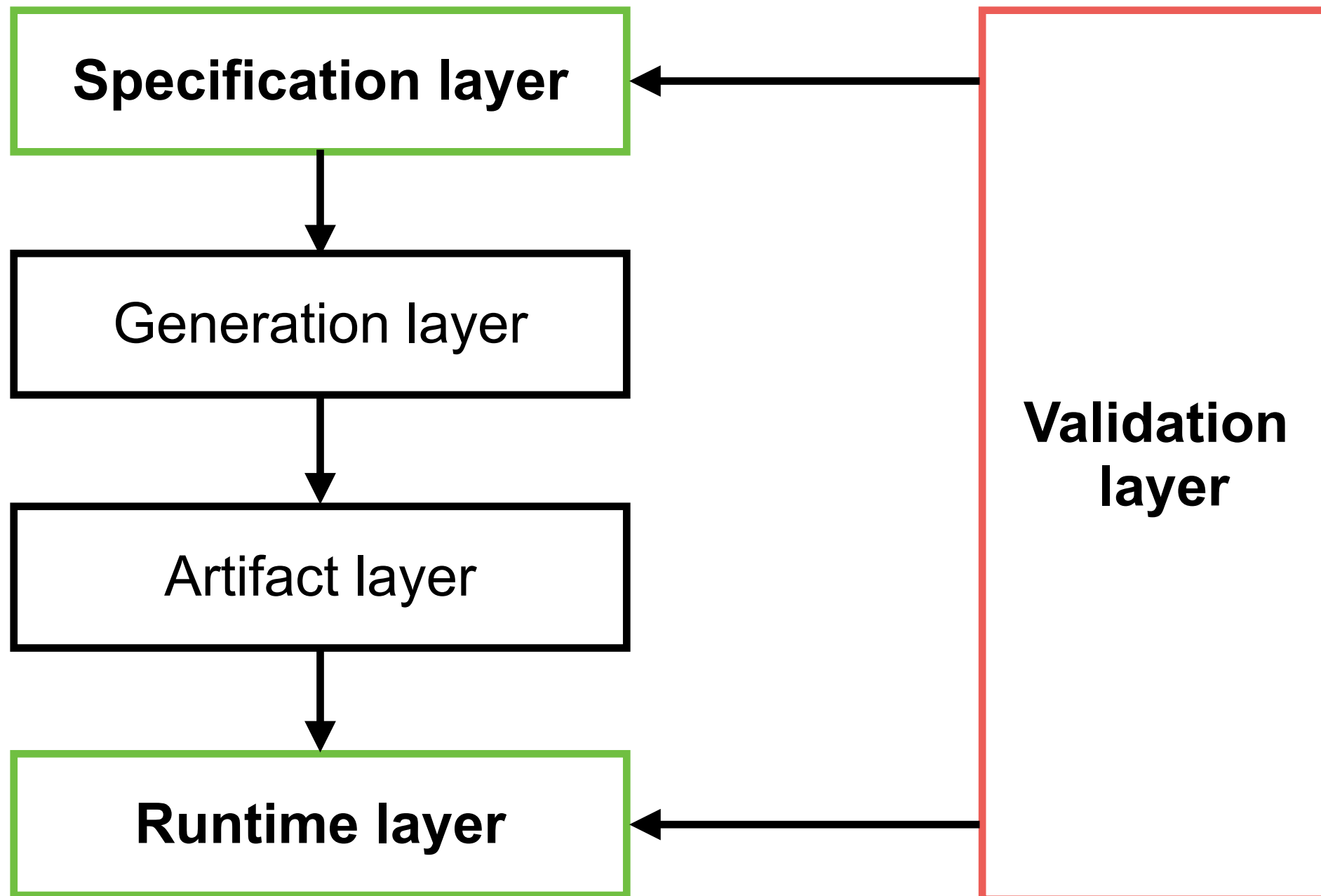


Types of Specification ?

User requirements
Design documents
Interface contracts (UI, API)
Data schemas
Validation rules
Data/business flows
Security policy and constraints
Resources and performance constraints



SSD Architecture



<https://www.infoq.com/articles/spec-driven-development>



MDD + SDD == Better result



When to use SDD ?

No one-size-fit-all !!

Low complexity

High complexity

<https://medium.com/google-cloud/benefits-and-challenges-of-spec-driven-development-and-how-antigravity-is-changing-the-game-3343a6942330>



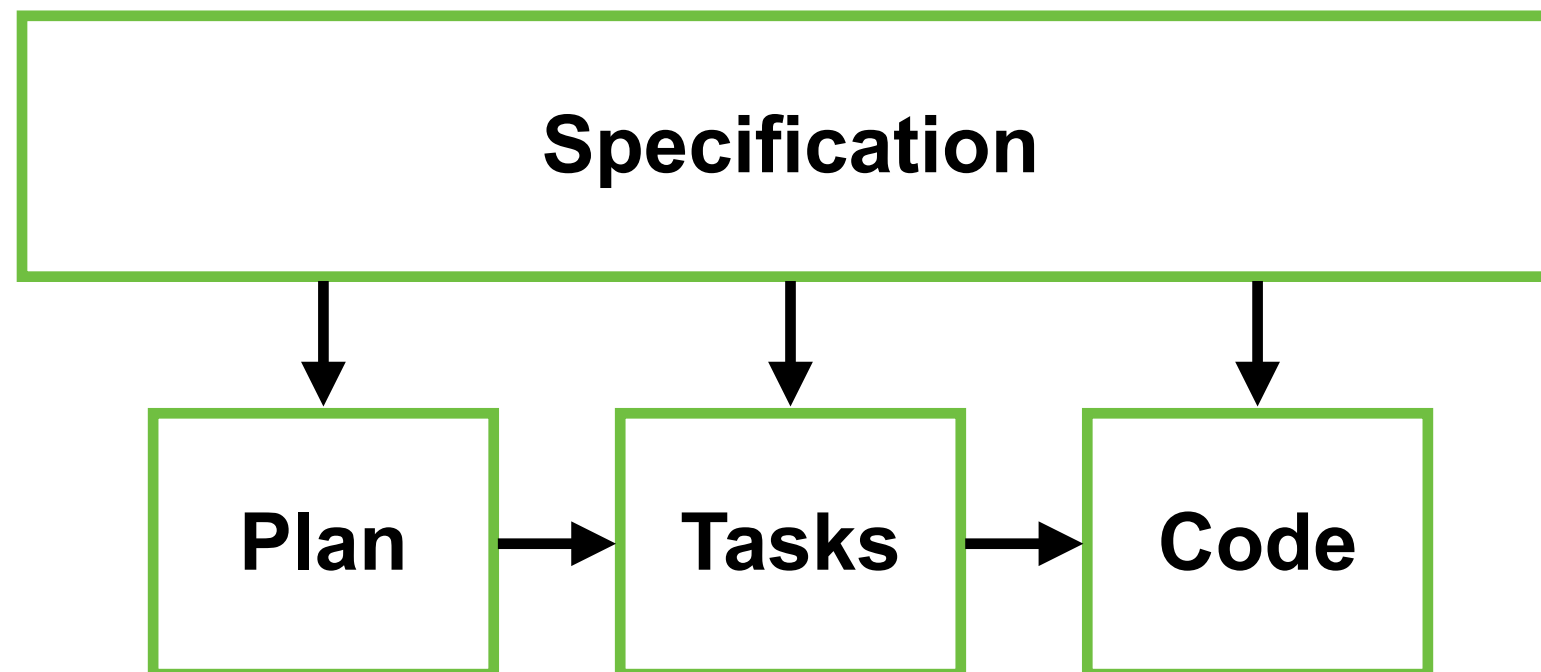
SDD Tools ?

GitHub Spec-kit

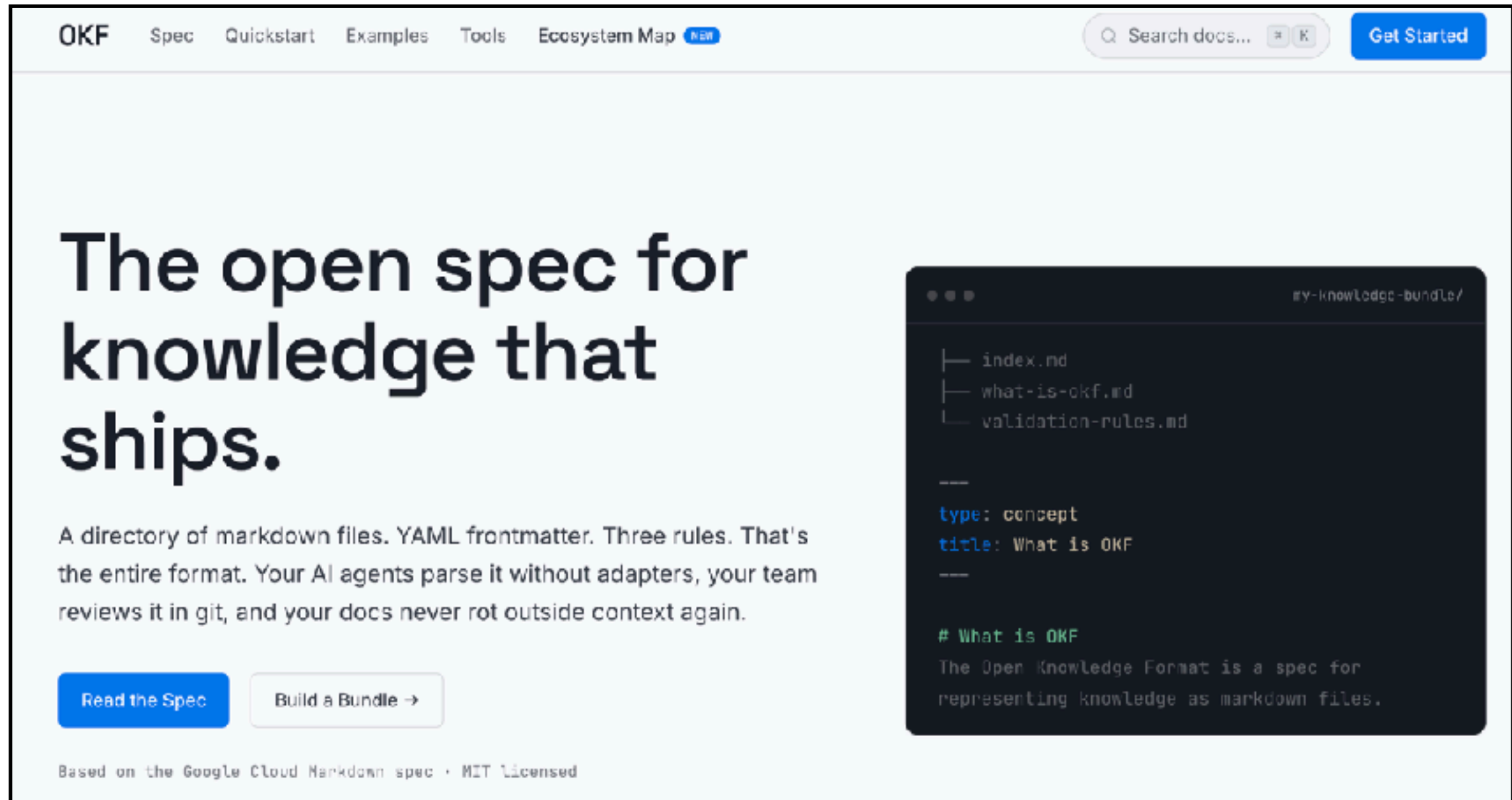
Open Spec

Kiro.dev

Tessl



Open Knowledge Format (OKF)



OKF Spec Quickstart Examples Tools Ecosystem Map NEW

Search docs... # K [Get Started](#)

The open spec for knowledge that ships.

A directory of markdown files. YAML frontmatter. Three rules. That's the entire format. Your AI agents parse it without adapters, your team reviews it in git, and your docs never rot outside context again.

[Read the Spec](#) [Build a Bundle →](#)

Based on the Google Cloud Markdown spec • MIT Licensed

```
my-knowledge-bundle/  
├── index.md  
├── what-is-okf.md  
└── validation-rules.md  
  
---  
type: concept  
title: What is OKF  
---  
  
# What is OKF  
The Open Knowledge Format is a spec for  
representing knowledge as markdown files.
```

<https://okf.md/>



Easy Approach to Requirement Syntax (EARS)



<https://alistairmavin.com/ears/>



Try by yourself !!



CLAUDE
CODE



Design your Workflow ?

Start with requirements

Frontend

Backend

Database

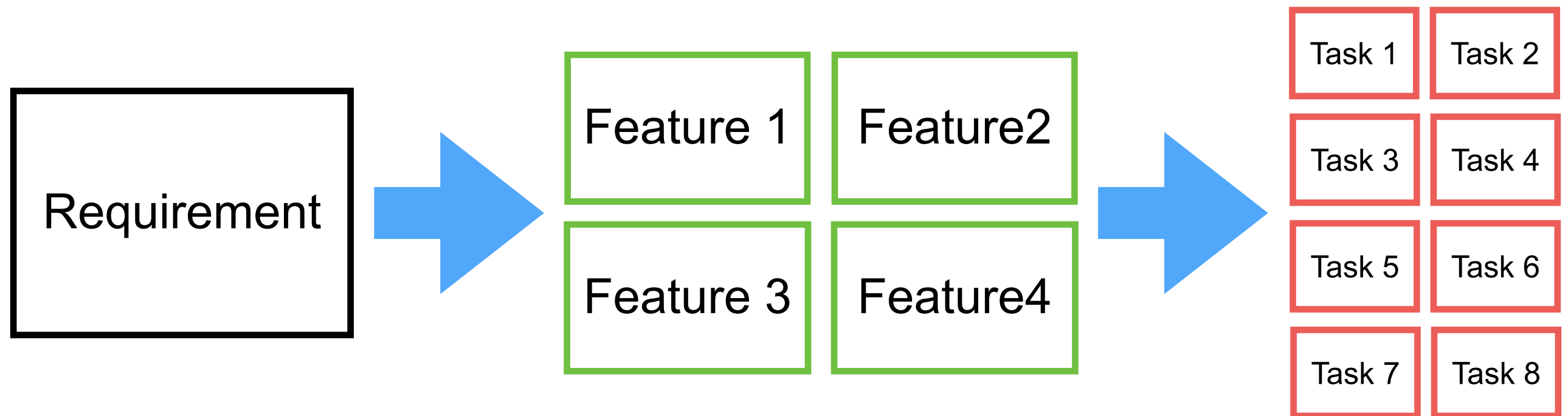


Requirement (Human + AI)

Understand overall scope of system

Clarify

Try to break down into features and tasks



Working with Grill-me skill



<https://github.com/mattpocock/skills/blob/main/skills/productivity/grill-me/SKILL.md>



Superpowers skill



<https://github.com/obra/Superpowers>



Working with Agent-Skills



<https://github.com/addyosmani/agent-skills>



Write specification !!

Frontend

Backend



Frontend ?

Mockup User Interface

Design System

User workflow

Input validation

API contract (spec)

Test cases



Backend ?

API contract (spec)

Business flow

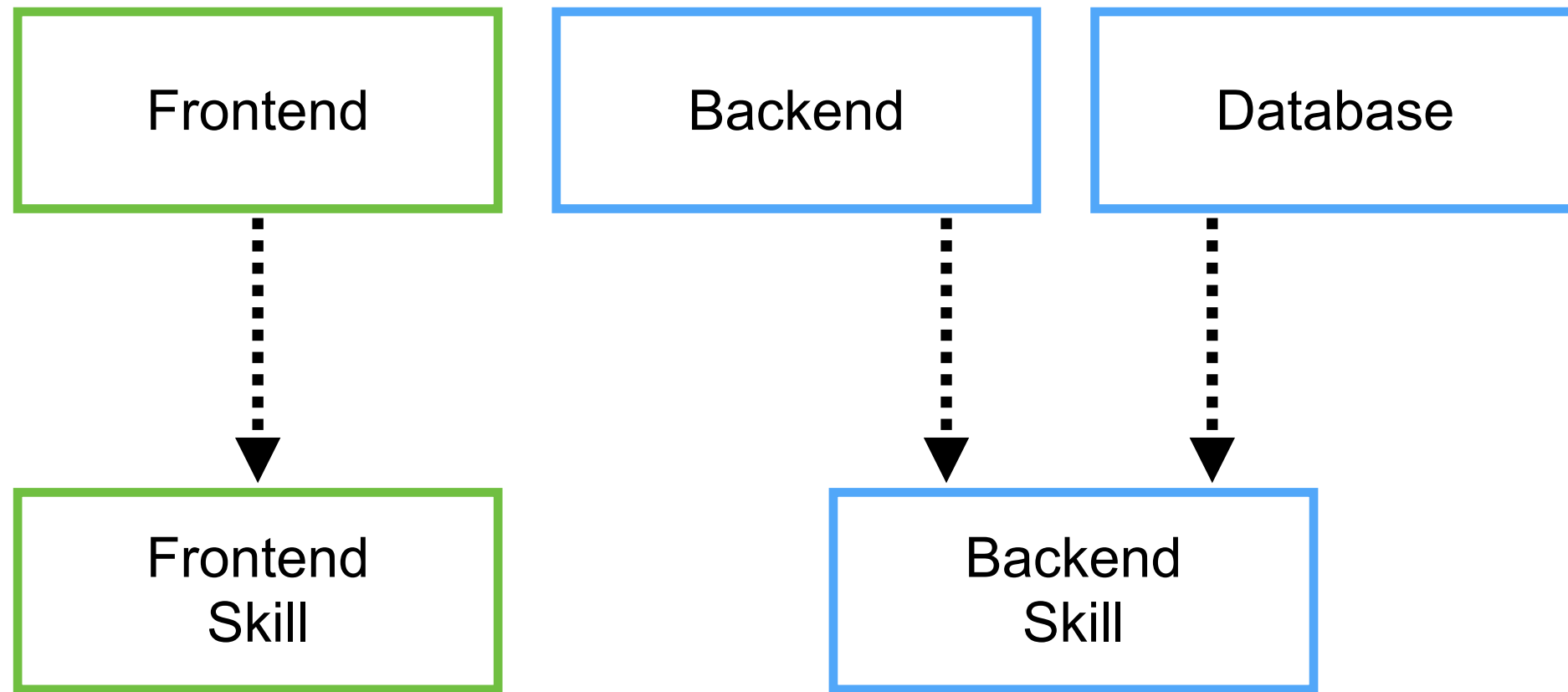
Input validation

Database design (ER diagram)

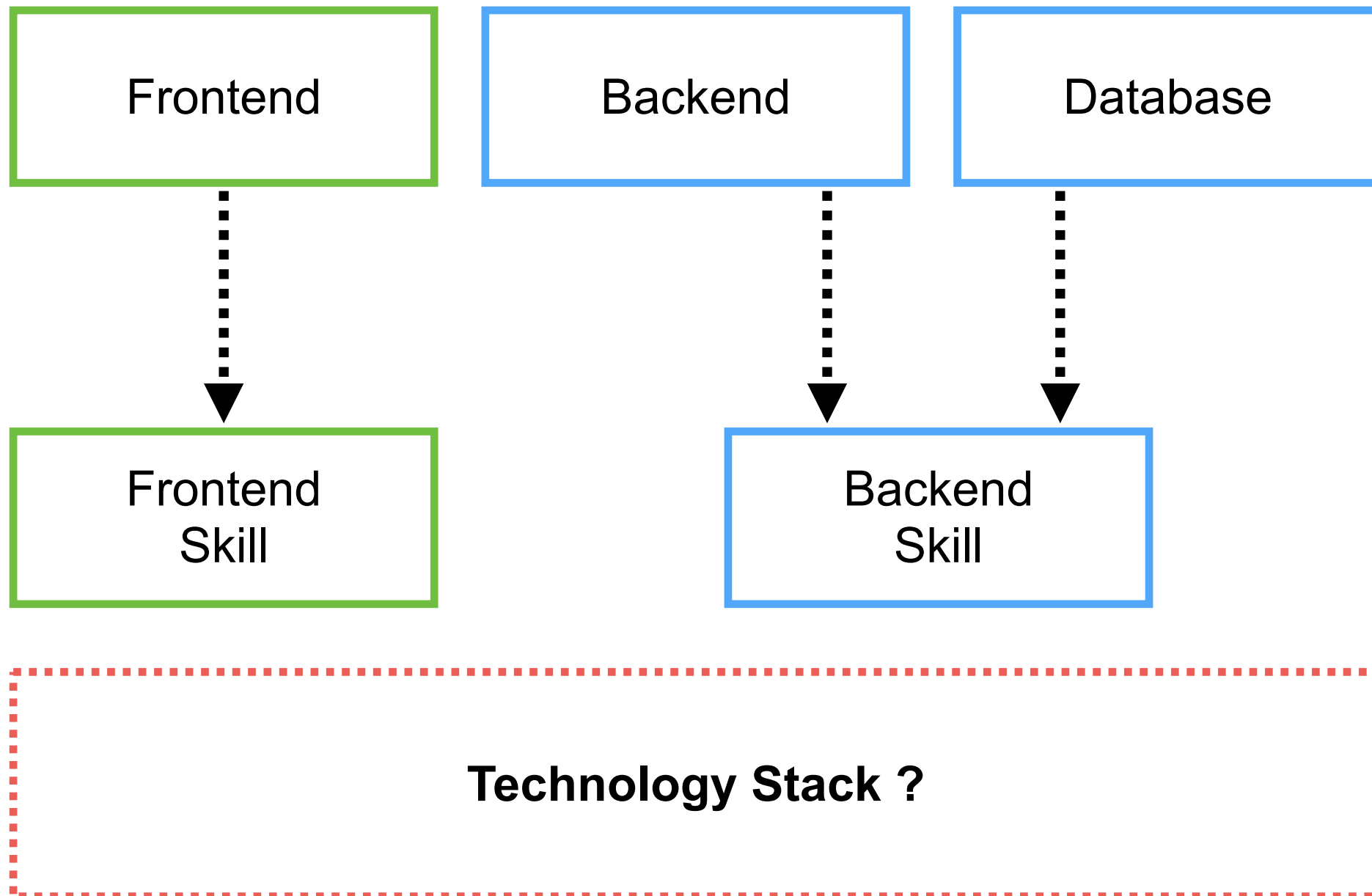
Test cases



Design AI Agent Skills ?



Technology Stack ?



Create AI Agent Skills !!

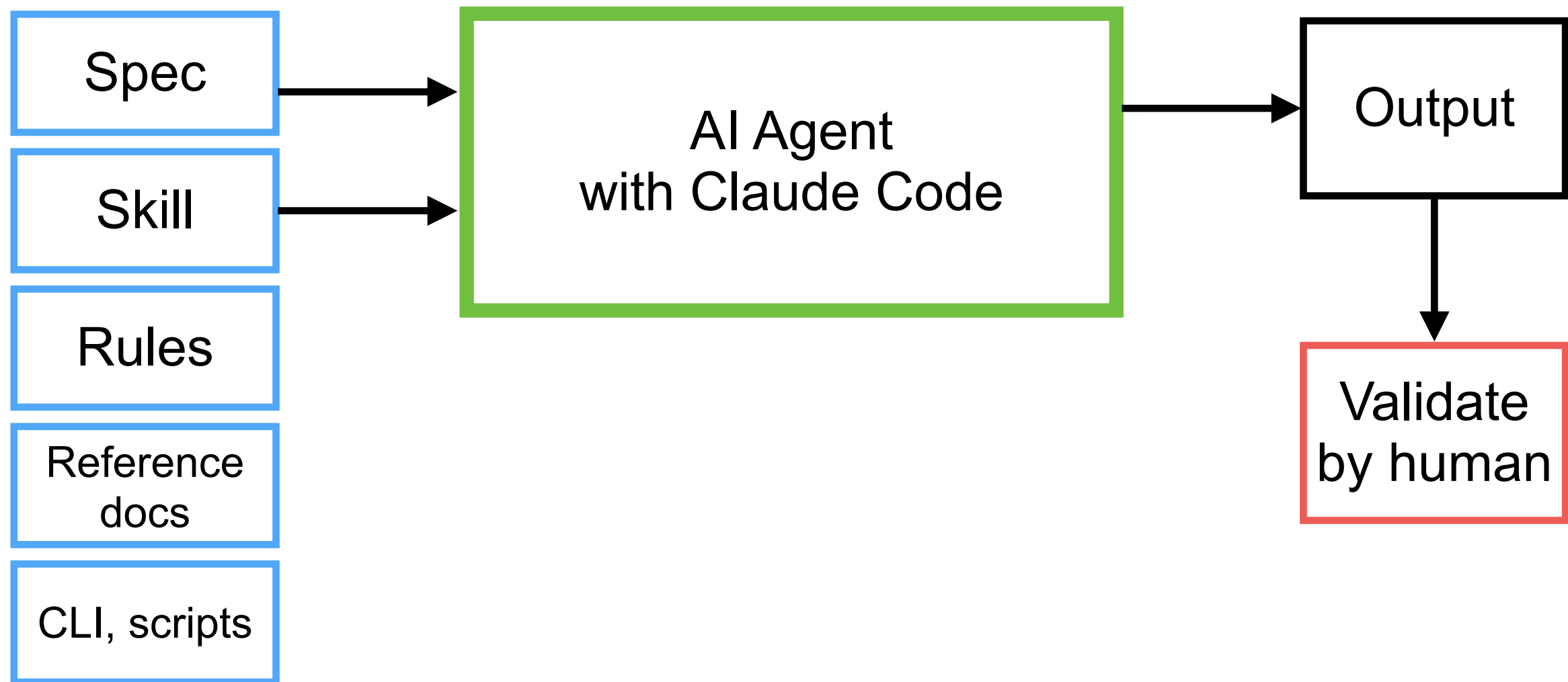


Let' start



Simple Workflow (1)

Provide **context** that feedforward to AI Agent

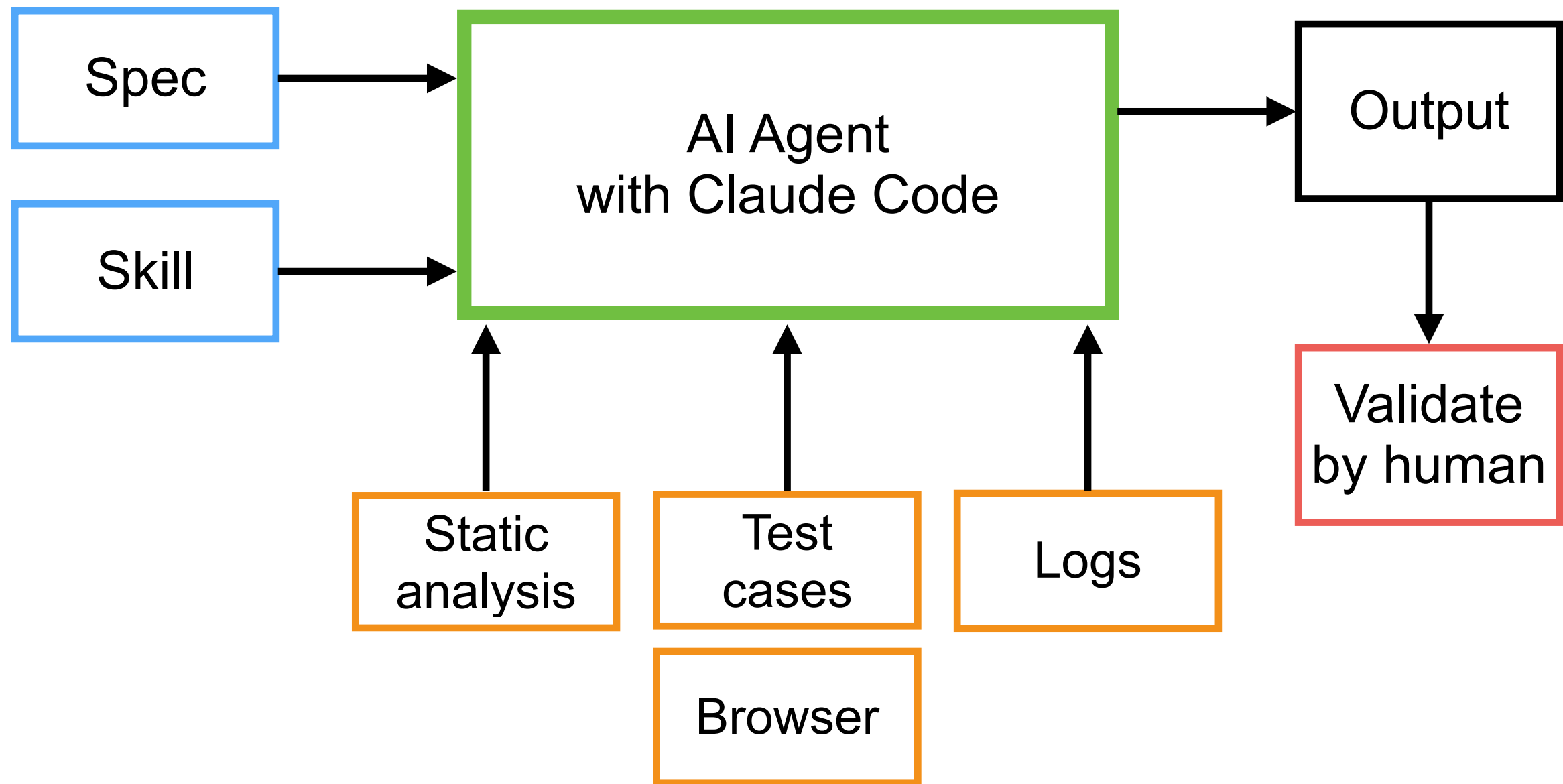


<https://martinfowler.com/articles/harness-engineering.html>



Simple Workflow (2)

Provide **feedback** to AI agent

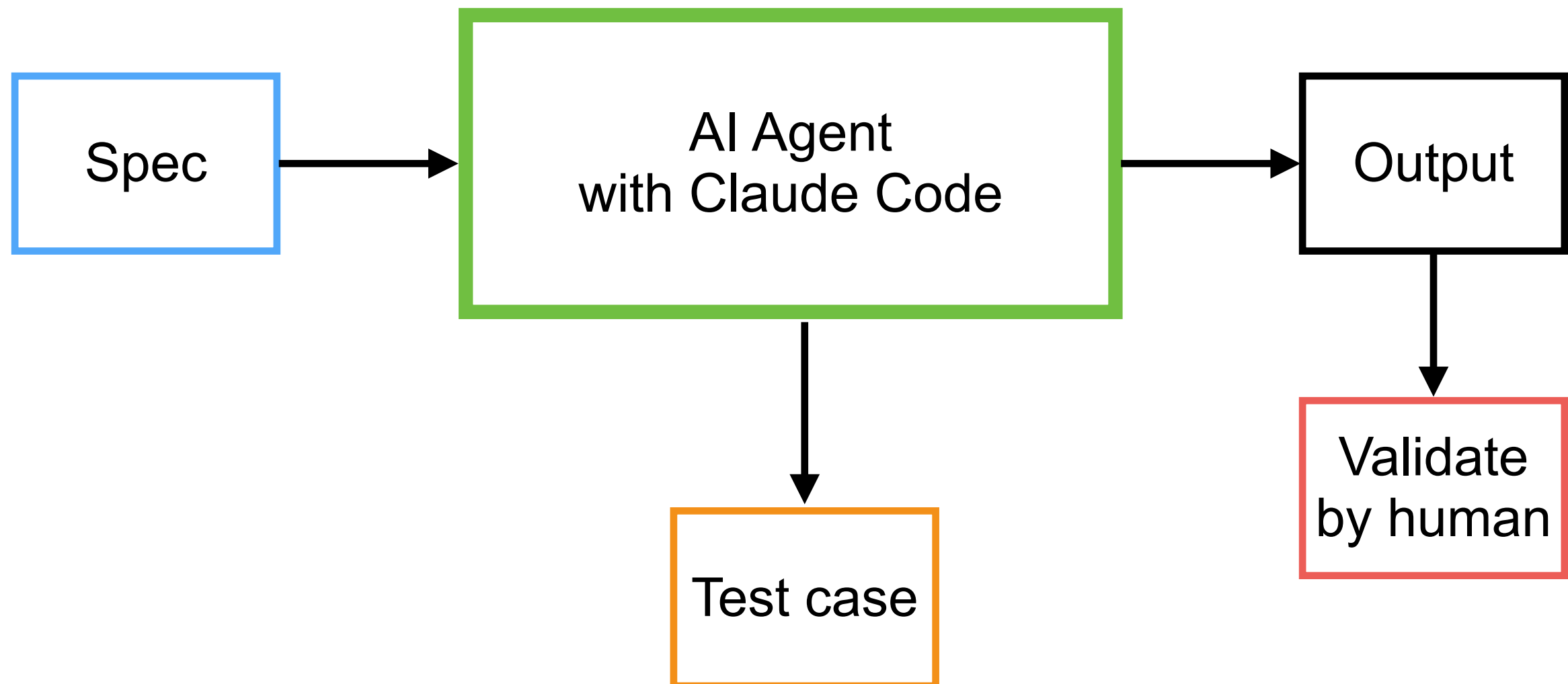


<https://martinfowler.com/articles/sensors-for-coding-agents.html>



Simple Workflow (3)

Start with **plan** mode !!



Plan before Coding



Verify by Human



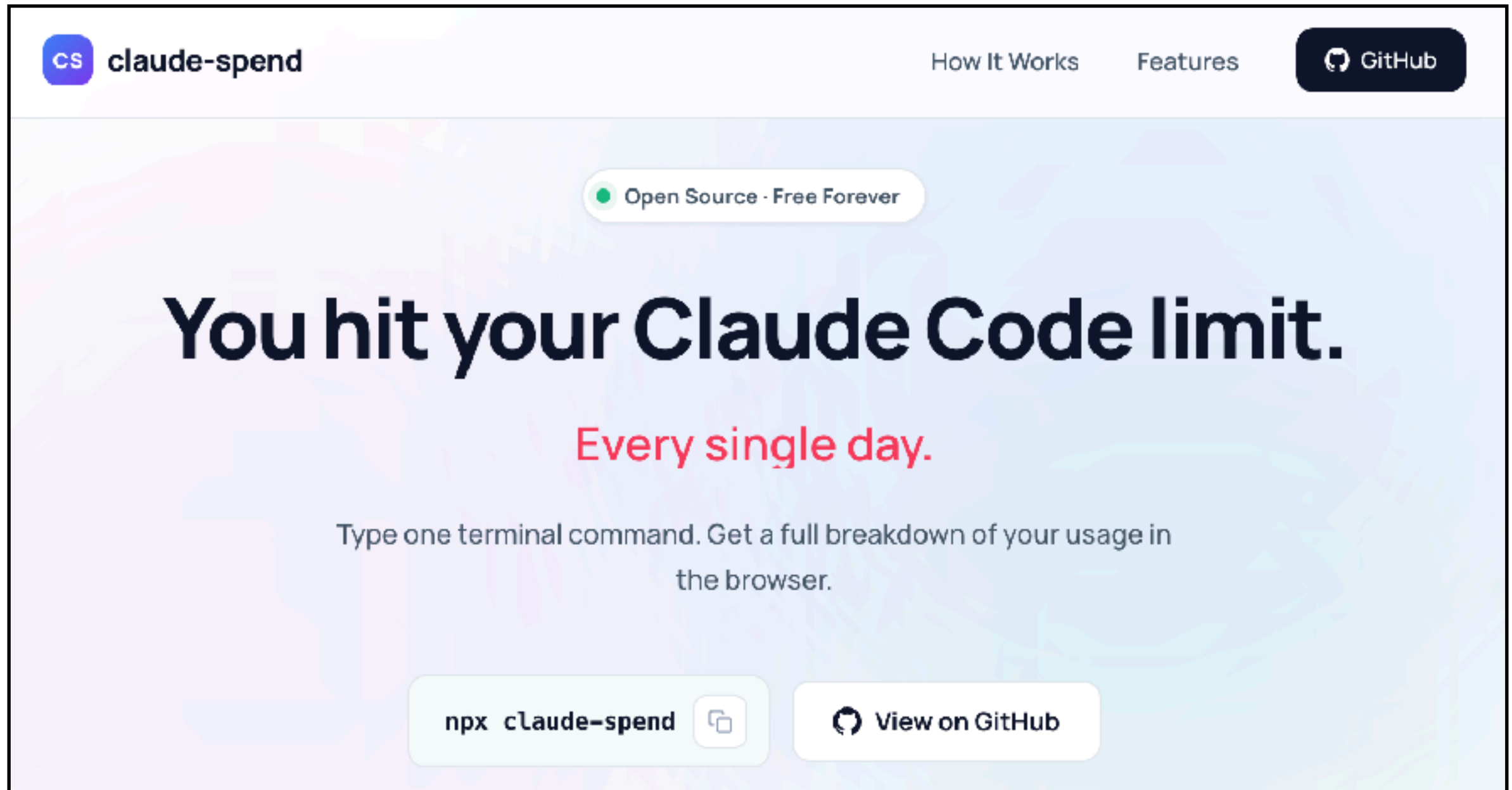
Token Usage ?



<https://github.com/phury/claude-usage>



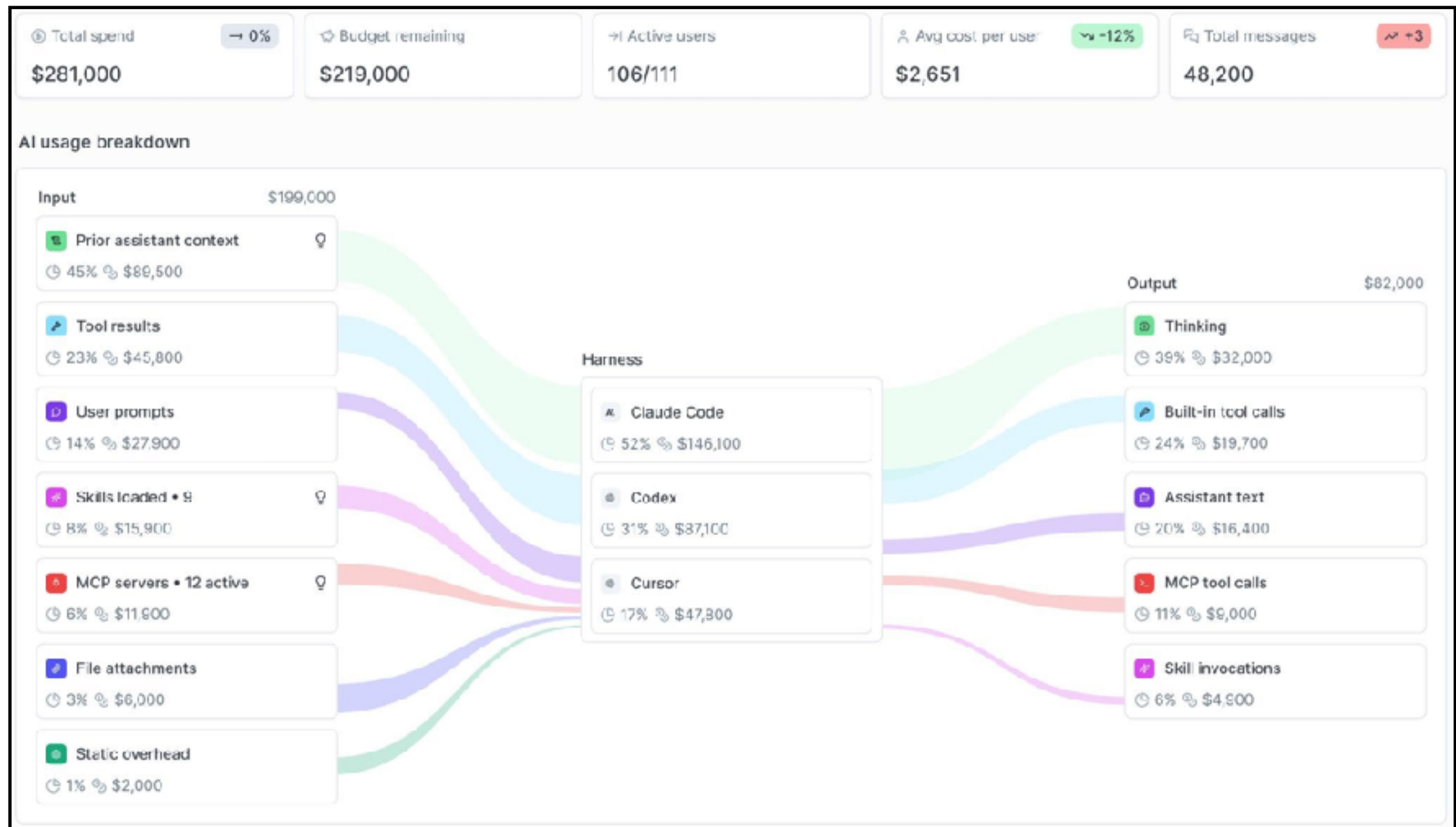
Token Usage ?



<https://claudespend.live/>



Token Usage ?



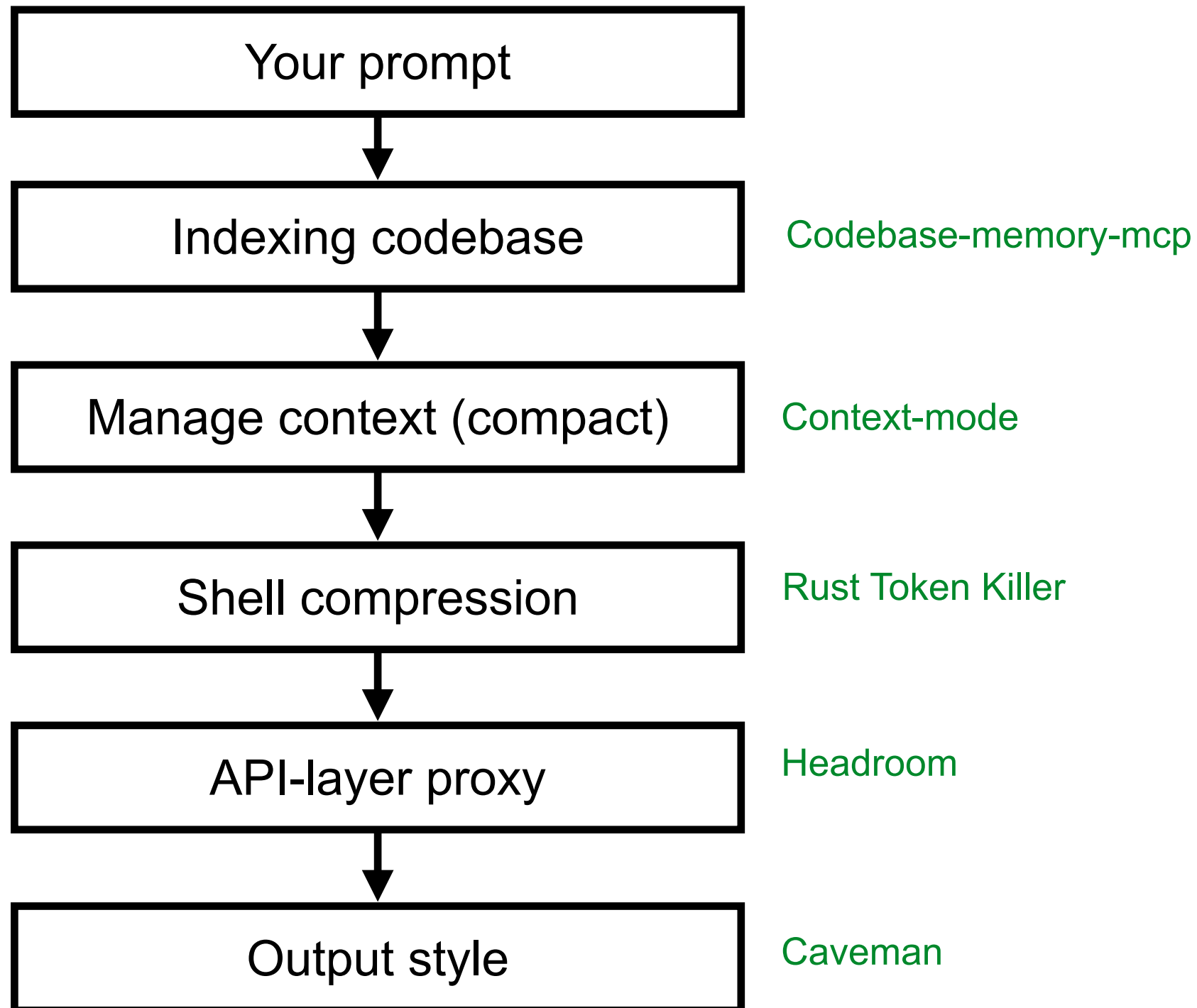
<https://www.comet.com/site/products/opik/features/ai-spend-tracker/>



Layer to reduce token usage ?



Working with Large Codebase



Token vs Accuracy ?



Q/A

